

Electrical circuits Lab

هذا العمل صدقة جارية عن أرواح موتى وشهداء المسلمين عامة وشهداء فلسطين خاصة اللهم إنهم في ذمتك وحبل جوارك، فقههم من فتنة القبر وعذاب النار، وأنت أهل الوفاء والحق، اللهم اغفر لهم وارحمهم، فإنك أنت الغفور الرحيم
اللهم اغفر لهم وارحمهم واعف عنهم وعافهم، وأكرم نزلهم ووسع مدخلهم، واغسلهم بالماء والثلج والبرد، ونقهم من الخطايا كما ينقى الثوب الأبيض من الدنس، وأبدلهم داراً خيراً من دارهم وأهلاً خيراً من أهلهم وأزواجاً خيراً من أزواجهم، وقهم فتنة القبر وعذاب النار

اللهم إنهم عبيدك وبناء عبيدك وبناء إماءك كانوا يشهدون أن لا إله إلا أنت، وأن محمداً عبدك ورسولك، وأنت أعلم بهم، اللهم إن كانوا محسنين فزدهم في إحسانهم، وإن كانوا مسيئين فتجاوز عن سيئاتهم، اللهم لا تحرمنا أجرهم ولا تفتنا بعدهم

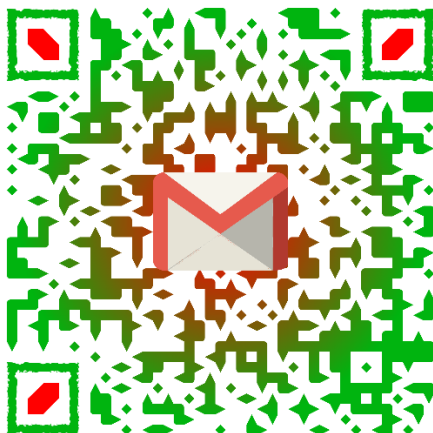
اللهم سدّد رمي المقاومة
اللهم لا ترفع لليهود راية ولا تحقق لهم غاية واجعلهم لمن خلفهم عبرة وآية

قال تعالى: ﴿سُبْحَانَ الَّذِي أَسْرَى بِعَبْدِهِ لَيْلًا مِنَ الْمَسْجِدِ الْحَرَامِ إِلَى الْمَسْجِدِ الْأَقْصَا الَّذِي بَارَكْنَا حَوْلَهُ لِنُرِيَهُ مِنْ آيَاتِنَا إِنَّهُ هُوَ السَّمِيعُ الْبَصِيرُ * وَآتَيْنَا مُوسَى الْكِتَابَ وَجَعَلْنَاهُ هُدًى لِّبَنِي إِسْرَائِيلَ أَلَّا تَتَّخِذُوا مِنْ دُونِي وَكِيلًا * ذُرِّيَّةً مِنْ حَمَلْنَا مَعَ نُوحٍ إِنَّهُ كَانَ عَبْدًا شَكُورًا * وَقَضَيْنَا إِلَى بَنِي إِسْرَائِيلَ فِي الْكِتَابِ لَتُفْسِدُنَّ فِي الْأَرْضِ مَرَّتَيْنِ وَلَتَعْلُنَّ عُلُوقًا كَبِيرًا * فَإِذَا جَاءَ وَعْدُ أُولَاهُمَا بَعَثْنَا عَلَيْكُمْ عِبَادًا لَنَا أُولِي بَأْسٍ شَدِيدٍ فَجَاسُوا خِلَالَ الدِّيَارِ وَكَانَ وَعْدًا مَفْعُولًا * ثُمَّ رَدَدْنَا لَكُمُ الْكَرَّةَ عَلَيْهِمْ وَأَمْدَدْنَاكُمْ بِأَمْوَالٍ وَبَنِينَ وَجَعَلْنَاكُمْ أَكْثَرَ نَفِيرًا * إِنْ أَحْسَنْتُمْ أَحْسَنْتُمْ أَنْفُسَكُمْ وَإِنْ أَسَأْتُمْ فَلَهَا فَإِذَا جَاءَ وَعْدُ الْآخِرَةِ لِيَسُوءُوا وُجُوهَكُمْ وَلِيَدْخُلُوا الْمَسْجِدَ كَمَا دَخَلُوهُ أَوَّلَ مَرَّةٍ وَلِيُتَبِّرُوا مَا عَلَوْا تَتْبِيرًا * عَسَى رَبُّكُمْ أَنْ يَرْحَمَكُمْ وَإِنْ عُذْتُمْ عُنَا وَجَعَلْنَا جَهَنَّمَ لِلْكَافِرِينَ حَصِيرًا﴾ {سورة الإسراء

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

والكمال لله وحده لهذا من الممكن أن تتواجد بعض الأخطاء في الحلول لذلك نرجو منكم الاستفسار عن الأسئلة التي تريدون السؤال عنها بالتواصل مع منشأ هذا العمل

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Previous Exams Questions

1. Electrical resistance is measured by:
 - a. Voltmeter
 - b. Ohmmeter
 - c. Ammeter

Correct answer: Ohmmeter

2. An ideal ammeter has equivalent resistance =
 - a. Zero
 - b. R Load
 - c. Infinity

Correct answer: Zero

Ammeter is connected in series with the element so the internal resistance of the ammeter must be equal zero theoretically, and practically, it's very small

3. An ideal voltmeter has equivalent resistance =
 - a. Zero
 - b. Infinity
 - c. R Load

Correct answer: Infinity

Voltmeter is connected in parallel with the element so the internal resistance of the voltmeter must be equal infinity theoretically, and practically, it's very high

4. Nodal analysis is a method depends on:
 - a. Kirchhoff's voltage law (KVL)
 - b. Ohm's law
 - c. Kirchhoff's current law (KCL)

Correct answer: Kirchhoff's current law (KCL)

5. Mesh analysis is a method depends on:
 - a. Kirchhoff's voltage law (KVL)
 - b. Kirchhoff's current law (KCL)
 - c. Ohm's law

Correct answer: Kirchhoff's voltage law (KVL)

6. The voltmeter is connected in:
 - a. Series
 - b. Parallel
 - c. Any of them

Correct answer: parallel

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7. The ammeter is connected in:
- Series
 - Parallel
 - Any of them

Correct answer: series

8. The Thevenin equivalent voltage for a given circuit found by:
- Opening the output terminals
 - Shorting the voltage source
 - Opening the voltage source
 - Shorting the output terminals

Correct answer: Opening the output terminals

9. When a sine wave has a frequency of 50Hz in 10 seconds it goes through:
- 5 cycles
 - 10 cycles
 - 1/5 cycle
 - 500 cycles

Correct answer: 500 cycles

In one second, the number of cycles is equal the frequency of a sine wave, so for 50Hz sine wave, the number of cycles per second is 50 cycles, therefore, the number of cycles in 10 seconds is:

$$10\text{second} * \frac{50\text{ cycles}}{\text{second}} = 500\text{ cycles}$$

10. Voltmeter on series will act as:
- OR
- Voltmeter on parallel will act as:
- OR
- Voltmeter will act as:
- Open circuit
 - Short circuit

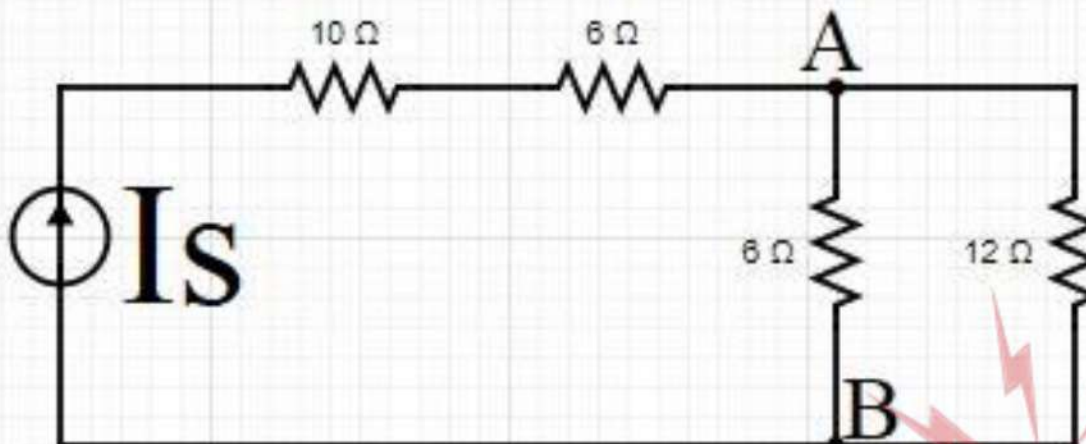
Correct answer: Open circuit

11. Ammeter on series will act as:
- OR
- Ammeter on parallel will act as:
- OR
- Ammeter will act as:
- Open circuit
 - Short circuit

Correct answer: Short circuit

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12. If $V_{AB} = 32V$, the source current is:



- a. 5.33 A
- b. 4 A
- c. 8 A
- d. 2.67 A

Correct answer: 8 A

The current source is the summation of currents in 6 Ω & 12 Ω resistors, and as the resistances and the voltage across them are known, the current is calculated using ohm's law:

$$I_{6\Omega} = \frac{V_{AB}}{R_{6\Omega}} = \frac{32V}{6\Omega} = \frac{16}{3}A$$

$$I_{12\Omega} = \frac{V_{AB}}{R_{12\Omega}} = \frac{32V}{12\Omega} = \frac{8}{3}A$$

$$I_S = I_{6\Omega} + I_{12\Omega} = \frac{16}{3}A + \frac{8}{3}A = 8A$$

13. If $V_{AB} = 32V$, the equivalent resistance is:



- a. 4 Ω
- b. 25 Ω

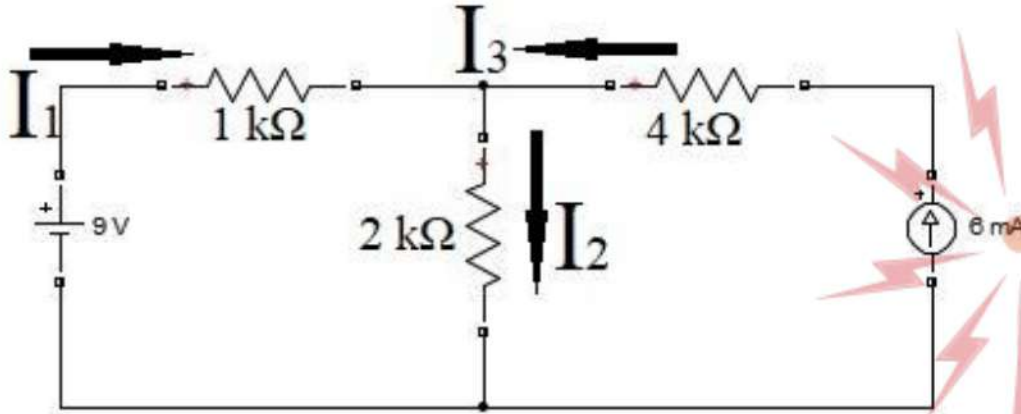
ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

- c. $20\ \Omega$
- d. None of them

Correct answer: $20\ \Omega$

$$R_{eq} = (6\ \Omega // 12\ \Omega) + 6\ \Omega + 10\ \Omega = \frac{6\ \Omega * 12\ \Omega}{6\ \Omega + 12\ \Omega} + 6\ \Omega + 10\ \Omega = 20\ \Omega$$

14. By using superposition theorem, the current I_1 , due to the voltage source only is:



- a. $6\ mA$
- b. $4.5\ mA$
- c. $2\ mA$
- d. $4\ mA$
- e. $-4\ mA$
- f. $0\ mA$
- g. $-2\ mA$
- h. $9\ mA$
- i. $3\ mA$

Correct answer: $3\ mA$

By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors $1\ k\Omega$ & $2\ k\Omega$ are in series:

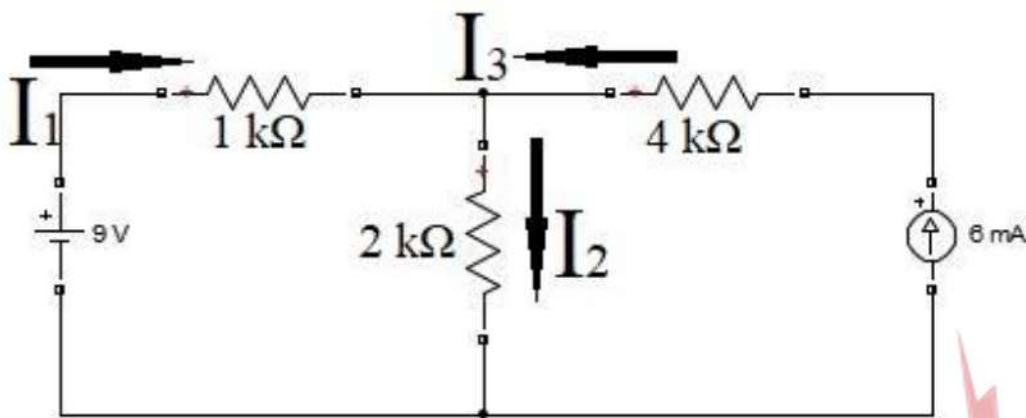
$$R_{series} = R_{1\ k\Omega} + R_{2\ k\Omega} = 1\ k\Omega + 2\ k\Omega = 3\ k\Omega$$

The current due to the voltage source is found by using ohm's law:

$$I = \frac{V_s}{R_{series}} = \frac{9\ V}{3\ k\Omega} = 3\ mA$$

15. By using superposition theorem, the current I_2 , due to the voltage source only is:

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



- a. 6 mA
- b. 9 mA
- c. 0 mA
- d. 4 mA
- e. -4 mA
- f. 3 mA
- g. -2 mA
- h. 2 mA
- i. 4.5 mA

Correct answer: 3 mA

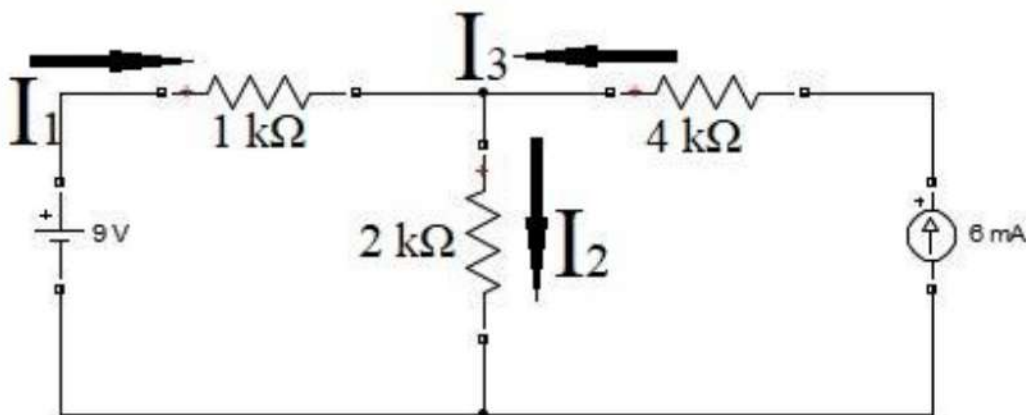
By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors 1 kΩ & 2 kΩ are in series:

$$R_{series} = R_{1\text{ k}\Omega} + R_{2\text{ k}\Omega} = 1\text{ k}\Omega + 2\text{ k}\Omega = 3\text{ k}\Omega$$

The current due to the voltage source is found by using ohm's law:

$$I = \frac{V_s}{R_{series}} = \frac{9\text{ V}}{3\text{ k}\Omega} = 3\text{ mA}$$

16. By using superposition theorem, the current I_3 , due to the voltage source only is:



- a. 8 mA
- b. 3 mA

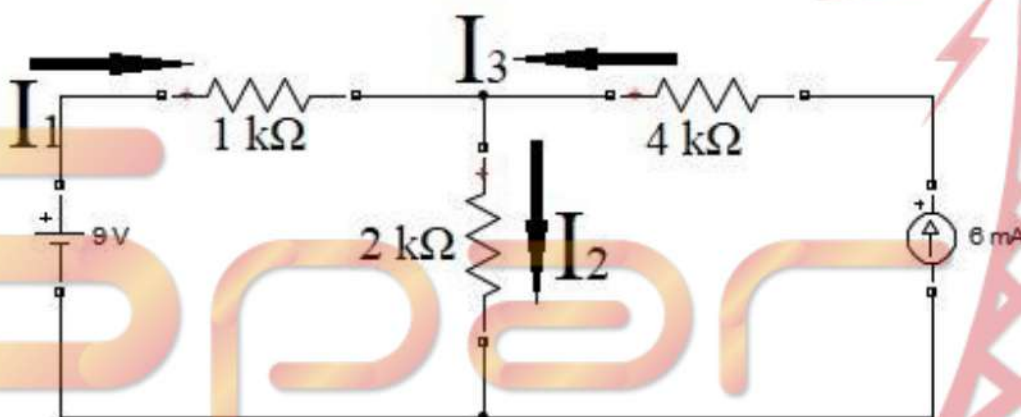
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- c. 2 mA
- d. 9 mA
- e. -4 mA
- f. 6 mA
- g. 0 mA
- h. -2 mA
- i. 4 mA
- j. 4.5 mA

Correct answer: 0 mA

By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the $4\text{ k}\Omega$ resistance is open circuited, so the current I_3 is zero

17. By using superposition theorem, the current I_1 , due to the current source only is:



- a. 3 mA
- b. 8 mA
- c. 0 mA
- d. 6 mA
- e. 2 mA
- f. -4 mA
- g. 9 mA
- h. 4.5 mA
- i. -2 mA
- j. 4 mA

Correct answer: -4 mA

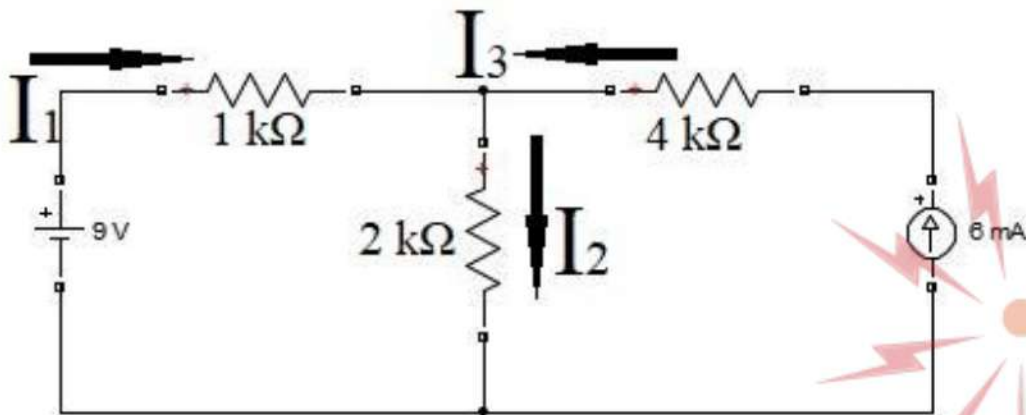
By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so using current divider rule the current I_1 is found, as the resistors $1\text{ k}\Omega$ & $2\text{ k}\Omega$ are in parallel:

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$$I_1 = -\frac{R_2 k\Omega I_S}{R_1 k\Omega + R_2 k\Omega} = -\frac{2 k\Omega * 6 A}{1 k\Omega + 2 k\Omega} = -4 mA$$

Note that, the negative sign for the current I_1 cause the actual direction for the current due to the current source only is opposite for that in the figure

18. By using superposition theorem, the current I_2 , due to the current source only is:



- a. 3 mA
- b. 4.5 mA
- c. 0 mA
- d. 2 mA
- e. 6 mA
- f. 4 mA
- g. -4 mA
- h. -2 mA
- i. 8 mA
- j. 9 mA

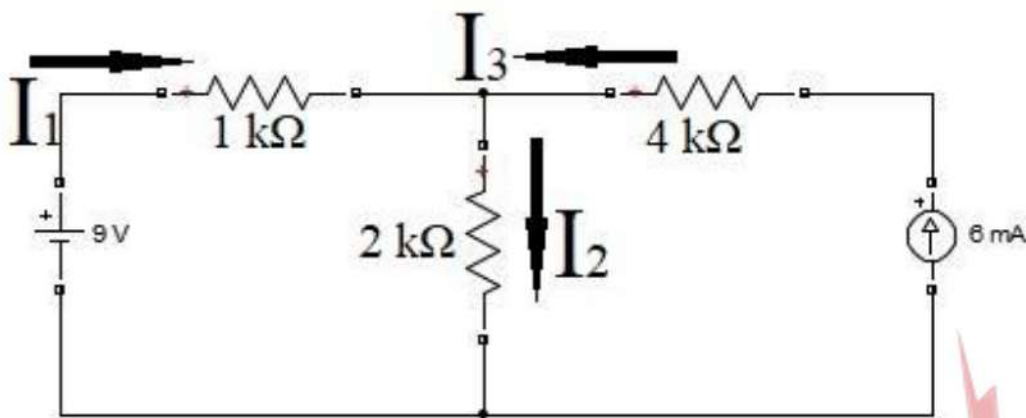
Correct answer: 2 mA

By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so using current divider rule the current I_1 is found, as the resistors 1 kΩ & 2 kΩ are in parallel:

$$I_1 = -\frac{R_2 k\Omega I_S}{R_1 k\Omega + R_2 k\Omega} = \frac{1 k\Omega * 6 A}{1 k\Omega + 2 k\Omega} = 2 mA$$

19. By using superposition theorem, the current I_3 , due to the current source only is:

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- a. 0 mA
- b. 4 mA
- c. 3 mA
- d. -4 mA
- e. -2 mA
- f. 9 mA
- g. 2 mA
- h. 4.5 mA
- i. 6 mA

Correct answer: 6 mA

By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the current I_3 is the same as the current of the current source

20. The equivalent resistance seen by the source (R_{eq}) is:



- a. $4\ \Omega$
- b. $22\ \Omega$
- c. $12\ \Omega$

d. 20Ω

Correct answer: 20Ω

$$R_{eq} = (6 \Omega // 12 \Omega) + 6 \Omega + 10 \Omega = 20 \Omega$$

21. A capacitor and a resistor are connected in series to a sine wave generator. The frequency is set so that the capacitive reactance is equal to the resistance and thus, an equal amount of voltage appears across each component. If the frequency was decreased:

- a. $V_C = V_R$
- b. $V_C < V_R$
- c. $V_C > V_R$

Correct answer: $V_C > V_R$

$$\uparrow X_C = \frac{1}{(\omega \downarrow)C} = \frac{1}{2\pi(f \downarrow)C}$$

$$\uparrow V_C = I(X_C \uparrow)$$

With the variation of frequency, the resistance is constant theoretically
And as $V_C = V_R$, and if the frequency was decreased:

$$V_C > V_R$$

22. An inductor and a resistor are connected in series to a sine wave generator. The frequency is set so that the inductive reactance is equal to the resistance and thus, an equal amount of voltage appears across each component. If the frequency was decreased:

- a. $V_L = V_R$
- b. $V_L > V_R$
- c. $V_L < V_R$

Correct answer: $V_L < V_R$

$$\downarrow X_L = (\omega \downarrow)C = 2\pi(f \downarrow)C$$

$$\downarrow V_L = I(X_L \downarrow)$$

With the variation of frequency, the resistance is constant theoretically
And as $V_L = V_R$, and if the frequency was decreased:

$$V_L < V_R$$

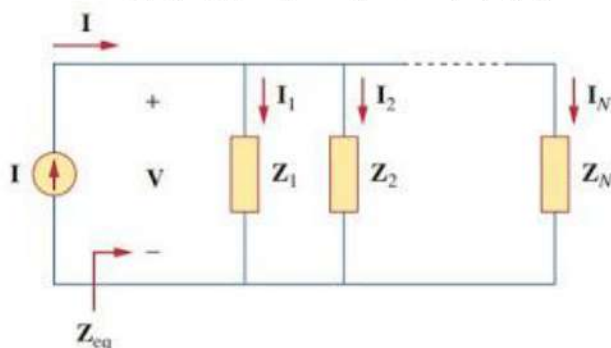
23. In the parallel connection:

- a. The Voltages applied on all the resistors are the same
- b. The currents applied on all the resistors are the same
- c. The equivalent resistance equals the summation of the resistances included in the circuit
- d. All the answers are right

Correct answer: The Voltages applied on all the resistors are the same

For parallel connected impedances:

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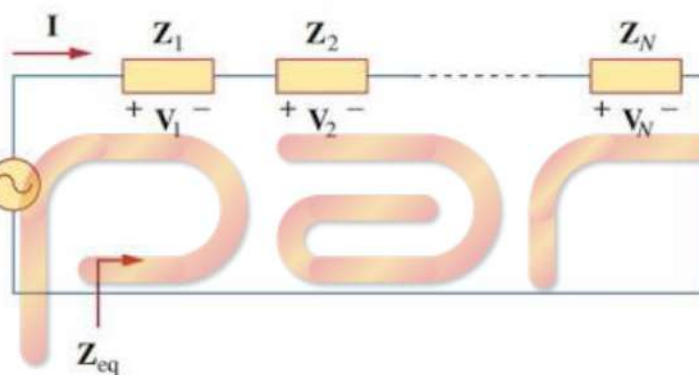


24. In the series connection:

- The voltage of the source equals the summation of the voltages on the included resistors in the circuit
- The equivalent resistance equals the summation of the resistances included in the circuit
- The current flowing in the circuit is constant along the circuit
- All the answers are right

Correct answer: All the answers are right

For series connected impedances:



$$Z_{eq} = Z_1 + Z_2 + Z_3 + \dots + Z_n$$

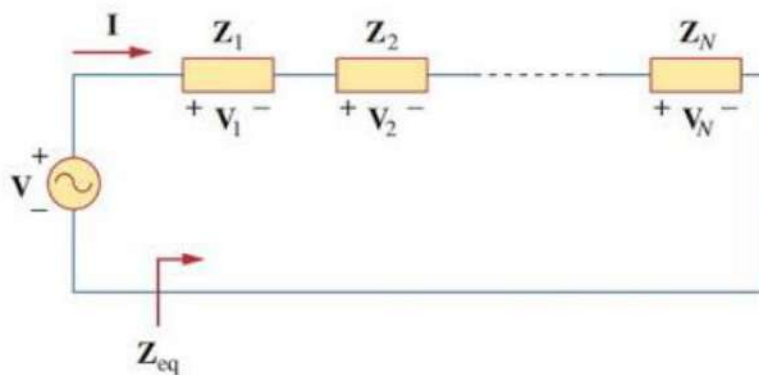
25. In the series connection:

- The value of the equivalent resistance equals the summation of the values of the resistances involved in the circuit
- The voltages applied on all the resistors are the same
- The equivalent resistance equals the summation of the resistances included in the circuit
- All the answers are right

Correct answer: The equivalent resistance equals the summation of the resistances included in the circuit

For series connected impedances:

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$$Z_{eq} = Z_1 + Z_2 + Z_3 + \dots + Z_n$$

26. In superposition theorem:

- Current source become open circuit, and the voltage source become short circuit
- Voltage source become open circuit, and the current source become short circuit
- Both voltage source and current source become short circuit
- Both voltage source and current source become open circuit

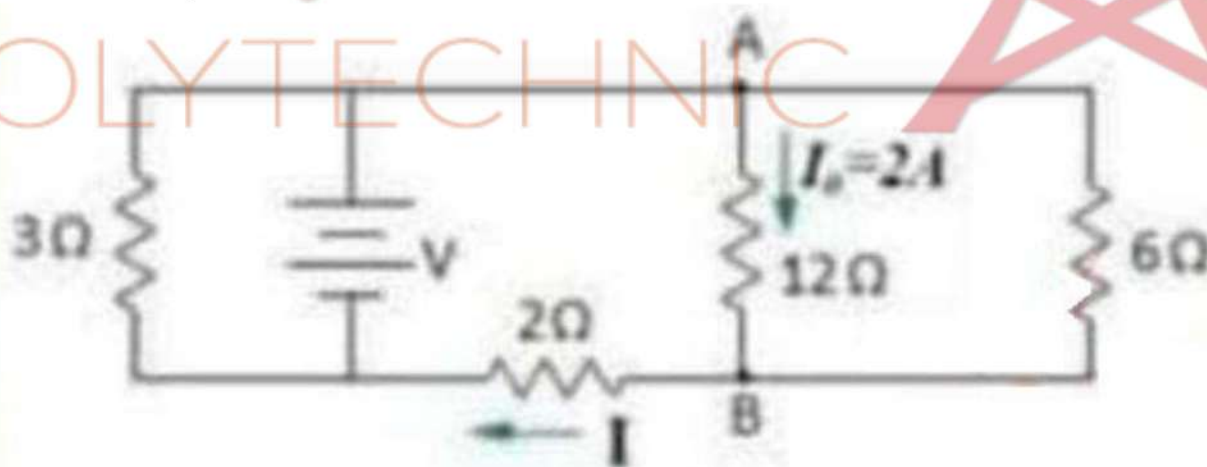
Correct answer: Current source become open circuit, and the voltage source become short circuit

27. In a network, maximum power transfer occurs when:

- $R_{Thevenin} = R_{load}$
- $R_{Thevenin} * R_{load} = 1$
- $R_{Thevenin} + R_{load} = 1$
- $R_{Thevenin}/R_{load} = 0$

Correct answer: $R_{Thevenin} = R_{load}$

28. Calculate the equivalent resistance of this circuit:



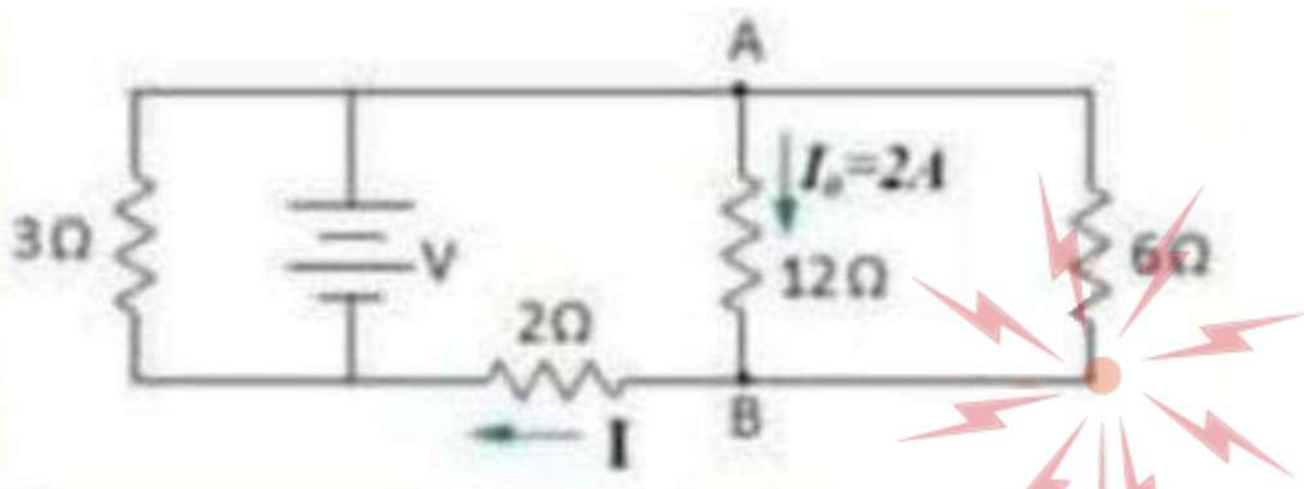
- 7.2Ω
- 2Ω
- 9Ω
- 3.4Ω

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Correct answer: $2\ \Omega$

$$R_{eq} = ((12\ \Omega // 6\ \Omega) + 2\ \Omega) // 3\ \Omega = 2\ \Omega$$

29. Calculate the current I in this circuit:



- a. $0.5\ A$
- b. $3\ A$
- c. $6\ A$
- d. $1.5\ A$
- e. $2\ A$

Correct answer: $6\ A$

The voltage across the $12\ \Omega$ resistor is calculated as follows:

$$V_{12\ \Omega} = I_{12\ \Omega} R_{12\ \Omega} = 2\ A * 12\ \Omega = 24\ V$$

And in case of parallel elements, the voltage is the same, so the voltage across the $6\ \Omega$ resistor is:

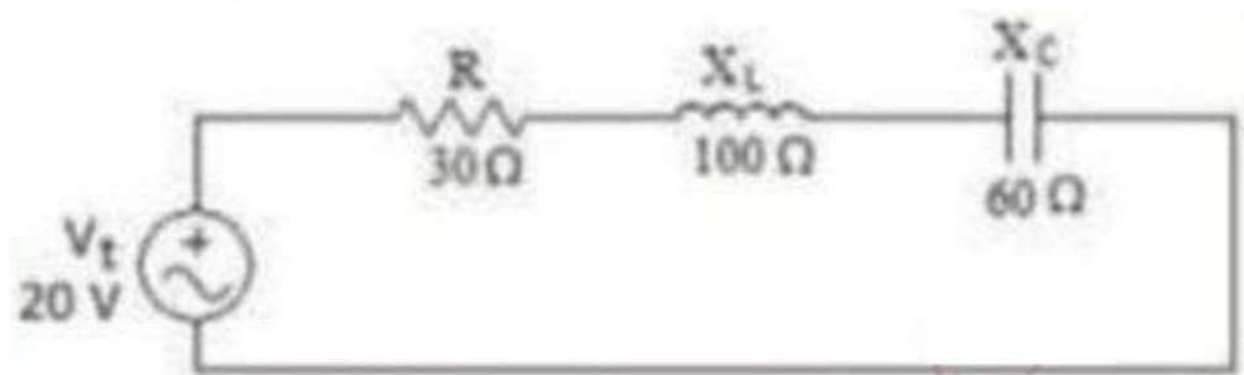
$$V_{6\ \Omega} = V_{12\ \Omega} = I_{6\ \Omega} R_{6\ \Omega}$$

$$I_{6\ \Omega} = \frac{V_{12\ \Omega}}{R_{6\ \Omega}} = \frac{24\ V}{6\ \Omega} = 4\ A$$

Therefore, the current I is the summation of the currents in $6\ \Omega$ & $12\ \Omega$ resistors

$$I = I_{6\ \Omega} + I_{12\ \Omega} = 4\ A + 2\ A = 6\ A$$

30. Total current is



ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

- a. 1.2 A
- b. 0.66 A
- c. 0.4 A
- d. 0.2 A

Correct answer: 0.4 A

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{30^2 + (100 - 60)^2} = \sqrt{30^2 + 40^2} = 50 \Omega$$

$$I = \frac{V}{Z} = \frac{20 V}{50 \Omega} = 0.4 A$$

31. If one of the resistors in a parallel circuit was removed, the total resistance:

- a. Increases
- b. Decreases by the value of the removed resistor
- c. Remains the same
- d. Decreases
- e. Increases by the value of the removed resistor

Correct answer: Increases

32. If one of the resistors in a parallel circuit was added, the total resistance:

- a. Decreases by the value of the removed resistor
- b. Decreases
- c. Remains the same
- d. Increases
- e. Increases by the value of the removed resistor

Correct answer: Decreases

33. If one of the resistors in a series circuit was removed, the total resistance:

- a. Decreases by the value of the removed resistor
- b. Remains the same
- c. Decreases
- d. Increases by the value of the removed resistor
- e. Increases

Correct answer: Decreases by the value of the removed resistor

34. If one of the resistors in a series circuit was added, the total resistance:

- a. Decreases by the value of the added resistor
- b. Remains the same
- c. Increases
- d. Decreases
- e. Increases by the value of the added resistor

Correct answer: Increases by the value of the added resistor

35. At the resonant frequency, the load will become:

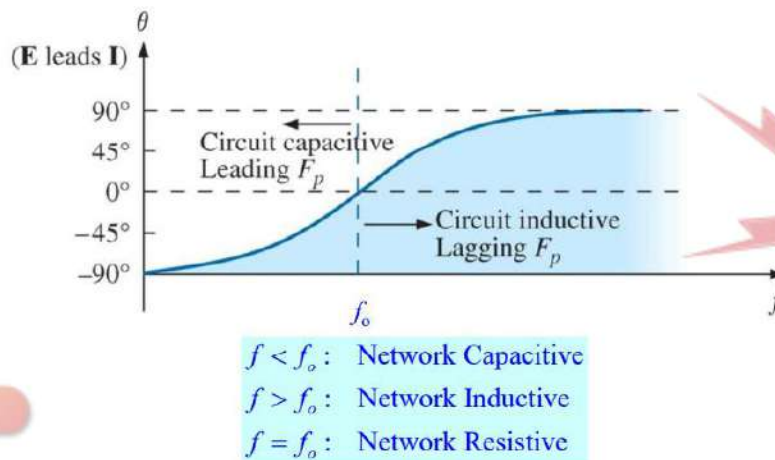
ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

- a. Capacitive
- b. Inductive
- c. Resistive

Correct answer: Resistive

And it is known that at resonant frequency $X_C = X_L$, so the reactive component is eliminated, therefore the load is purely resistive load

PHASE OF SERIES RESONANCE CIRCUIT



Phase plot for the series resonant circuit.

36. Below the resonant frequency, the load will become:

- a. Capacitive
- b. Inductive
- c. Resistive

Correct answer: Capacitive

Below the resonant frequency, the frequency decreases, so the capacitive reactance increases and the inductive reactance decreases:

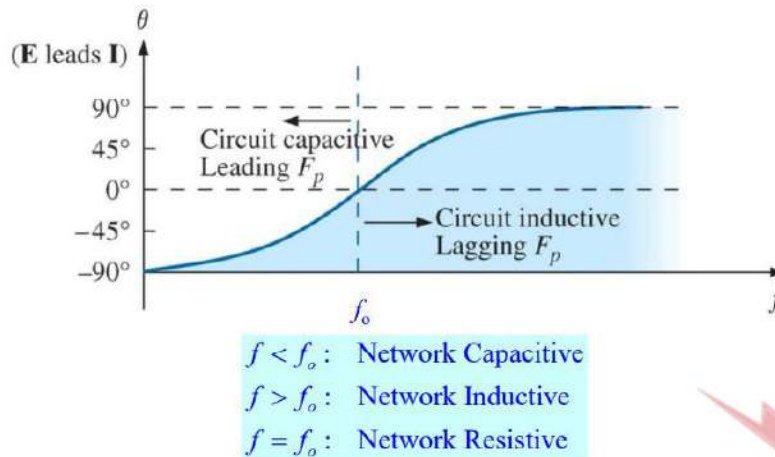
$$\uparrow X_C = \frac{1}{\omega C} = \frac{1}{2\pi(f \downarrow)C}$$

$$\downarrow X_L = 2\pi(f \downarrow)L$$

And it is known that at resonant frequency $X_C = X_L$, so when frequency decreases below resonant frequency, the capacitive reactance increases and becomes more than the inductive reactance which is also decreases, so the circuit becomes resistive capacitive, as shown in the following figure:

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك، ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

PHASE OF SERIES RESONANCE CIRCUIT



Phase plot for the series resonant circuit.

37. Above the resonant frequency, the load will become:

- Resistive
- Capacitive
- Inductive

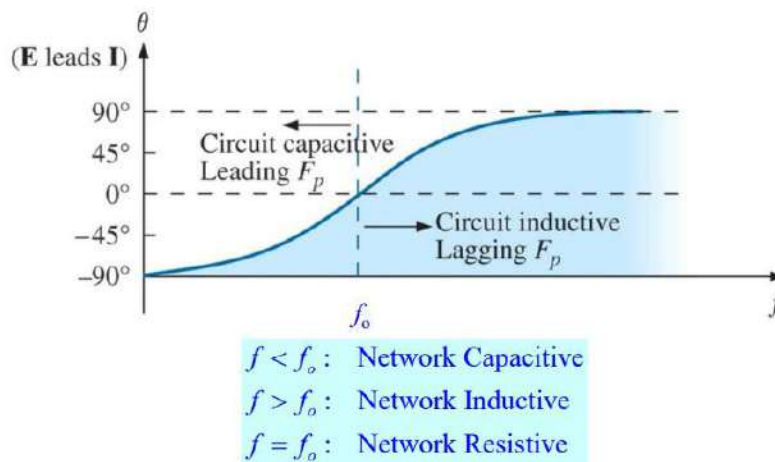
Correct answer: Inductive

Above the resonant frequency, the frequency increases, so the capacitive reactance decreases and the inductive reactance increases:

$$\downarrow X_C = \frac{1}{\omega C} = \frac{1}{2\pi(f \uparrow)C}$$

$$\uparrow X_L = 2\pi(f \uparrow)L$$

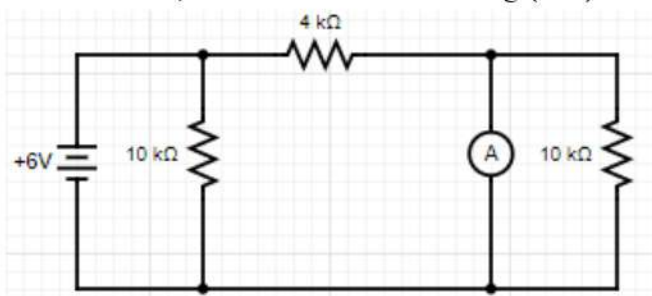
And it is known that at resonant frequency $X_C = X_L$, so when frequency increases above resonant frequency, the capacitive reactance decreases and becomes less than the inductive reactance which is also increases, so the circuit becomes resistive inductive, as shown in the following figure:



Phase plot for the series resonant circuit.

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

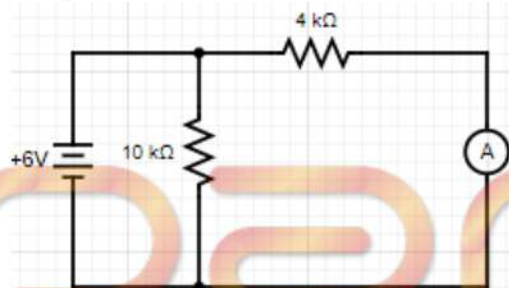
38. For the circuit shown when $V_s = 6V$, then the ammeter reading (mA) is:



- a. 2.5 mA
- b. 3 mA
- c. 2 mA
- d. 1.5 mA

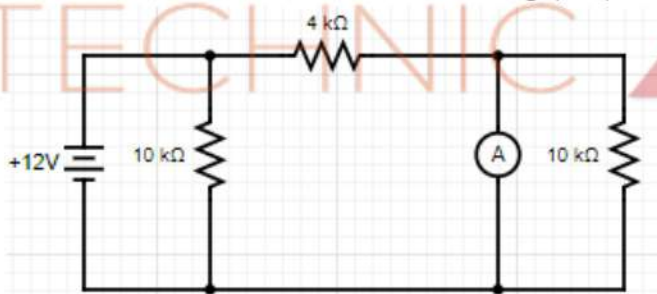
Correct answer: 1.5 mA

The ammeter internal resistance equals zero theoretically, and it is very low practically, which means that the 10 kΩ resistance is shorted by the ammeter, so the circuit becomes as follows:



$$I_{Ammeter} = \frac{V}{R} = \frac{6V}{4k\Omega} = 1.5mA$$

39. For the circuit shown when $V_s = 12V$, then the ammeter reading (mA) is:

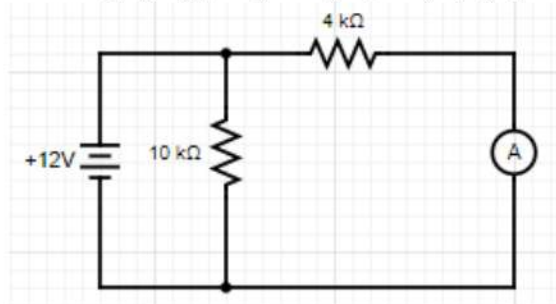


- a. 2.5 mA
- b. 3 mA
- c. 2 mA
- d. 1.5 mA

Correct answer: 3 mA

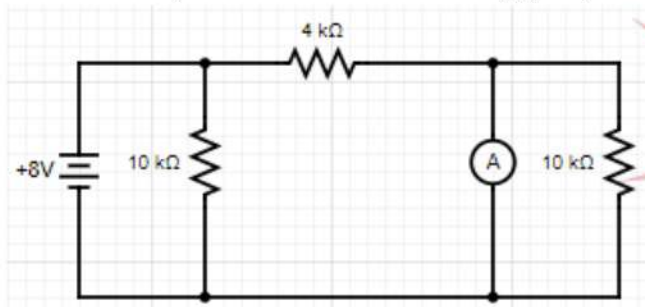
The ammeter internal resistance equals zero theoretically, and it is very low practically, which means that the 10 kΩ resistance is shorted by the ammeter, so the circuit becomes as follows:

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



$$I_{Ammeter} = \frac{V}{R} = \frac{12 V}{4 k\Omega} = 3 mA$$

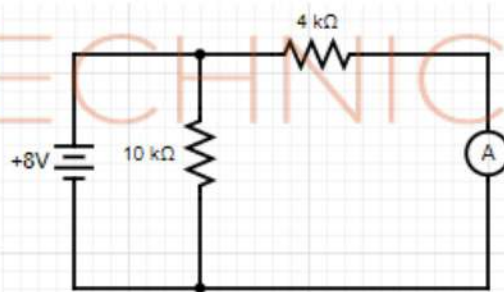
40. For the circuit shown when $V_s = 8V$, then the ammeter reading (mA) is:



- a. 2.5 mA
- b. 3 mA
- c. 2 mA
- d. 1.5 mA

Correct answer: 2 mA

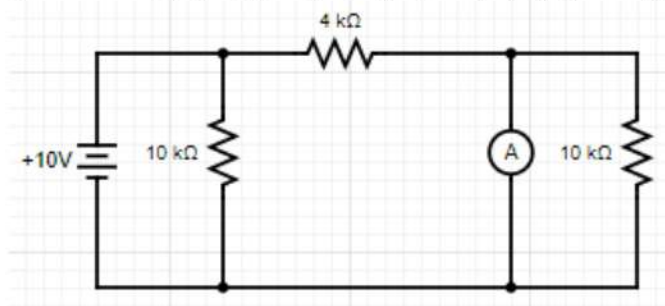
The ammeter internal resistance equals zero theoretically, and it is very low practically, which means that the 10 kΩ resistance is shorted by the ammeter, so the circuit becomes as follows:



$$I_{Ammeter} = \frac{V}{R} = \frac{8 V}{4 k\Omega} = 2 mA$$

41. For the circuit shown when $V_s = 10V$, then the ammeter reading (mA) is:

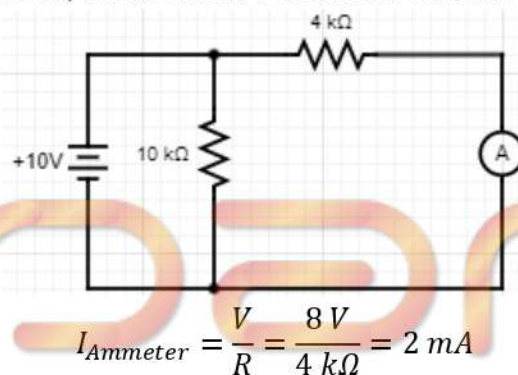
ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



- a. 1.5 mA
- b. 3 mA
- c. 2 mA
- d. 2.5 mA

Correct answer: 2.5 mA

The ammeter internal resistance equals zero theoretically, and it is very low practically, which means that the 10 kΩ resistance is shorted by the ammeter, so the circuit becomes as follows:



42. The current is measured by:

- a. Ohmmeter
- b. Voltmeter
- c. Ammeter

Correct answer: Ammeter

43. The voltage is measured by:

- a. Ammeter
- b. Ohmmeter
- c. Voltmeter

Correct answer: Voltmeter

44. The resistance is measured by:

- a. Ammeter
- b. Ohmmeter
- c. Voltmeter

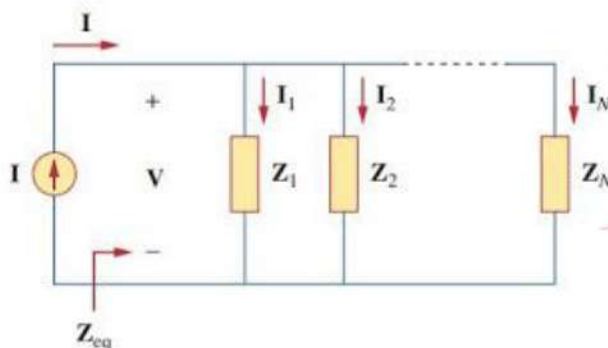
ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

Correct answer: Ohmmeter

45. Where current division occurs?
- Both series and parallel resistors
 - When resistors are equal
 - Series connected resistors
 - Parallel connected resistors

Correct answer: Parallel connected resistors

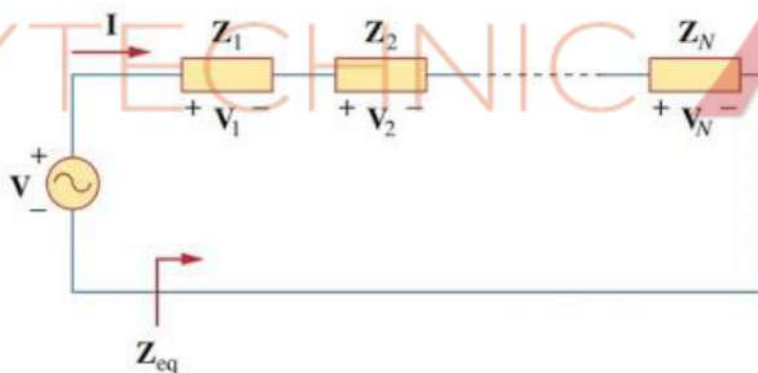
In series the voltage across the resistors is the same, but the current is divided



46. Where voltage division occurs?
- Both series and parallel resistors
 - Series connected resistors
 - When resistors are equal
 - Parallel connected resistors

Correct answer: Series connected resistors

In series the current passing through the resistors is the same, but the voltage is divided



47. A (voltmeter/ammeter) connected in an A.C circuit reads voltage in the form of:
- RMS voltage
 - Mean square voltage
 - Peak voltage
 - Average voltage
 - Transient voltage
 - All of them

Correct answer: RMS voltage

The voltmeter & ammeter are always reads voltage in the form of RMS voltage in an A.C circuit

48. The reactance (X) of an inductor of 15 mH at a frequency of 60 HZ is:

- a. 5.65Ω
- b. $5.65 k\Omega$
- c. $0.17 k\Omega$
- d. $0.17 k\Omega$

Correct answer: 5.65Ω

$$X_L = 2\pi fL = 2\pi * 60Hz * 15mH = 5.65 \Omega$$

49. Which one of the following statement is NOT true:

- a. The voltage across an element can be measured while it is connected in the circuit
- b. The electrical power of an element can be measured while it is connected in the circuit
- c. The current through an element can be measured while it is connected in the circuit
- d. The resistance of an element can be measured while it is connected in the circuit

Correct answer: The electrical power of an element can be measured while it is connected in the circuit

50. The unit of measurement of the power is:

- a. Volt. Ampere
- b. Volt / Ampere
- c. Watt. Sec
- d. Joule

Correct answer: Volt. Ampere

51. The unit of the charge is:

- a. Ohm
- b. Watt
- c. Coulomb
- d. Ampere

Correct answer: Coulomb

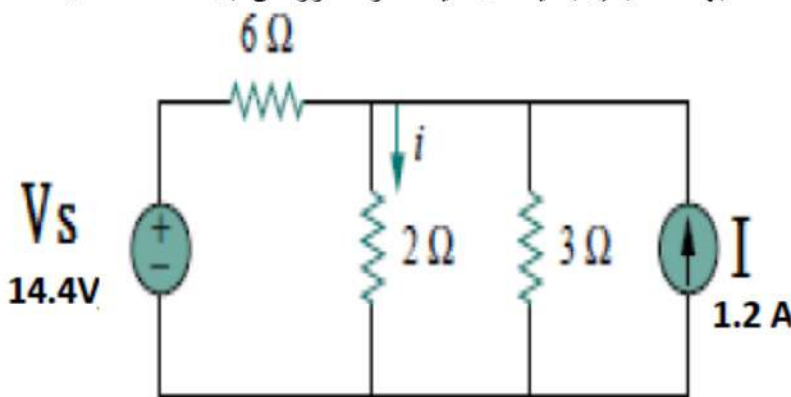
52. The unit of the (resistance/reactance/inductive reactance/capacitive reactance) is:

- a. Ampere
- b. Coulomb
- c. Watt
- d. Ohm

Correct answer: Ohm

53. The current (in the branch of voltage source/ in the branch of 6Ω resistor) due to voltage source only is: (superposition)

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- a. 0.8 A
- b. 0.2 A
- c. 2 A
- d. 1 A
- e. 0.6 A
- f. 2.4 A
- g. 1.2 A
- h. 0.4 A

Correct answer: 2 A

By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors 2 Ω & 3 Ω are in parallel:

$$R_{parallel} = \frac{R_2 \Omega * R_3 \Omega}{R_2 \Omega + R_3 \Omega} = \frac{2 \Omega * 3 \Omega}{2 \Omega + 3 \Omega} = \frac{6 \Omega^2}{5 \Omega} = 1.2 \Omega$$

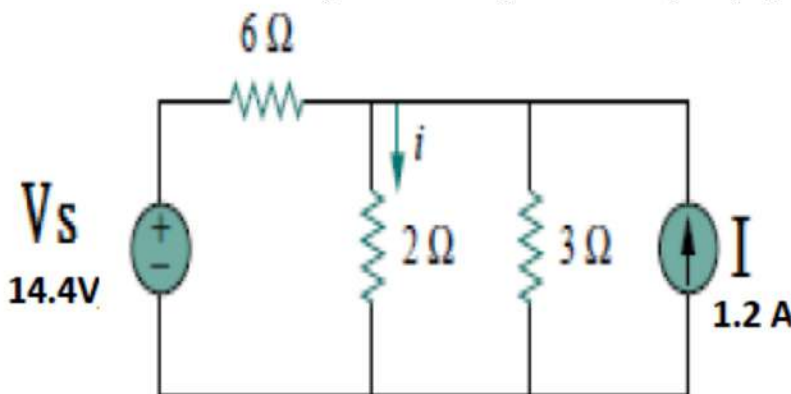
The equivalent resistance is:

$$R_{eq} = R_{parallel} + R_{6 \Omega} = 1.2 \Omega + 6 \Omega = 7.2 \Omega$$

The current due to the voltage source in the branch of voltage source:

$$I = \frac{V_s}{R_{eq}} = \frac{14.4V}{7.2 \Omega} = 2 A$$

54. The current (i/ in the branch of 2 Ω resistor) due to voltage source only is: (superposition)



- a. 2 A
- b. 0.4 A

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

- c. 0.6 A
- d. 1 A
- e. 2.4 A
- f. 0.2 A
- g. 1.2 A
- h. 0.8 A

Correct answer: 1.2 A

By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors 2 Ω & 3 Ω are in parallel:

$$R_{parallel} = \frac{R_2 \Omega * R_3 \Omega}{R_2 \Omega + R_3 \Omega} = \frac{2 \Omega * 3 \Omega}{2 \Omega + 3 \Omega} = \frac{6 \Omega^2}{5 \Omega} = 1.2 \Omega$$

The equivalent resistance is:

$$R_{eq} = R_{parallel} + R_6 \Omega = 1.2 \Omega + 6 \Omega = 7.2 \Omega$$

The current due to the voltage source in the branch of voltage source:

$$I = \frac{V_s}{R_{eq}} = \frac{14.4V}{7.2 \Omega} = 2 A$$

Now, the current in the branch of 2 Ω resistor is found by using current divider rule:

$$I_2 \Omega + I_3 \Omega = I$$

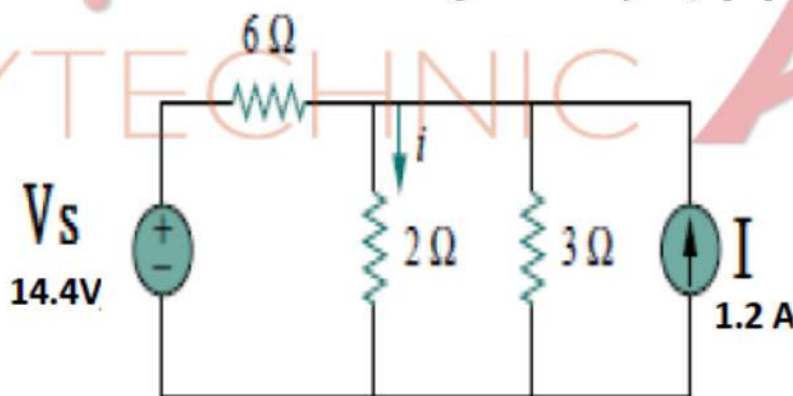
$$I_3 \Omega = I - I_2 \Omega$$

$$R_2 \Omega I_2 \Omega = R_3 \Omega I_3 \Omega$$

$$R_2 \Omega I_2 \Omega = R_3 \Omega (I - I_2 \Omega)$$

$$I_2 \Omega = \frac{R_3 \Omega I}{R_2 \Omega + R_3 \Omega} = \frac{3 \Omega * 2A}{2 \Omega + 3 \Omega} = 0.6 \Omega * 2A = 1.2 A$$

55. The current in the branch of 3 Ω resistor due to voltage source only is: (superposition)



- a. 2 A
- b. 0.2 A
- c. 1 A
- d. 2.4 A
- e. 0.8 A
- f. 0.4 A
- g. 1.2 A

h. 0.6 A

Correct answer: 0.8 A

By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors 2 Ω & 3 Ω are in parallel:

$$R_{parallel} = \frac{R_2 \Omega * R_3 \Omega}{R_2 \Omega + R_3 \Omega} = \frac{2 \Omega * 3 \Omega}{2 \Omega + 3 \Omega} = \frac{6 \Omega^2}{5 \Omega} = 1.2 \Omega$$

The equivalent resistance is:

$$R_{eq} = R_{parallel} + R_6 \Omega = 1.2 \Omega + 6 \Omega = 7.2 \Omega$$

The current due to the voltage source in the branch of voltage source:

$$I = \frac{V_s}{R_{eq}} = \frac{14.4V}{7.2 \Omega} = 2 A$$

Now, the current in the branch of 3 Ω resistor is found by using current divider rule:

$$I_2 \Omega + I_3 \Omega = I$$

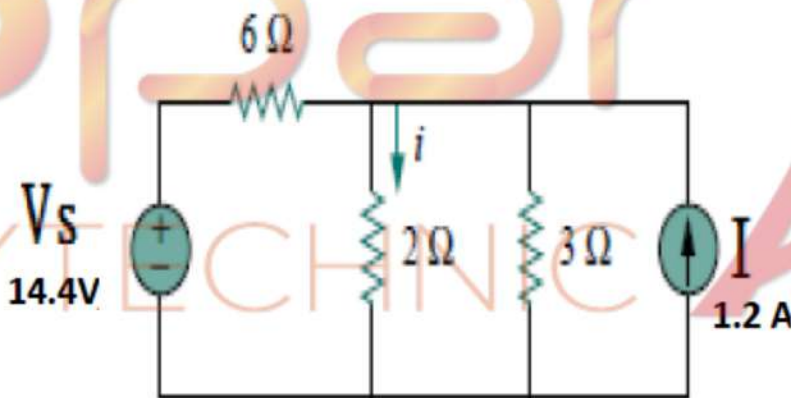
$$I_2 \Omega = I - I_3 \Omega$$

$$R_2 \Omega I_2 \Omega = R_3 \Omega I_3 \Omega$$

$$R_2 \Omega (I - I_3 \Omega) = R_3 \Omega I_3 \Omega$$

$$I_3 \Omega = \frac{R_2 \Omega I}{R_2 \Omega + R_3 \Omega} = \frac{2 \Omega * 2A}{2 \Omega + 3 \Omega} = 0.4 \Omega * 2A = 0.8 A$$

56. The current (in the branch of voltage source/ in the branch of 6 Ω resistor) due to current source only is: (superposition)



- a. 0.6 A
- b. 0.8 A
- c. 0.2 A
- d. 1.2 A
- e. 2 A
- f. 0.4 A
- g. 2.4 A
- h. 1 A

Correct answer: 0.2 A

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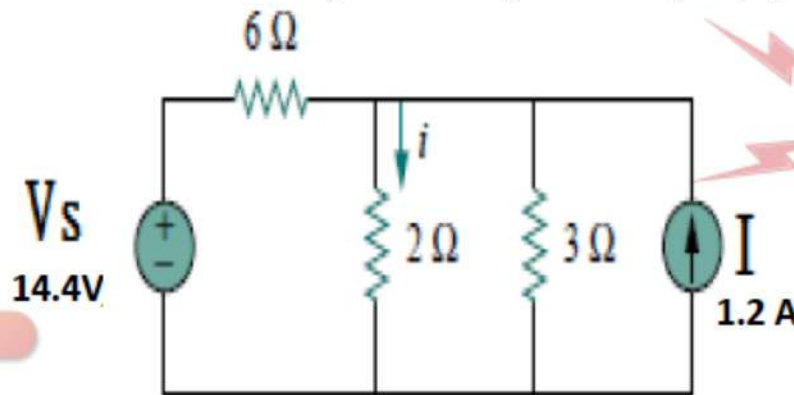
By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors $2\ \Omega$, $3\ \Omega$ & $6\ \Omega$ are in parallel, the current in the required branch is found by using current divider rule, but to use current divider rule, the equivalent resistance of other two resistances has to be calculated:

$$R_{parallel} = \frac{R_2 \Omega * R_3 \Omega}{R_2 \Omega + R_3 \Omega} = \frac{2\ \Omega * 3\ \Omega}{2\ \Omega + 3\ \Omega} = \frac{6\ \Omega^2}{5\ \Omega} = 1.2\ \Omega$$

The current due to the current source in the branch of voltage source is found now by using current divider rule:

$$I_{6\ \Omega} = \frac{R_{parallel} I_{current\ source}}{R_6 \Omega + R_{parallel}} = \frac{1.2\ \Omega * 1.2A}{6\ \Omega + 1.2\ \Omega} = 0.1667\ \Omega * 1.2A = 0.2\ A$$

57. The current (i) in the branch of $2\ \Omega$ resistor) due to voltage source only is: (superposition)



- a. $1.2\ A$
- b. $2\ A$
- c. $0.6\ A$
- d. $0.4\ A$
- e. $1\ A$
- f. $0.8\ A$
- g. $2.4\ A$
- h. $0.2\ A$

Correct answer: $0.6\ A$

By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors $2\ \Omega$, $3\ \Omega$ & $6\ \Omega$ are in parallel, the current in the required branch is found by using current divider rule, but to use current divider rule, the equivalent resistance of other two resistances has to be calculated:

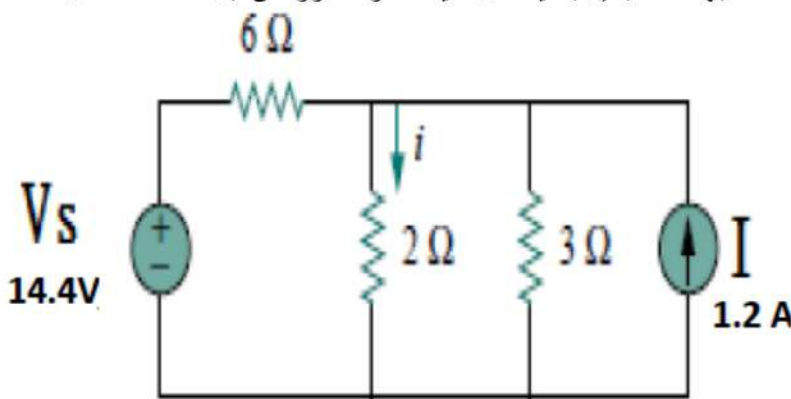
$$R_{parallel} = \frac{R_6 \Omega * R_3 \Omega}{R_6 \Omega + R_3 \Omega} = \frac{6\ \Omega * 3\ \Omega}{6\ \Omega + 3\ \Omega} = \frac{18\ \Omega^2}{9\ \Omega} = 2\ \Omega$$

The current due to the current source in the branch of voltage source is found now by using current divider rule:

$$I_{6\ \Omega} = \frac{R_{parallel} I_{current\ source}}{R_2 \Omega + R_{parallel}} = \frac{2\ \Omega * 1.2A}{2\ \Omega + 2\ \Omega} = 0.5\ \Omega * 1.2A = 0.6\ A$$

58. The current in the branch of $3\ \Omega$ resistor due to voltage source only is: (superposition)

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- a. 2 A
- b. 2.4 A
- c. 1.2 A
- d. 0.6 A
- e. 0.2 A
- f. 1 A
- g. 0.8 A
- h. 0.4 A

Correct answer: 0.4 A

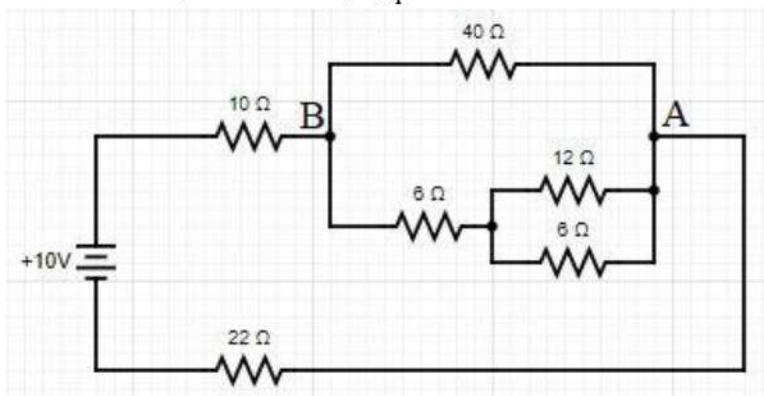
By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors 2 Ω, 3 Ω & 6 Ω are in parallel, the current in the required branch is found by using current divider rule, but to use current divider rule, the equivalent resistance of other two resistances has to be calculated:

$$R_{parallel} = \frac{R_{2\Omega} * R_{6\Omega}}{R_{2\Omega} + R_{6\Omega}} = \frac{2\Omega * 6\Omega}{2\Omega + 6\Omega} = \frac{12\Omega^2}{8\Omega} = 1.5\Omega$$

The current due to the current source in the branch of voltage source is found now by using current divider rule:

$$I_{6\Omega} = \frac{R_{parallel} I_{current\ source}}{R_{3\Omega} + R_{parallel}} = \frac{1.5\Omega * 1.2A}{3\Omega + 1.5\Omega} = 0.3333\Omega * 1.2A = 0.4\ A$$

59. The equivalent resistance seen by the source (R_{eq}) is:



- a. 26 Ω
- b. 40 Ω

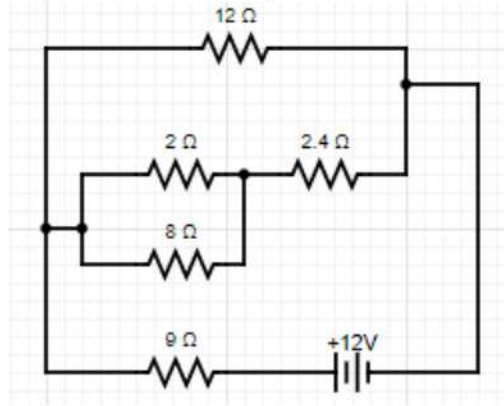
ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك، ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

- c. 8Ω
- d. 18Ω

Correct answer: 40Ω

$$R_{eq} = 22 \Omega + 10 \Omega + (((6 \Omega // 12 \Omega) + 6 \Omega) // 40 \Omega) = 40 \Omega$$

60. The equivalent resistance seen by the source (R_{eq}) is:



- a. 15Ω
- b. 12Ω
- c. 9Ω
- d. 3Ω

Correct answer: 12Ω

$$R_{eq} = 8 \Omega + (((8 \Omega // 2 \Omega) + 2.4 \Omega) // 12 \Omega) = 12 \Omega$$

61. Which of the following combinations of components represents an impedance of $110 + j314 \Omega$ at a frequency of 100 Hz :

- a. A resistor of 110Ω in series with an inductor of 500 mH
- b. An inductor of 50 mH in series with a capacitor of $5 \mu\text{F}$
- c. A resistor of 314Ω in series with an inductor of 5 mH
- d. A resistor of 110Ω in series with a capacitor of $5 \mu\text{F}$

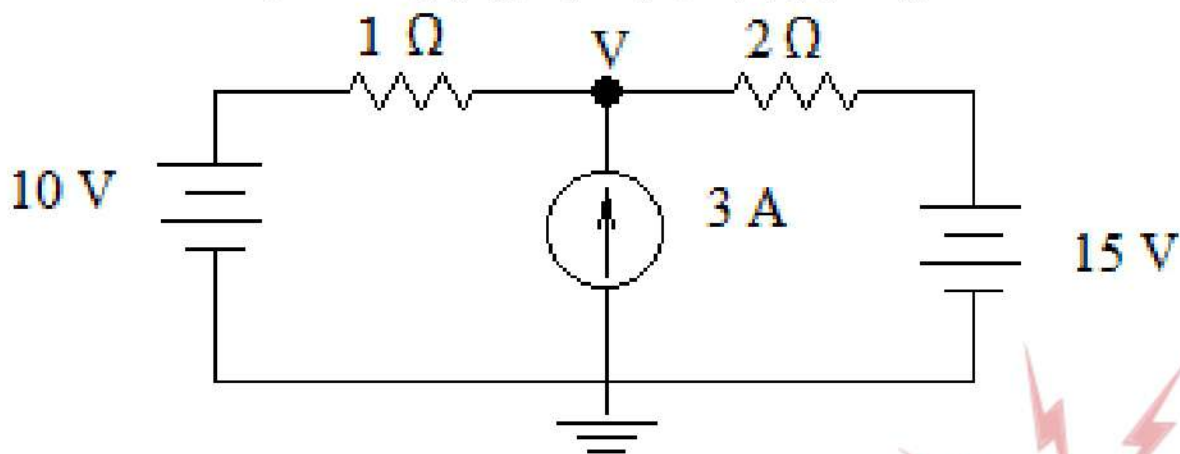
Correct answer: A resistor of 110Ω in series with an inductor of 500 mH

$$X_L = 2\pi fL$$

$$L = \frac{X_L}{2\pi f} = \frac{314 \Omega}{2\pi * 100 \text{ Hz}} = 500 \text{ mH}$$

62. Choose the right equation:

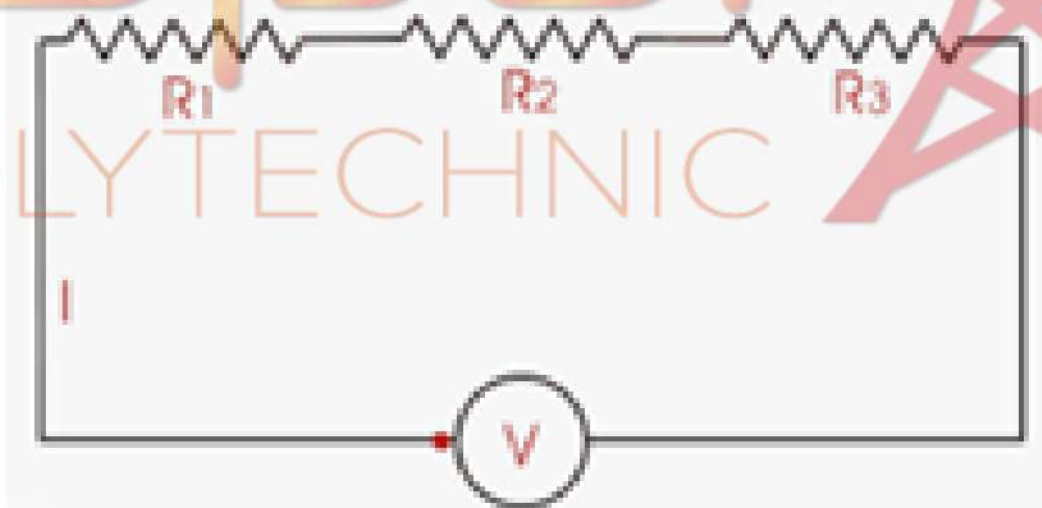
ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



- a. $V + 10 + 3 + \frac{V-15}{2} = 0$
- b. $V - 10 - 3 + \frac{V-15}{2} = 0$
- c. $V - 10 + 3 + \frac{V-15}{2} = 0$
- d. $V + 10 + 3 - \frac{V-15}{2} = 0$

Correct answer: $V - 10 - 3 + \frac{V-15}{2} = 0$

63. If the Battery = 36V, $R_1 = 1 \text{ MOhms}$, $R_2 = 3 \text{ MOhms}$, $R_3 = 5 \text{ MOhms}$, then find V_{R_2} :



- a. 12V
- b. 5V
- c. 15V
- d. 20V
- e. 4V

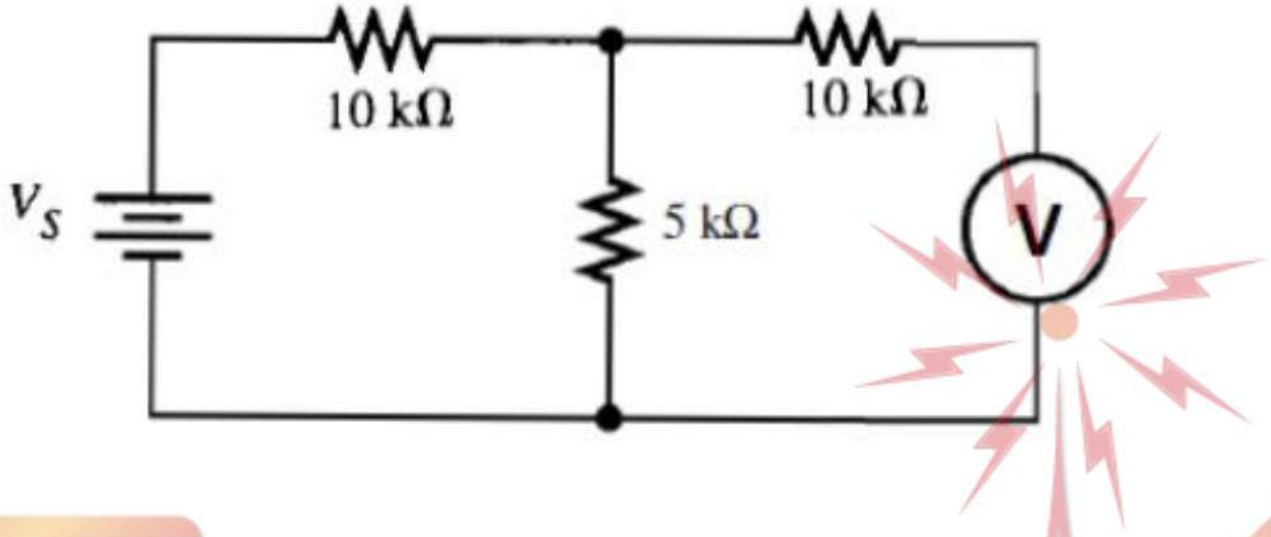
ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

Correct answer: 12V

By using voltage divider rule, the voltage across R2 is calculated:

$$V_{R2} = \frac{R2}{R1 + R2 + R3} V_{Battery} = \frac{3M\Omega}{1M\Omega + 3M\Omega + 5M\Omega} * 36V = 12V$$

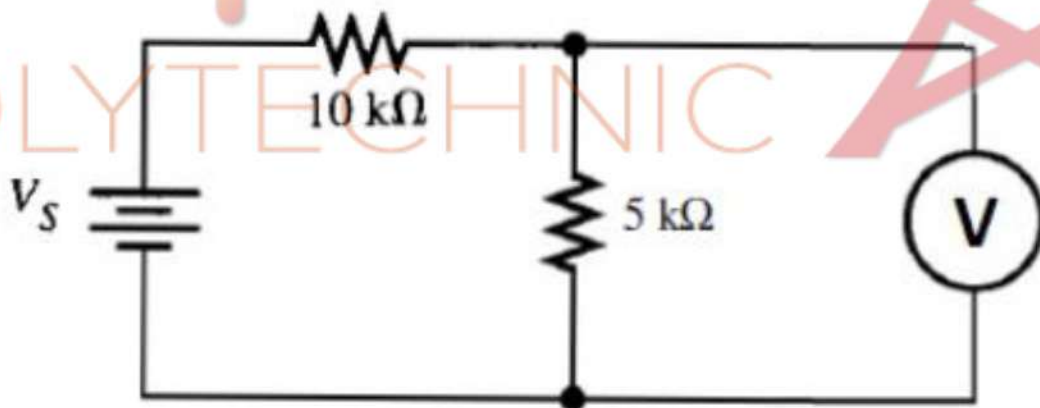
64. For the circuit shown when $V_s = 12V$, then the voltmeter reading (in V) is:



- a. 4
- b. 5
- c. 3
- d. 2

Correct answer: 4

The voltmeter is connected in series with the 10 kΩ resistance so the current doesn't pass through it cause the equivalent circuit to the voltmeter is open circuit, and according to this the circuit is equivalent to the following one:

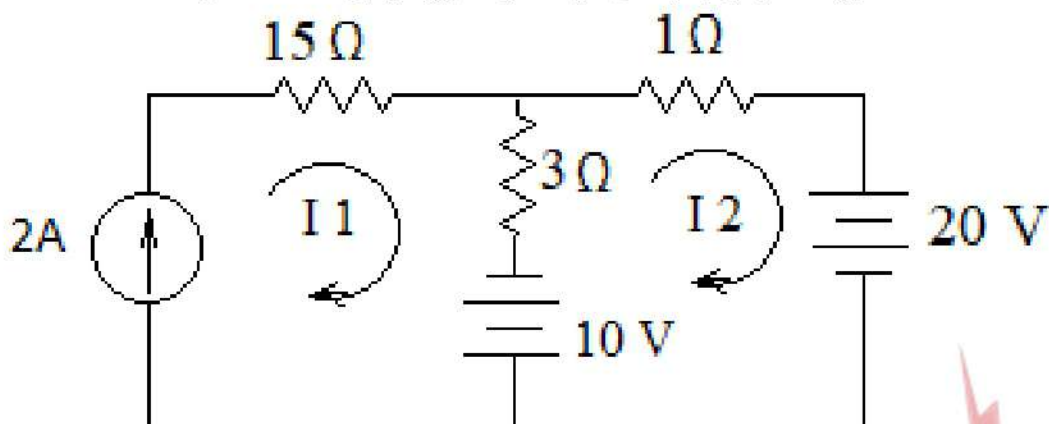


So the voltage reading (in V) is calculated by using voltage divider rule (VDR):

$$V_{5\ k\Omega} = \frac{R_{5\ k\Omega}}{R_{5\ k\Omega} + R_{10\ k\Omega}} V_s = \frac{5\ k\Omega}{5\ k\Omega + 10\ k\Omega} * 12V = 4V$$

65. Calculate the mesh currents:

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



- a. 3.5 A, 2 A
- b. 2 A, -6 A
- c. 2 A, 3.8 A
- d. -2 A, 10 A

Correct answer: 2 A, -6 A

For a current source that is not contained between two meshes, the mesh current will take the positive if the mesh current is in the same direction of the current source or negative value if the mesh current is in the opposite direction of the current source, so:

$$I_1 = 2A$$

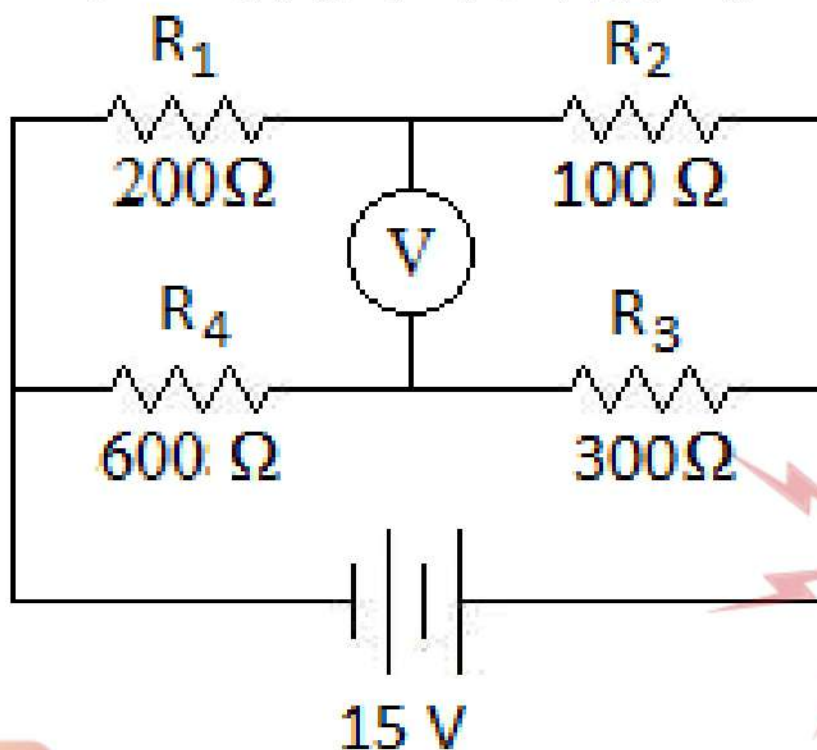
The current I_2 is found by applying KVL on the right mesh:

$$+10 + 3(I_2 - I_1) + 1 * I_2 + 20 = 0$$

$$I_2 = -6A$$

66. The reading of the voltmeter is equal to:

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- a. 15 V
- b. 0 V
- c. 7.5 V
- d. 10 V

Correct answer: 0 V

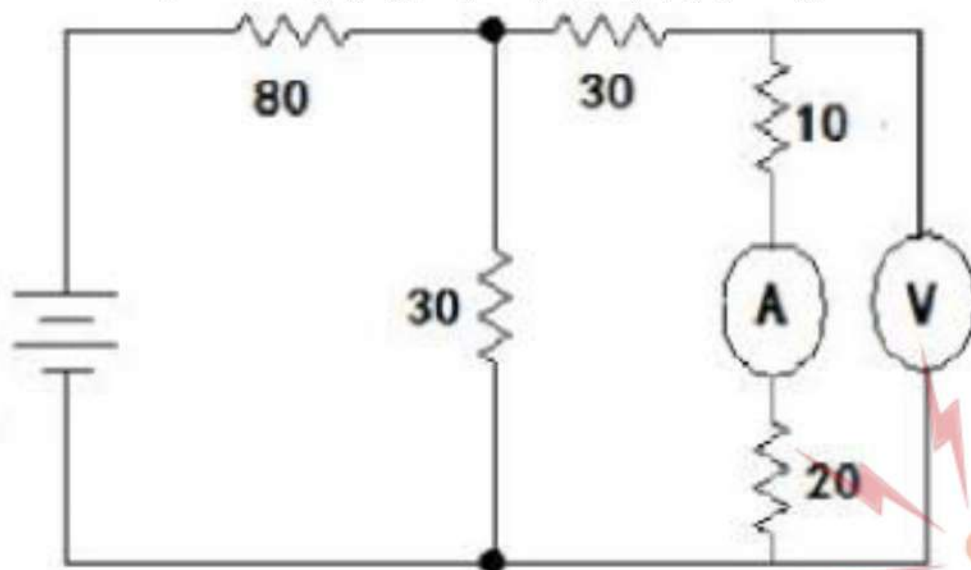
The reading of the voltmeter is the voltage difference between two points and note that the ratio between resistances on each side is the same:

$$\frac{100\ \Omega}{300\ \Omega} = \frac{200\ \Omega}{600\ \Omega} = \frac{1}{3}$$

Then, the drop voltage is the same on each branch, so there is no voltage difference between the two points which are connected with voltmeter

67. For the circuit shown below, consider the voltage source is 50 V, then the voltmeter and ammeter readings are:

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- a. 10 V, 0.33 A
- b. 5V, 0.16 A
- c. 10 V, 0.16 A
- d. 5V, 0.33 A

Correct answer: 5V, 0.16 A

The equivalent resistance is:

$$R_{eq} = 80 \, \Omega + 30 \, \Omega // (30 \, \Omega + 10 \, \Omega + 20 \, \Omega) = 80 \, \Omega + 30 \, \Omega // 60 \, \Omega$$

$$R_{eq} = 80 \, \Omega + \frac{30 \, \Omega * 60 \, \Omega}{30 \, \Omega + 60 \, \Omega} = 100 \, \Omega$$

The current source is calculated using ohm's law:

$$I_s = \frac{V_s}{R_{eq}} = \frac{50 \, V}{100 \, \Omega} = \frac{1}{2} \, A$$

Using current divider rule (CDR) the ammeter current is calculated:

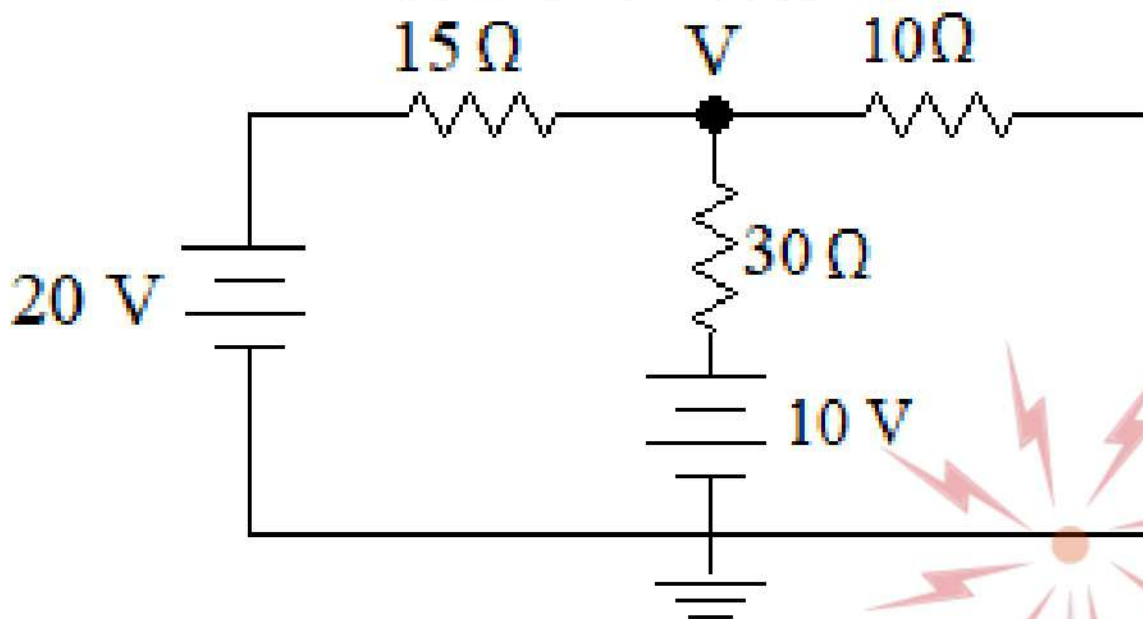
$$I_{ammeter} = \frac{30 \, \Omega}{30 \, \Omega + 30 \, \Omega + 10 \, \Omega + 20 \, \Omega} * I_s = \frac{30 \, \Omega}{90 \, \Omega} * \frac{1}{2} \, A = \frac{1}{6} \, A$$

The voltmeter reading is calculated using ohm's law:

$$V_{voltmeter} = (10 \, \Omega + 20 \, \Omega) I_{ammeter} = (10 \, \Omega + 20 \, \Omega) * \frac{1}{6} \, A = 5 \, V$$

68. Find the value of the node voltage V:

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- a. 2.4 V
- b. 3.8 V
- c. 10 V
- d. 8.33 V

Correct answer: 8.33 V

$$\frac{V - 20}{15} + \frac{V - 10}{30} + \frac{V}{10} = 0$$

$$V = 8.33 \text{ V}$$

69. Mesh current method is a method based on:

- a. Kirchhoff Current Law
- b. Kirchhoff Voltage Law

Correct answer: Kirchhoff Voltage Law

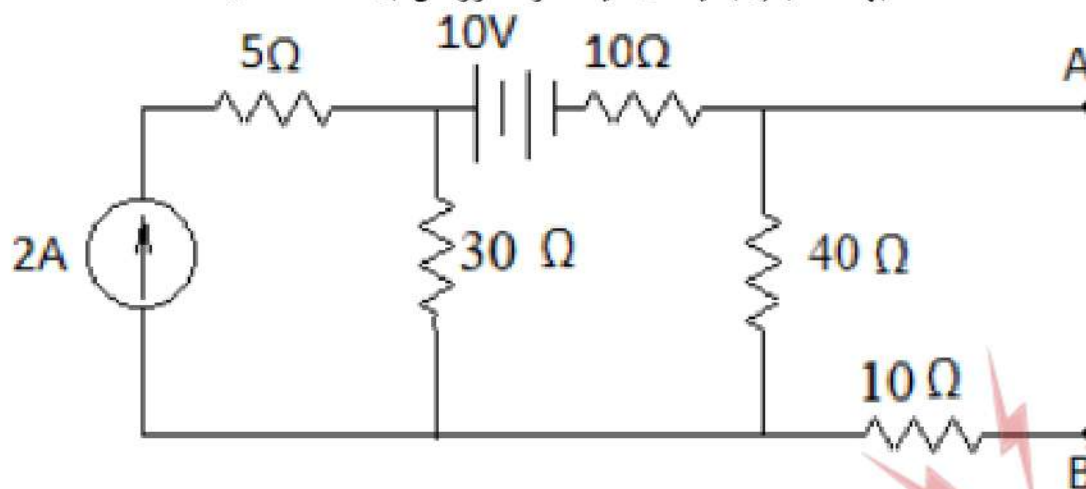
70. Nodal current method is a method based on:

- a. Kirchhoff Current Law
- b. Kirchhoff Voltage Law

Correct answer: Kirchhoff Current Law

71. Find Thevenin resistance R_{th} between terminals A and B in the given network:

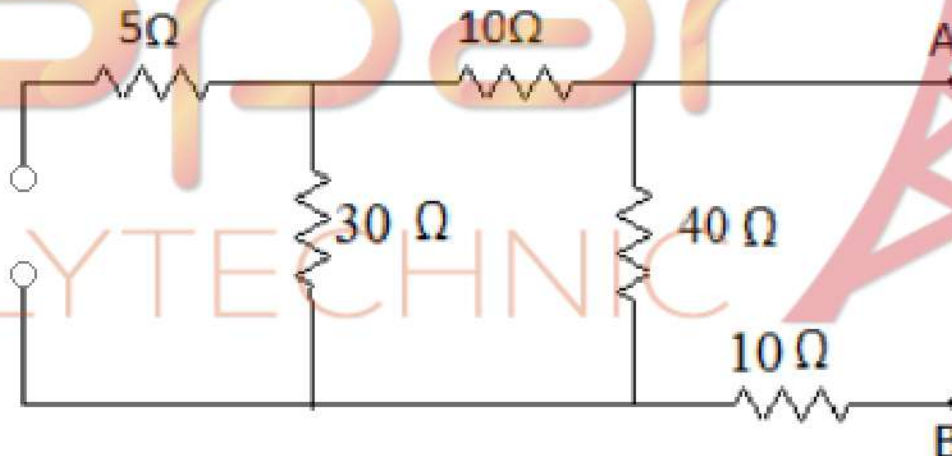
ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



- a. $20\ \Omega$
- b. $10\ \Omega$
- c. $30\ \Omega$
- d. $45\ \Omega$

Correct answer: $30\ \Omega$

To find thevenin's resistance R_{th} between terminals A and B in an network, the independent sources are deactivated where the voltage sources are replaced by a short circuit and the current sources are replaced by an open circuit. It is worth noting that dependent sources should not be deactivated at all.



Now the thevenin's resistance R_{th} is calculated:

$$R_{th} = (30\ \Omega + 10\ \Omega) // 40\ \Omega + 10\ \Omega$$

$$R_{th} = 30\ \Omega$$

72. Mesh analysis is generally used to determine:

- a. Current
- b. Voltage
- c. Power
- d. Resistor

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

Correct answer: Current

73. Nodal analysis is generally used to determine:

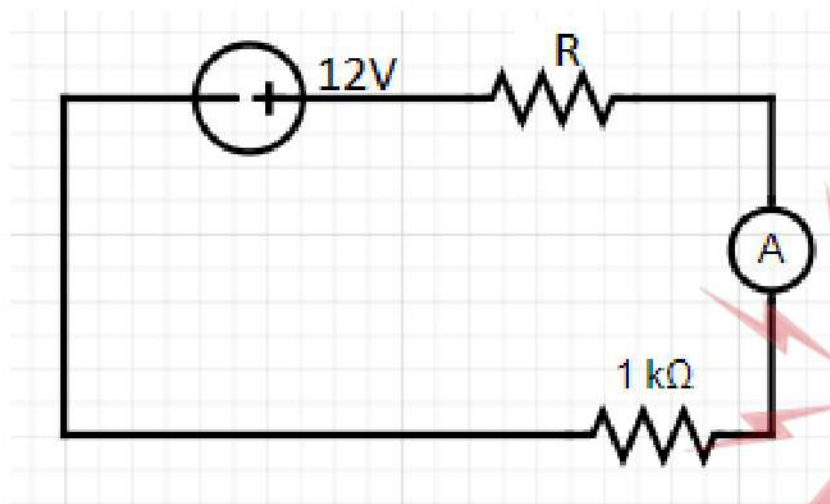
- a. Resistor
- b. Power
- c. Voltage
- d. Current

Correct answer: Voltage



Additional Questions

1. Consider the circuit shown in figure bellow. When the ammeter reading is 2mA then the value of R in (K Ω) is:



- a. 1
- b. 5
- c. 2
- d. 3

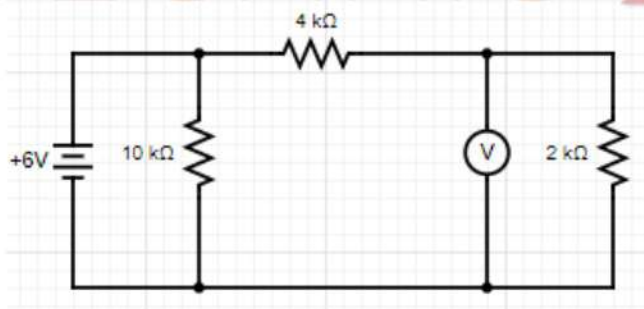
Correct answer: 5

$$I_{\text{ammeter}} = \frac{V_S}{R + R_1 \text{ k}\Omega}$$

$$2 \text{ mA} = \frac{12 \text{ V}}{R + 1 \text{ k}\Omega}$$

$$R = \frac{12 \text{ V}}{2 \text{ mA}} - 1 \text{ k}\Omega = 5 \text{ k}\Omega$$

2. For the circuit shown when $V_S = 6\text{V}$, then the voltmeter reading (V) is:



- a. 6 V
- b. 0 V
- c. 2 V
- d. 4 V

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

Correct answer: 2 V

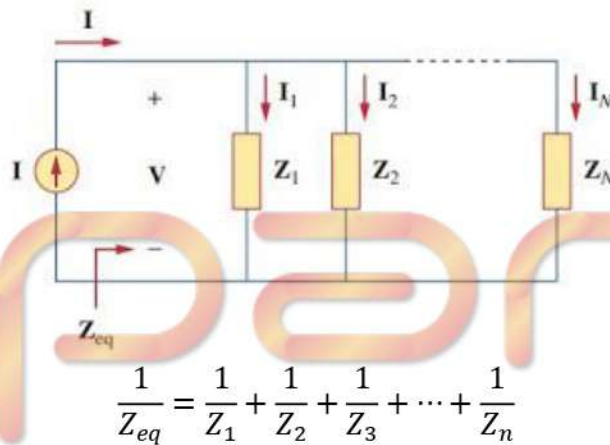
By using voltage divider rule:

$$V_{\text{voltmeter}} = \frac{R_{2\text{ k}\Omega}}{R_{4\text{ k}\Omega} + R_{2\text{ k}\Omega}} V_S = \frac{2\text{ k}\Omega}{4\text{ k}\Omega + 2\text{ k}\Omega} * 6\text{ V} = 2\text{ V}$$

3. When equivalent impedance equals one of the impedances divided by the number of them:
 - a. Parallel connected impedances
 - b. When impedances are equal and connected in parallel
 - c. Both series and parallel impedances
 - d. Series connected impedances
 - e. Both series and parallel impedances, in case of equal impedances
 - f. When impedances are equal and connected in series

Correct answer: When impedances are equal and connected in parallel

For parallel connected impedances:



And for equal impedances:

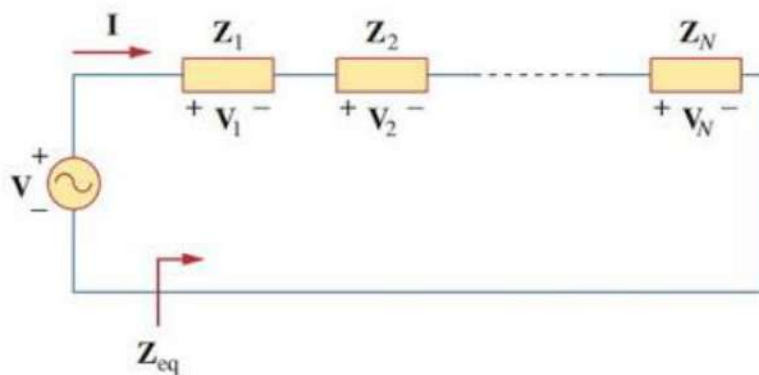
$$Z_{eq} = \frac{Z}{n}$$

4. When equivalent impedance equals one of the impedances multiplied by the number of them:
 - a. Parallel connected impedances
 - b. When impedances are equal and connected in parallel
 - c. Series connected impedances
 - d. Both series and parallel impedances
 - e. When impedances are equal and connected in series
 - f. Both series and parallel impedances, in case of equal impedances

Correct answer: When impedances are equal and connected in series

For series connected impedances:

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$$Z_{eq} = Z_1 + Z_2 + Z_3 + \dots + Z_n$$

And for equal impedances:

$$Z_{eq} = nZ$$

5. Ohm is the unit of measurement of:

- a. Current
- b. Capacitance
- c. Resistance
- d. Voltage

Correct answer: Resistance

6. Farad is the unit of measurement of:

- a. Capacitance
- b. Reactance
- c. Power
- d. Inductance

Correct answer: Capacitance

7. Henry is the unit of measurement of:

- a. Capacitance
- b. Reactance
- c. Power
- d. Inductance

Correct answer:

8. Ampere is the unit of measurement of:

- a. Current
- b. Voltage
- c. Reactance
- d. Inductance

Correct answer: Current

9. Volt is the unit of measurement of:

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- Capacitance
- Reactance
- Current
- Voltage

Correct answer: Voltage

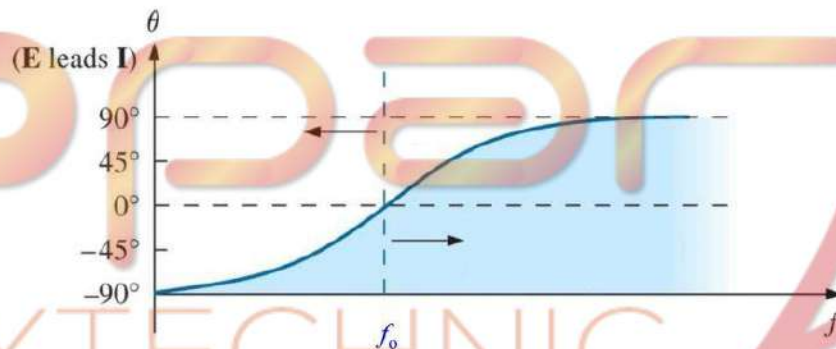
10. At the resonant frequency in a series RLC circuit:

- Impedance is minimum
- The voltage on the capacitor is equal to the voltage on the inductor (coil)
- The current is as high as possible
- All of them

Correct answer: All of them

At the resonant frequency in a series RLC circuit, the impedance of the circuit is minimum, as the capacitive reactance eliminates the inductive reactance, therefore the current is as high as possible, and as the capacitive reactance equals the inductive reactance, and the current passing through them is the same, so the voltage on the capacitor is equal to the voltage on the inductor (coil)

11. The figure below shows phase plot for the series resonant circuit, as the frequency increases above resonant frequency, the power factor becomes:



- Lagging
- Remains unity
- May be Lagging or Leading according to the values of the equivalent capacitance and equivalent inductance
- Leading

Correct answer: Lagging

As the frequency increases above resonant frequency, the capacitive reactance decreases and the inductive reactance increases:

$$\downarrow X_C = \frac{1}{\omega C} = \frac{1}{2\pi(f \uparrow)C}$$

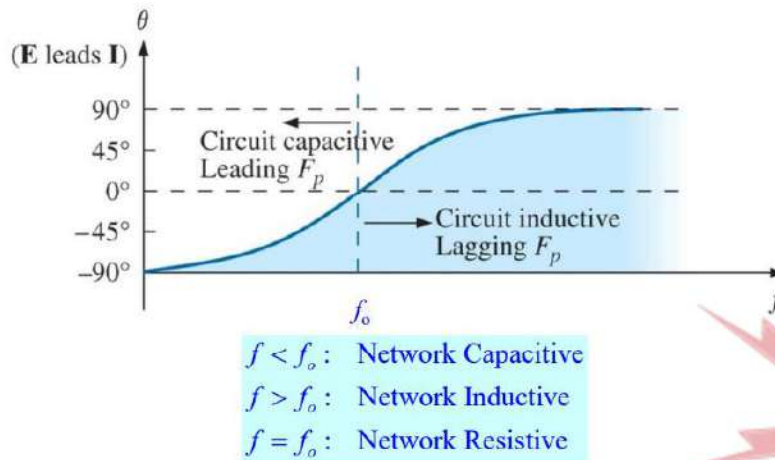
$$\uparrow X_L = 2\pi(f \uparrow)L$$

And it is known that at resonant frequency $X_C = X_L$, so when frequency increases above resonant frequency, the capacitive reactance decreases and becomes less than the inductive reactance which is also increases, so the

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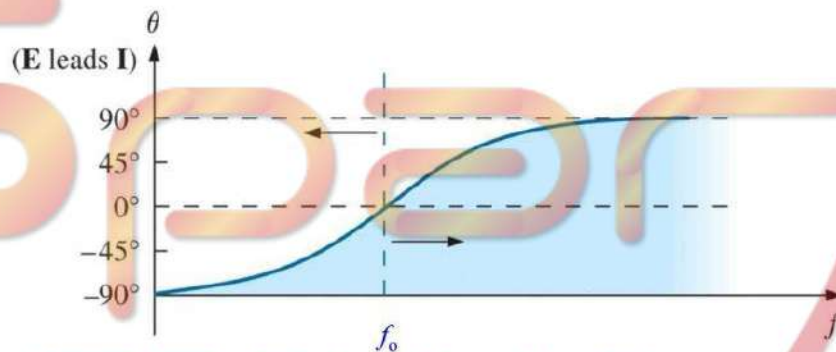
circuit becomes resistive inductive, and so the current lags the voltage, and therefore the power factor is lagging power factor, as shown in the following figure:

PHASE OF SERIES RESONANCE CIRCUIT



Phase plot for the series resonant circuit.

12. The figure below shows phase plot for the series resonant circuit, as the frequency decreases below resonant frequency, the power factor becomes:



- Leading
- May be Lagging or Leading according to the values of the equivalent capacitance and equivalent inductance
- Remains unity
- Lagging

Correct answer: Leading

As the frequency decreases below resonant frequency, the capacitive reactance increases and the inductive reactance decreases:

$$\uparrow X_C = \frac{1}{\omega C} = \frac{1}{2\pi(f \downarrow)C}$$

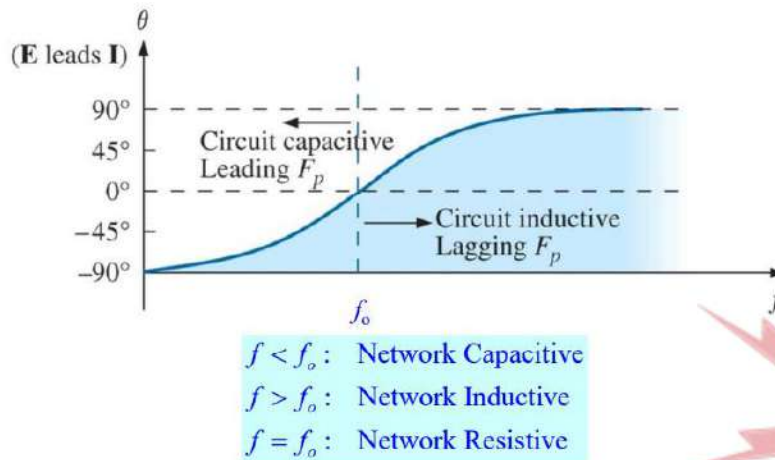
$$\downarrow X_L = 2\pi(f \downarrow)L$$

And it is known that at resonant frequency $X_C = X_L$, so when frequency decreases below resonant frequency, the capacitive reactance increases and becomes more than the inductive reactance which is also decreases, so

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the circuit becomes resistive capacitive, and so the current leads the voltage, and therefore the power factor is leading power factor, as shown in the following figure:

PHASE OF SERIES RESONANCE CIRCUIT



Phase plot for the series resonant circuit.

13. Which one of the following statements is true:

- At resonance, the voltage and current are in phase
- Resonance occurs in a circuit that has at least one inductor and one capacitor
- Since the resonant circuit is adjusted to select a band of frequencies it is called a selectivity curve
- Inductor and capacitor voltages can be much more than (Q times) source voltage
- At resonance, impedance is purely resistive
- All of them

Correct answer: All of them:

- Resonance occurs in a circuit that has at least one inductor and one capacitor
- At resonance, impedance is purely resistive
- At resonance, the voltage and current are in phase
- Inductor and capacitor voltages can be much more than (Q times) source voltage, where at resonance impedance is purely resistive, so:

$$I = \frac{V}{R}$$

$$|V_C| = I * X_C = \frac{V_m}{R} * \frac{1}{\omega_o C} = QV$$

$$|V_L| = I * X_L = \frac{V_m}{R} * \omega_o L = QV$$

$$Q = \frac{\omega_o L}{R} = \frac{1}{\omega_o CR}$$

- Since the resonant circuit is adjusted to select a band of frequencies it is called a selectivity curve

14. Dissipated power is half of the maximum value at:

- Half Power Frequencies
- Band frequencies

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- c. Cutoff frequencies
- d. All of them

Correct answer: All of them

15. Which one of the following statements is not true:

- a. Resonance frequency can't be obtained from the half-power frequencies
- b. QUALITY FACTOR is the ratio of the resonance frequency to the bandwidth. The higher the Q the smaller is the bandwidth
- c. BANDWIDTH is the difference between the half-power frequencies
- d. The SHARPNESS of the resonance is measured by the QUALITY FACTOR

Correct answer: Resonance frequency can't be obtained from the half-power frequencies

- a. BANDWIDTH is the difference between the half-power frequencies

$$B = \omega_2 - \omega_1$$

- b. The SHARPNESS of the resonance is measured by the QUALITY FACTOR

$$\uparrow Q = \frac{\omega_o}{B \downarrow}$$

- c. QUALITY FACTOR is the ratio of the resonance frequency to the bandwidth. The higher the Q the smaller is the bandwidth

$$Q = \frac{\omega_o}{B}$$

- d. Resonance frequency can be obtained from the half-power frequencies

$$\omega_o = \sqrt{\omega_1 \omega_2}$$

16. If resistance is made smaller with a fixed inductance and capacitance, the bandwidth _____ and the selectivity _____:

- a. Decreases, increases
- b. No relation between them
- c. Increases, increases

Correct answer: Decreases, increases

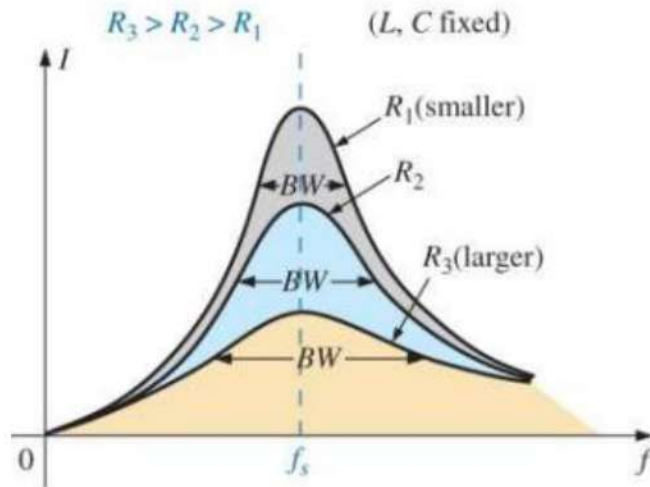
From equation in [appendix I](#):

$$\uparrow Q = \frac{1}{R \downarrow} \sqrt{\frac{L}{C}}$$

$$\downarrow B = \frac{\omega_o}{Q \uparrow}$$

Therefore, as resistance is made smaller with a fixed inductance and capacitance, the quality factor increases, and so the bandwidth decreases and selectivity increases, and this is illustrated in the following figure:

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



Effect of R on selectivity

17. If the ratio L/C increases with fixed resistance, the bandwidth again _____ with an _____ in selectivity:

- No relation between them
- Decreases, increase
- Increase, increase

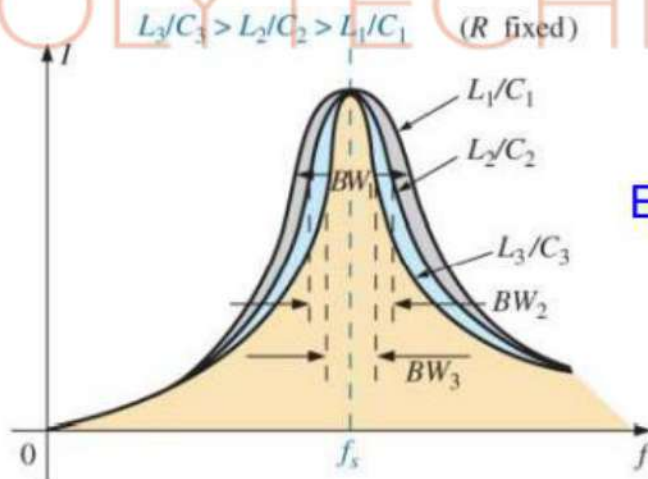
Correct answer: Decreases, increase

From equation in [appendix I](#):

$$\uparrow Q = \frac{1}{R} \sqrt{\frac{L}{C}} \uparrow$$

$$\downarrow B = \frac{\omega_o}{Q} \uparrow$$

Therefore, as the ratio L/C increases with fixed resistance, the quality factor increases, and so the bandwidth decreases and selectivity increases, and this is illustrated in the following figure:



Effect of L and C on selectivity

18. The current at resonant frequency in series RLC circuits is:

- Does not change

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

- b. Minimum
- c. Depends on the circuit
- d. Maximum

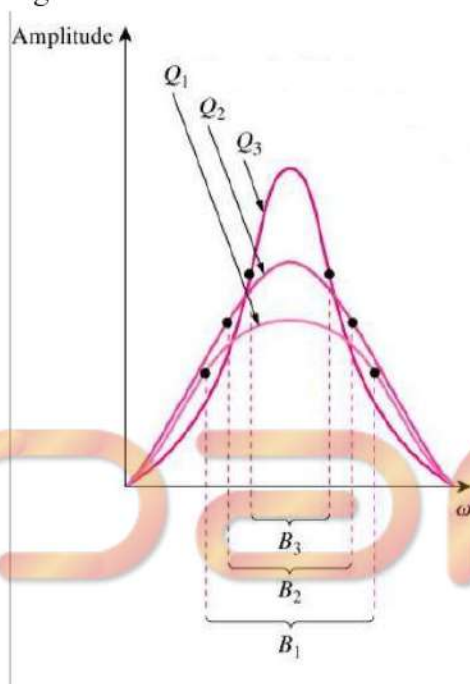
Correct answer: Maximum

As $X_L = X_C$, then $Z = \sqrt{R^2 + X^2} = \sqrt{R^2 + (X_L - X_C)^2} = R$

So,

$$I(\text{Maximum}) = \frac{V}{Z} = \frac{V}{R(\text{Minimum})}$$

19. The selectivity for the following curve is:



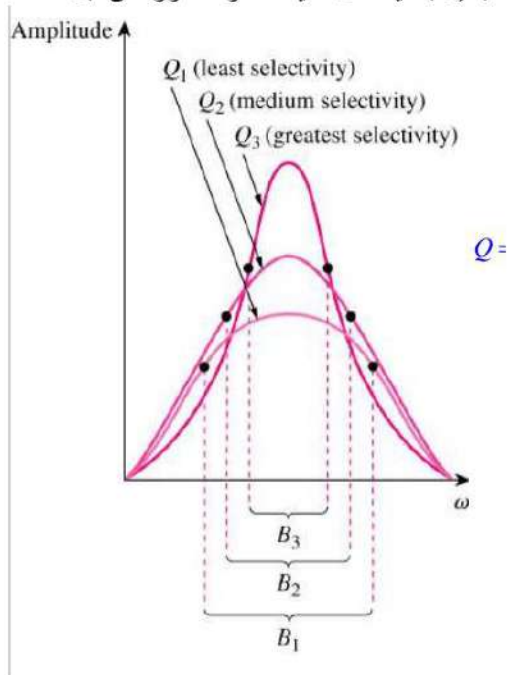
- a. $Q_1 > Q_2 > Q_3$
- b. $Q_3 > Q_2 > Q_1$
- c. None of them

Correct answer: $Q_3 > Q_2 > Q_1$

As the band width decreases, the quality factor decreases, as shown in the following equation and figure:

$$\uparrow Q = \frac{\omega_o}{B} \downarrow$$

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



20. What is the only method we can use to analysis the circuits which contain different frequencies sources?

- Norton theorem
- Nodal analysis method
- Source transformations
- Max. Power transfer
- None of them
- Thevenin theorem
- Mesh analysis method
- Superposition theorem
- Cannot be analysis at all

Correct answer: Superposition theorem

The superposition theorem must be used if the circuit contains different frequencies sources

Note that, the circuit which contains AC and DC sources is a circuit which has different frequencies sources as the DC source has zero frequency

21. Which of these methods called universal method?

- Norton theorem
- Nodal analysis method
- Source transformations
- Max. Power transfer
- Thevenin theorem
- Mesh analysis method
- Superposition theorem
- None of them

Correct answer: Nodal analysis method

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

Universal method is a method that can be applied to any electrical circuit, whether it is two-dimensional (planar) or three-dimensional (nonplanar) electrical circuit.

This comparison is conducted between the basic methods of analysis, which is the basis of electrical circuit theorems (Thevenin theorem, Norton theorem, Superposition theorem, Source transformations, Max. Power transfer), which are the Nodal analysis method (NAM) and the mesh analysis method (MAM).

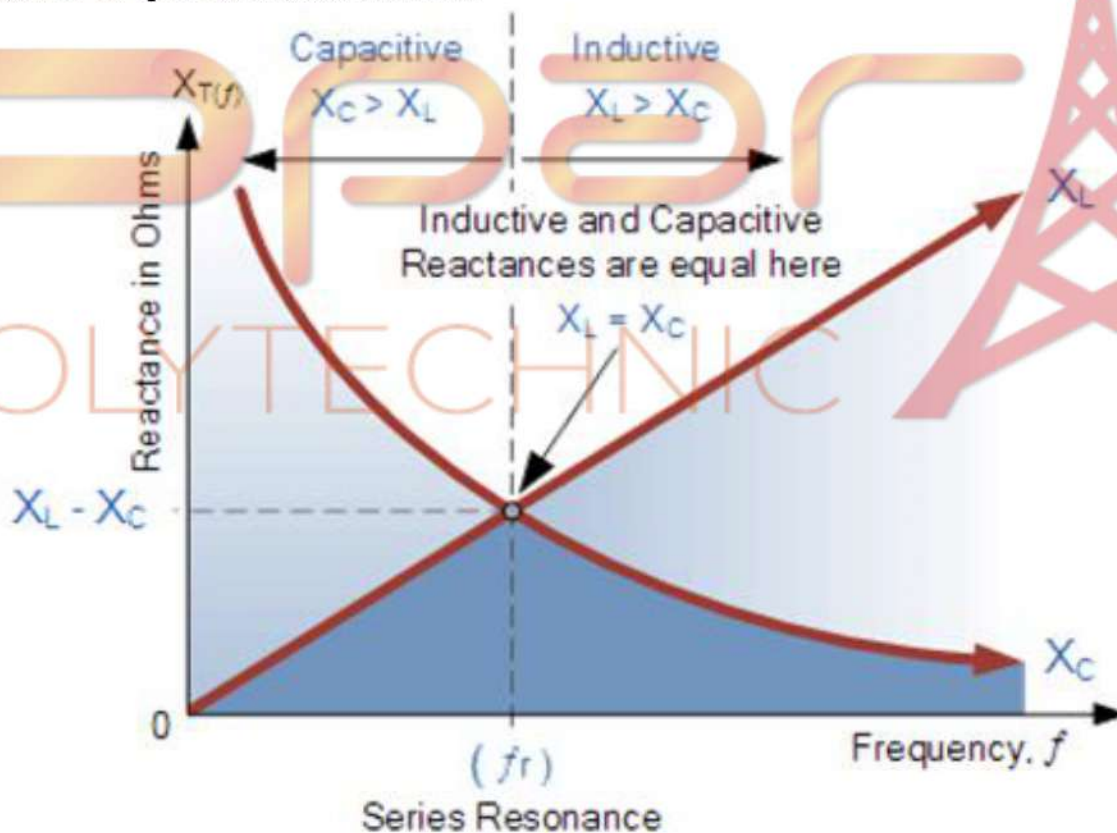
The method of the Mesh Analysis can only be applied to planar circuits since they depend on loops, so if we applied the MAM on a nonplanar circuit, literally a too large number of loops would be considered and the method would become impractical at all.

The method of the Nodal Analysis can be applied to planar and nonplanar circuits because it depends on the number of nodes and it is easy to apply the KCL even if the node is at a three-dimensional level (because you only have wires attached to the node and not loops)

22. Under DC steady state Conditions, the capacitor behaves as _____ and the inductor behaves as _____.

- A short circuit, a short circuit
- An open circuit, a short circuit
- A short circuit, an open circuit
- An open circuit, an open circuit
- None of them

Correct answer: an open circuit, a short circuit



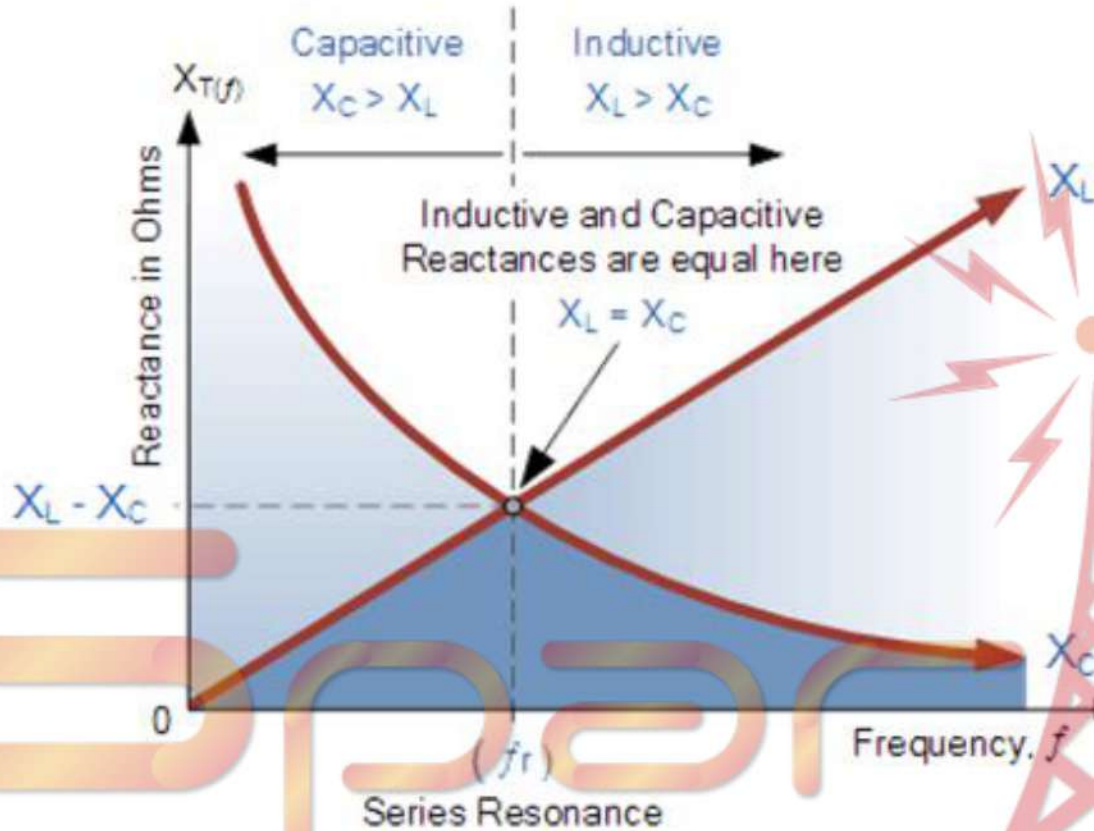
23. Under very high frequencies, the capacitor behaves as _____ and the inductor behaves as _____.

- A short circuit, a short circuit
- An open circuit, an open circuit

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

- c. An open circuit, a short circuit
- d. A short circuit, an open circuit
- e. None of them

Correct answer: a short circuit, an open circuit



24. Resonant frequency in series and parallel RLC circuits is:

- a. $\omega_o = \sqrt{LC}$
- b. $\omega_o = \frac{1}{LC}$
- c. $\omega_o = \frac{1}{\sqrt{LC}}$
- d. None of these

Correct answer: $\omega_o = \frac{1}{\sqrt{LC}}$

As $X_L = X_C$ at resonance frequency, then:

$$\begin{aligned}\omega_o L &= \frac{1}{\omega_o C} \\ \omega_o^2 &= \frac{1}{LC} \\ \omega_o &= \frac{1}{\sqrt{LC}}\end{aligned}$$

25. At resonance frequency, the magnitude of impedance in the series RLC circuit is:

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- a. Maximum
- b. Minimum
- c. Does not change
- d. Depends on the circuit

Correct answer: Minimum

As $X_L = X_C$, then $Z = \sqrt{R^2 + X^2} = \sqrt{R^2 + (X_L - X_C)^2} = R$

26. In series resonance, the supply voltage V_s and the current are:

- a. In-phase
- b. Out-of-phase by 90°
- c. Out-of-phase by 180°
- d. Depends on the circuit

Correct answer: In-phase

As $X_L = X_C$, then $Z = \sqrt{R^2 + X^2} = \sqrt{R^2 + (X_L - X_C)^2} = R$, as the impedance is purely resistive, then the voltage and current are in phase $\theta = 0^\circ \rightarrow \cos \theta = \cos 0^\circ = 1$

27. In series resonance circuit, the inductor voltage and capacitor voltage can be much more than the source voltage:

- a. True
- b. False

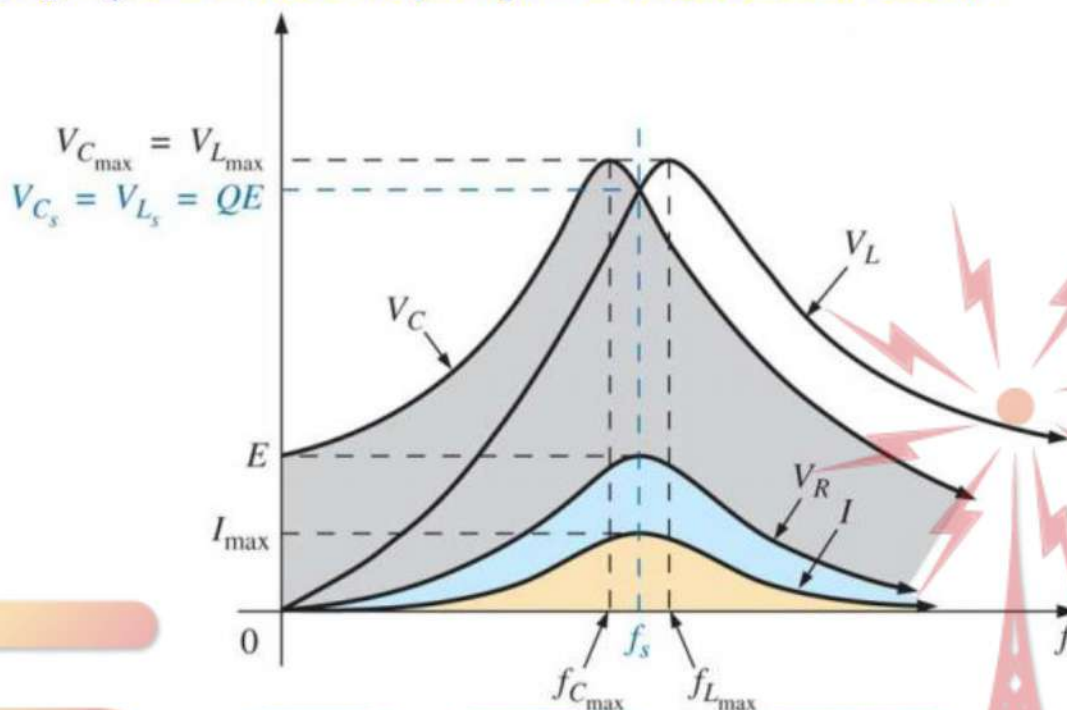
Correct answer: True

POLYTECHNIC

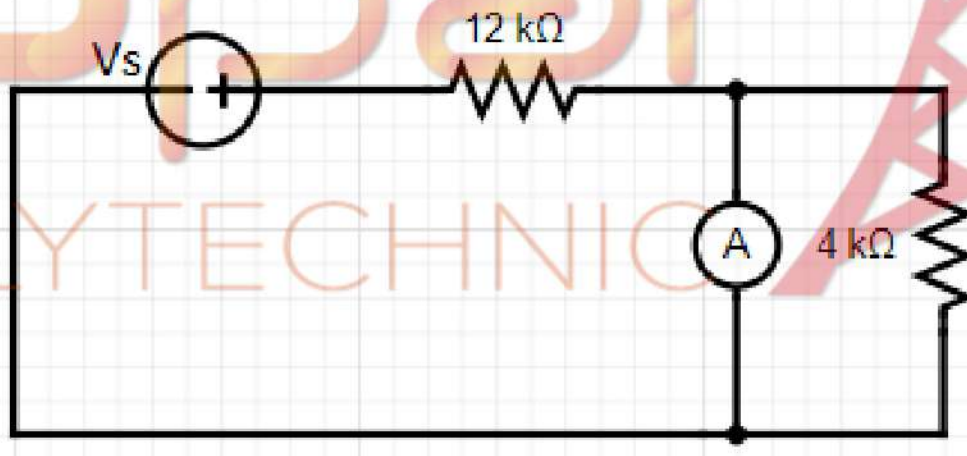
ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

SERIES RESONANCE

➤ V_R , V_L , V_C , and I versus frequency for a series resonant circuit.



28. Consider the circuit shown when $V_s = 9V$, then the ammeter reading is:



- a. 2.25 mA
- b. 0.75 mA
- c. 0 mA
- d. 0.0625 mA

Correct answer: 0.75 mA

The ammeter connected in parallel, so the resistance 4 kΩ is shorted

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



Therefore, the current is calculated as follows:

$$I = \frac{V_S}{R} = \frac{9\text{ V}}{12\text{ k}\Omega} = 0.75\text{ mA}$$

29. The ammeter is an instrument used to measure the _____ the element:

- a. Voltage across
- b. Power in
- c. Energy in
- d. Current through

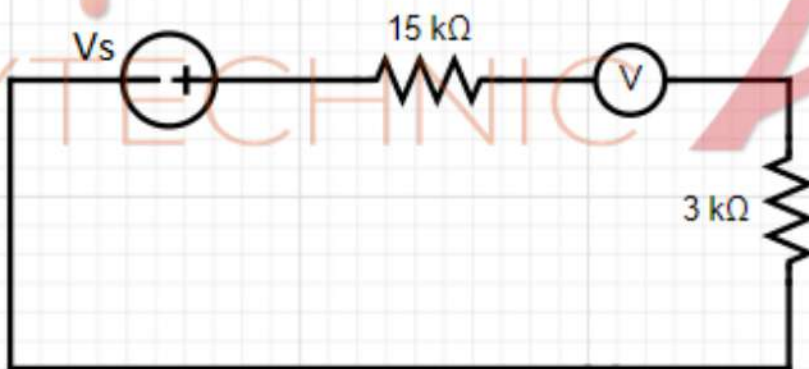
Correct answer: Current through

30. The voltmeter is an instrument used to measure the _____ the element:

- a. Current through
- b. Energy in
- c. Power in
- d. Voltage across

Correct answer: Voltage across

31. For the circuit shown when $V_S = 18\text{V}$, then the voltmeter reading is:



- a. 0V
- b. 18V
- c. 3V
- d. 15V

Correct answer: 18V

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

The voltmeter is connected in series, so the circuit is opened, and so the voltmeter measures the voltage of the source

32. When a practical ammeter is used to measure the current in an element then the percentage of error is increased as the internal resistance of the ammeter is (for the same circuit):

- a. Independent
- b. Increased
- c. Deceased
- d. Depends on the circuit connection

Correct answer: Increased

For ideal ammeter, the internal resistance equals zero, but for practical ammeter, the internal resistance is not zero (very low but not equal to zero), and it is known that the internal resistance of ammeter is in series with the element, so as the internal resistance of ammeter increases, the equivalent resistance also increases, and so the current decreases below its actual value, so the percentage error increases

33. When a practical voltmeter is used to measure the voltage across an element then the percentage of error is increased as the internal resistance of the voltmeter is (for the same circuit):

- a. Independent
- b. Depends on the circuit connection
- c. Increased
- d. Deceased

Correct answer: Deceased

For ideal voltmeter, the internal resistance equals infinity, but for practical voltmeter, the internal resistance is not infinity (very high but not equal to infinity), and it is known that the internal resistance of voltmeter is in parallel with the element, so as the internal resistance of voltmeter decreases, the equivalent resistance also decreases, and so the current increases upon its actual value, so the percentage error increases

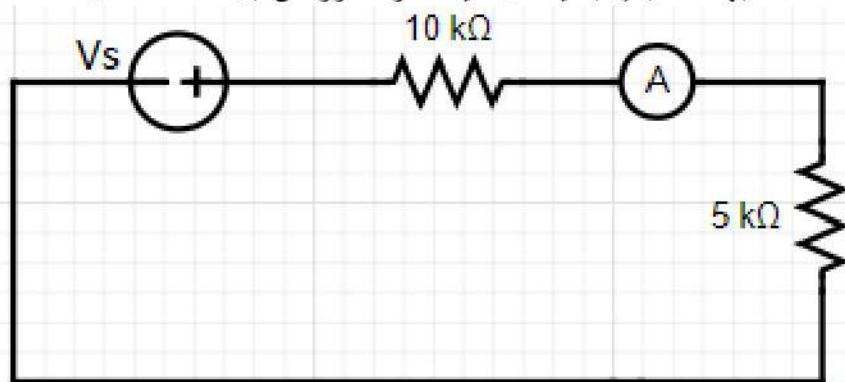
34. The equivalent circuit of the ammeter is:

- a. is a short circuit
- b. depends on the circuit connection
- c. is an open circuit

Correct answer: is a short circuit

35. For the circuit shown when $V_S = 9V$, then the ammeter reading is:

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

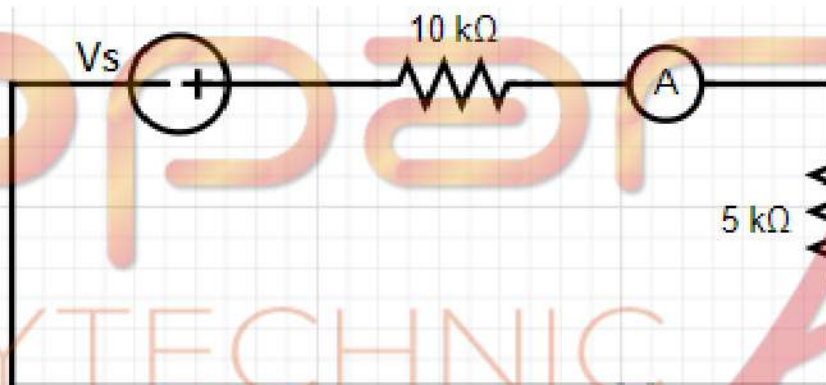


- a. 1.2 mA
- b. 0.6 mA
- c. 1.0 mA
- d. 0.8 mA

Correct answer: 0.6 mA

$$I = \frac{V_S}{R_T} = \frac{9V}{10\text{ k}\Omega + 5\text{ k}\Omega} = 0.6\text{ mA}$$

36. Consider the circuit shown. If $V_S = 18V$, then the percentage of error when the ammeter reading is 1.05 mA equal to:



- a. 5%
- b. 8%
- c. 12.5%

Correct answer: 12.5%

$$I_{theoretically} = \frac{V_S}{R_T} = \frac{18V}{10\text{ k}\Omega + 5\text{ k}\Omega} = 1.2\text{ A}$$

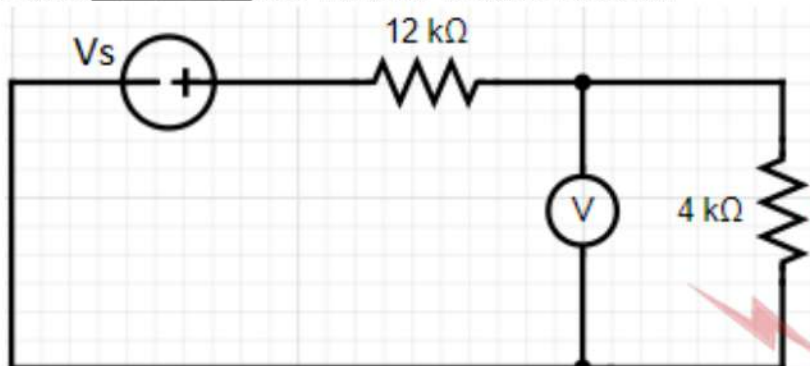
$$\text{Percentage Error}\% = \frac{|I_{theoretically} - I_{practically}|}{I_{theoretically}} * 100\% = \frac{|1.2\text{ A} - 1.05\text{ A}|}{1.2\text{ A}} * 100\% = 12.5\%$$

37. The equivalent circuit of the voltmeter is:

- a. is a short circuit
- b. is an open circuit
- c. depends on the circuit connection

Correct answer: is an open circuit

38. Consider the circuit shown. If the voltmeter is practical (has a large internal resistance) then current from the source becomes _____ that when the voltmeter is ideal:



- a. larger than
- b. the same as
- c. smaller than

Correct answer: larger than

For ideal voltmeter, the internal resistance equals infinity, so:

$$R_T = 12 \text{ k}\Omega + (4 \text{ k}\Omega // \text{infinity resistance}) = 12 \text{ k}\Omega + 4 \text{ k}\Omega = 16 \text{ k}\Omega$$

$$I = \frac{V_S}{R_T} = \frac{V_S}{16 \text{ k}\Omega}$$

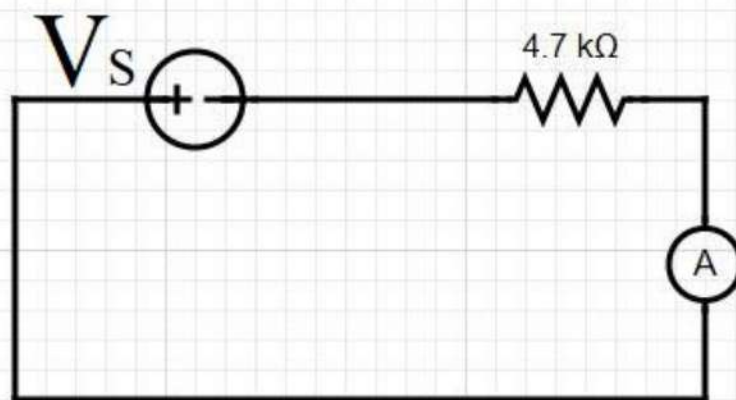
And for practical voltmeter, the internal resistance is not infinity (very high but not equal to infinity)

$$R_T = 4.7 \text{ k}\Omega + R_{\text{ammeter}}$$

$$I = \frac{V_S}{R_T} = \frac{V_S}{12 \text{ k}\Omega + (4 \text{ k}\Omega // \text{very high resistance})}$$

According to previous, the total resistance in the case of practical voltmeter is smaller than the total resistance in the case of ideal voltmeter, so the current in the case of practical voltmeter is larger than the current in the case of ideal voltmeter

39. Consider the circuit shown. If the ammeter is practical (has a small internal resistance) then current from the source becomes _____ that when the voltmeter is ideal:



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- a. larger than
- b. smaller than
- c. the same as

Correct answer: smaller than

For ideal ammeter, the internal resistance equals zero, so:

$$R_T = 4.7 \text{ k}\Omega$$

$$I = \frac{V_S}{R_T} = \frac{V_S}{4.7 \text{ k}\Omega}$$

And for practical ammeter, the internal resistance is not zero (very small but not equal to zero)

$$R_T = 4.7 \text{ k}\Omega + R_{\text{ammeter}}$$

$$I = \frac{V_S}{R_T} = \frac{V_S}{4.7 \text{ k}\Omega + R_{\text{ammeter}}}$$

According to previous, the total resistance in the case of practical ammeter is larger than the total resistance in the case of ideal ammeter, so the current in the case of practical ammeter is smaller than the current in the case of ideal ammeter

40. To measure the voltage across any element in a circuit. The voltmeter is connected:

- a. in series with element
- b. in parallel with element
- c. in parallel or in series both method can be used

Correct answer: in parallel with element

41. The unit of measurement of **current** is:

OR

The unit of measurement of **flow of electrical charge carriers** is:

- a. Watt
- b. Ampere
- c. Joule
- d. Coulomb

Correct answer: Ampere

42. The unit of measurement of voltage is:

- a. Ampere
- b. Volt
- c. Watt
- d. Joule

Correct answer: Volt

43. To measure the current through any element in a circuit the ammeter is connected:

- a. in parallel with element
- b. in parallel or in series both method can be used
- c. in series with element

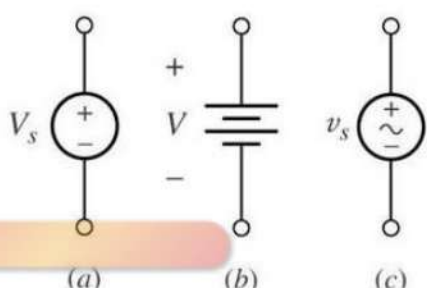
Correct answer: in series with element

44. Which of the following symbol is used for representing the Independent sources?

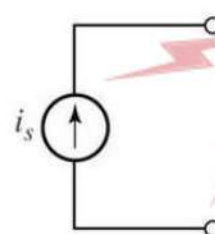
- Diamond
- Triangle
- Circle
- Square

Correct answer: Circle

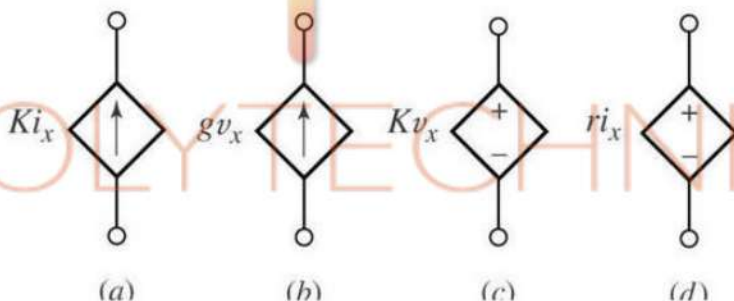
Sources:



Symbol for independent voltage source: (a) DC voltage source; (b) battery; (c) ac voltage source.



Symbol for an independent current source.



The four different types of dependent sources:

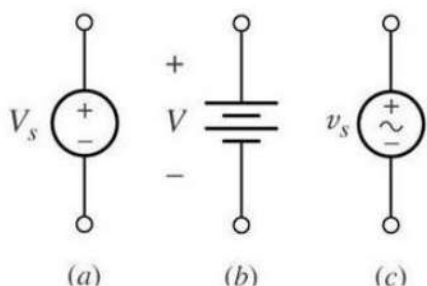
- current-controlled current source; (CCC)
- voltage-controlled current source; (VCC)
- voltage-controlled voltage source; (VCV)
- current-controlled voltage source. (CCV)

45. Which of the following symbol is used for representing the Dependent sources?

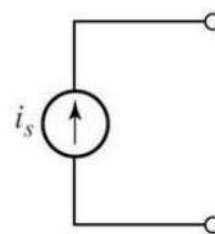
- Diamond
- Circle
- Triangle
- Square

Correct answer: Diamond

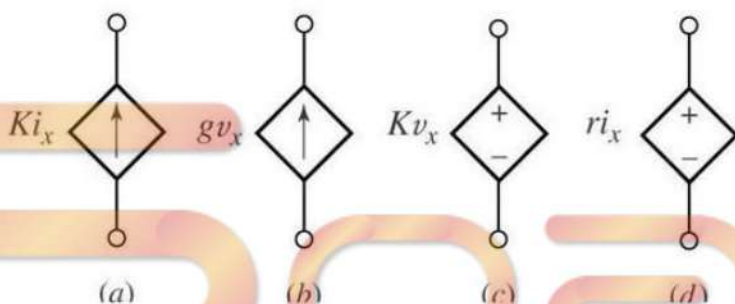
Sources:



Symbol for independent voltage source: (a) DC voltage source; (b) battery; (c) ac voltage source.



Symbol for an independent current source.



The four different types of dependent sources:

(a) current-controlled current source; (CCC)

(b) voltage-controlled current source; (VCC)

(c) voltage-controlled voltage source; (VCV)

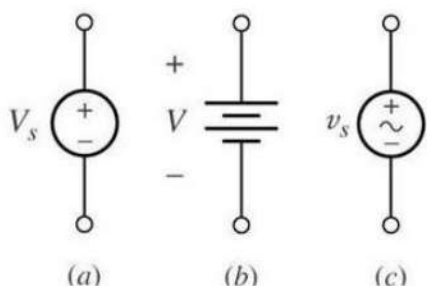
(d) current-controlled voltage source. (CCV)

46. How many types are the dependent sources?

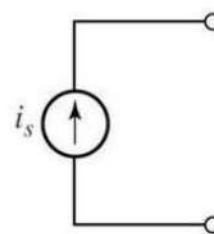
- 6
- 8
- 2
- 4

Correct answer: 4

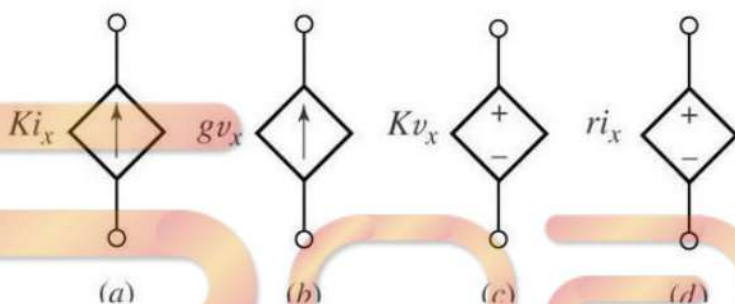
Sources:



Symbol for independent voltage source: (a) DC voltage source; (b) battery; (c) ac voltage source.



Symbol for an independent current source.



The four different types of dependent sources:

(a) current-controlled current source; (CCC)

(b) voltage-controlled current source; (VCC)

(c) voltage-controlled voltage source; (VCV)

(d) current-controlled voltage source. (CCV)

47. A supernode is situated between

- Two reference nodes
- Two essential nodes
- Essential node and reference node
- Essential node and neutral path

Correct answer: Two essential nodes

48. There are 13 branches in a complicated network and nearly 8 nodes. equations are required to solve the circuit in node-voltage method?

- 7
- 8
- 5
- 6

Correct answer: 7

To number of required equations needed to solve the circuit by using node-voltage method is obtained by the following equation:

$$\# \text{ of equations in NAM} = N - 1 = 8 - 1 = 7$$

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49. There are 13 branches in a complicated network and nearly 8 nodes. equations are required to solve the circuit in mech-current method?

- a. 6
- b. 5
- c. 8
- d. 7

Correct answer: 6

To number of required equations needed to solve the circuit by using mech-current method is obtained by the following equation:

$$\# \text{ of equations in NAM} = B - N + 1 = 13 - 8 + 1 = 6$$

50. Mesh analysis is the suitable for

- a. Current sources
- b. Voltage sources
- c. Complex elements
- d. Unilateral elements

Correct answer: Current sources

51. Which of the following is the Source Transformation?

- a. Complicated
- b. Bilateral
- c. Unilateral
- d. Unique

Correct answer: Bilateral

52. Which of the following is equal to thevenin's voltage?

- a. Short circuit current
- b. Open circuit voltage
- c. Short circuit voltage
- d. Open circuit current

Correct answer: Open circuit voltage

53. Network is designed with to obtain high efficiency.

- a. Zero Q factor
- b. High Q factor
- c. Unity Q factor
- d. Low Q factor

Correct answer: High Q factor

54. To which of the following is the superposition theorem applicable?

- a. Power
- b. Current

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- c. Voltage & voltage
- d. Voltage
- e. All of the above

Correct answer: Voltage & voltage

55. Which among the following is also regarded as 'Dual of Thevenin's Theorem'?

- a. Maximum Power Transfer Theorem
- b. Superposition Theorem
- c. Norton's Theorem
- d. Source Transformation Theorem

Correct answer: Norton's Theorem

56. In nodal analysis, if there are N nodes in the circuit then how many equations will be written to solve the network?

- a. $N^2 - 1$
- b. $N + 1$
- c. $N - 2$
- d. $N + 2$
- e. $N - 1$
- f. N

Correct answer: $N - 1$

57. In mesh analysis, if there are N nodes and B meshes in the circuit then how many equations will be written to solve the network?

- a. $B - 1$
- b. $B - N - 1$
- c. $B + N - 1$
- d. $B - N$
- e. $B + N + 1$
- f. $B - N + 1$

Correct answer: $B - N + 1$

58. Two ideal voltage sources of unequal output voltages cannot be placed in.....

- a. Parallel
- b. Series
- c. Both series and parallel
- d. None of them

Correct answer: Parallel

59. Two ideal current sources of unequal output voltages cannot be placed in.....

- a. Parallel

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- b. Series
- c. Both series and parallel
- d. None of them

Correct answer: Series

60. Which one is the network theorem in electrical engineering?

- a. Mesh analysis
- b. Nodal analysis
- c. Norton's
- d. All of them

Correct answer: Norton's

61. Kirchhoff's current law states that algebraic sum of the currents meeting at the junction is zero:

- a. True
- b. False

Correct answer: True

62. According to Kirchhoff's voltage law, the algebraic sum of all IR drops and e.m.fs. in any closed loop of a network is always ____.

- a. positive
- b. negative
- c. determined by battery e.m.fs.
- d. Zero

Correct answer: Zero

63. Kirchhoff's current law is applicable to only ____.

- a. electronic circuits
- b. closed loops in a network
- c. electric circuits
- d. junction in a network

Correct answer: junction in a network

64. Kirchhoff's voltage law is related to ____.

- a. junction currents
- b. IR drops
- c. battery e.m.fs and IR drops
- d. battery e.m.fs.

Correct answer: both B and C

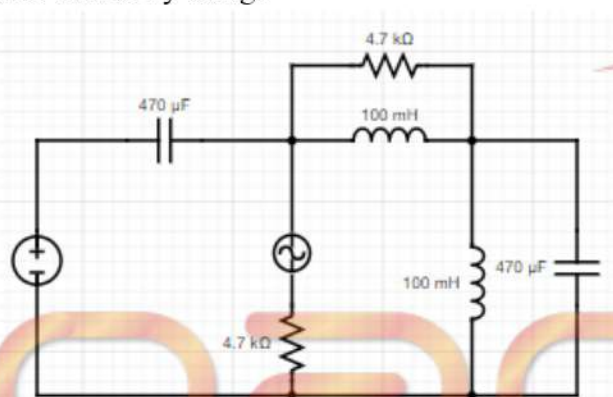
65. What does Thevenin's Theorem states?

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- any two terminal, linear bilateral circuit consisting of linear elements and active sources may contain independent and/or dependent sources connected across the terminal of Z_L can be replaced by a single equivalent voltage source of V_{th} in series with a single impedance Z_{th} across the two terminals of Z_L
- any two terminal, linear bilateral circuit consisting of linear elements and active sources may contain independent and/or dependent sources connected across the terminal of Z_L can be replaced by a single equivalent current source of I_N in parallel with a single impedance Z_N across the two terminals of Z_L
- Both of them
- None of them

Correct answer: any two terminal, linear bilateral circuit consisting of linear elements and active sources may contain independent and/or dependent sources connected across the terminal of Z_L can be replaced by a single equivalent voltage source of V_{th} in series with a single impedance Z_{th} across the two terminals of Z_L

66. The following circuit can be solved by using:



- Source transformations
- Thevenin theorem
- Superposition theorem
- Cannot be analysis at all
- Norton theorem
- All of them
- None of them

Correct answer: Superposition theorem

The superposition theorem must be used in the case of different frequencies sources, and it's know that the dc source has frequency equals zero

67. What do you mean by maximum power transfer theorem?

- Maximum power is transferred to the load when the load impedance equals zero
- Maximum power is transferred to the load when the load impedance equals the source impedance or the thevenin's impedance squared as seen from the load terminals
- Maximum power is transferred to the load when the load impedance equals the source impedance or the thevenin's impedance as seen from the load terminals
- None of them

Correct answer: Maximum power is transferred to the load when the load impedance equals the source impedance or the thevenin's impedance as seen from the load terminals

68. Which one is the limitations for Thevenin's theorem:are:

- a. It greatly simplifies the portion of the circuit of lesser importance and enable us to view the action of output part directly
- b. It reduces complex circuit to a simple circuit viz a single source of emf in series with a single resistance
- c. This theorem is applicable only for linear, bilateral networks
- d. None of the above

Correct answer: This theorem is applicable only for linear, bilateral networks

69. What are the advantages of thevenin's theorem?

- a. It reduces complex circuit to a simple circuit viz a single source of emf in series with a single resistance and it greatly simplifies the portion of the circuit of lesser importance and enable us to view the action of output part directly
- b. This theorem is valid only for a certain range, because it is applicable for linear circuits only
- c. It greatly simplifies the portion of the circuit of lesser importance and enable us to view the action of output part directly
- d. It reduces complex circuit to a simple circuit viz a single source of emf in series with a single resistance

Correct answer: Both A and C (It reduce complex circuit to a simple circuit viz a single source of emf in series with a single resistance & It greatly simplifies the portion of the circuit of lesser importance and enable us to view the action of output part directly)

70. Which one is the limitation of Norton's Theorem?

- a. The value of direct current which produces the same heating effect in a given resistor as is produced by the given alternating current when passed for the same time
- b. Voltage
- c. It's not for such modules which are not linear like diodes, the transistor
- d. None of the above

Correct answer: It's not for such modules which are not linear like diodes, the transistor

71. For series circuits:

- a. The current flowing in the circuit is constant along the circuit
- b. The equivalent resistance equals the summation of the resistances included in the circuit.
- c. circuit.
- d. The voltage of the source equals the summation of the voltages on the included resistors in the circuit
- e. All of them

Correct answer: All of them

72. For parallel circuit:

- a. The upset of the equivalent resistance equals the summation of the upsets of the resistances involved in the circuit
- b. The source current equals the summation of the individual currents in the circuit

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- c. The voltages applied on all the resistors are the same
- d. All of them

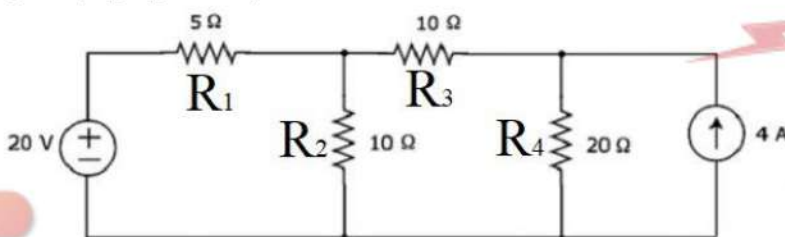
Correct answer: All of them

73. difference between the experimental values and calculated values results from:

- a. The presence of losses on the measuring devices and so a percentage error in measurement process
- b. losses of electrical energy in the form of heat generated in the wires
- c. losses of electrical energy in the form of heat generated in the resistors
- d. All of them

Correct answer: All of them

74. The current (in the branch of voltage source/ in the branch of $5\ \Omega$ resistor/ in the branch of R_1) due to voltage source only is: (superposition)



- a. 1.2 A
- b. $\frac{80}{39}$ A
- c. 1.6 A
- d. $\frac{40}{13}$ A
- e. $\frac{40}{39}$ A
- f. 0.4 A
- g. $\frac{20}{13}$ A

Correct answer: 1.6 A

By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors R_3 & R_4 are in series, and they are in parallel with R_2 , and they are in series with R_1 :

$$R_{series} = R_3 + R_4 = 10\ \Omega + 20\ \Omega = 30\ \Omega$$

The parallel resistance for R_{series} & R_2 is:

$$R_{parallel} = \frac{R_2 * R_{series}}{R_2 + R_{series}} = \frac{10\ \Omega * 30\ \Omega}{10\ \Omega + 30\ \Omega} = 7.5\ \Omega$$

The equivalent resistance is:

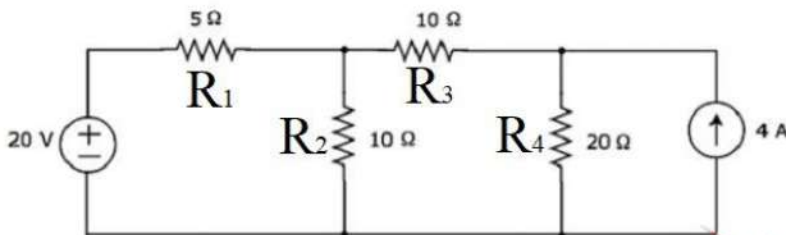
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$$R_{eq} = R_{parallel} + R_1 = 7.5 \Omega + 5 \Omega = 12.5 \Omega$$

The current due to the voltage source in the branch of voltage source:

$$I = \frac{V_S}{R_{eq}} = \frac{20 V}{12.5 \Omega} = 1.6 A$$

75. The current in the branch of R_2 due to voltage source only is: (superposition)



- a. $\frac{40}{39} A$
- b. $1.2 A$
- c. $1.6 A$
- d. $\frac{80}{39} A$
- e. $\frac{20}{13} A$
- f. $\frac{40}{13} A$
- g. $0.4 A$

Correct answer: $1.2 A$

By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors R_3 & R_4 are in series, and they are in parallel with R_2 , and they are in series with R_1 :

$$R_{series} = R_3 + R_4 = 10 \Omega + 20 \Omega = 30 \Omega$$

The parallel resistance for R_{series} & R_2 is:

$$R_{parallel} = \frac{R_2 * R_{series}}{R_2 + R_{series}} = \frac{10 \Omega * 30 \Omega}{10 \Omega + 30 \Omega} = 7.5 \Omega$$

The equivalent resistance is:

$$R_{eq} = R_{parallel} + R_1 = 7.5 \Omega + 5 \Omega = 12.5 \Omega$$

The current due to the voltage source in the branch of voltage source:

$$I = \frac{V_S}{R_{eq}} = \frac{20 V}{12.5 \Omega} = 1.6 A$$

Now, the current in the branch of R_2 resistor is found by using current divider rule:

$$I_{R_2} + I_{R_3+R_4} = I$$

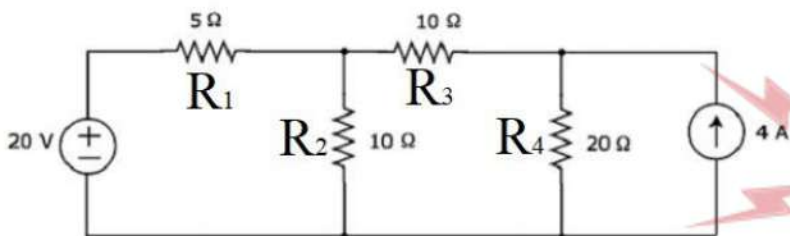
$$I_{R_3+R_4} = I - I_{R_2}$$

$$R_2 I_{R_2} = (R_3 + R_4) I_{R_3+R_4}$$

$$R_2 I_{R_2} = (R_3 + R_4)(I - I_{R_2})$$

$$I_{R_2} = \frac{(R_3 + R_4)I}{R_2 + R_3 + R_4} = \frac{(10 \Omega + 20 \Omega) * 1.6 A}{10 \Omega + 10 \Omega + 20 \Omega} = 1.2 A$$

76. The current in the branch of R_3 due to voltage source only is: (superposition)



a. $\frac{40}{39} A$

b. $1.6 A$

c. $\frac{20}{13} A$

d. $0.4 A$

e. $\frac{40}{13} A$

f. $\frac{80}{39} A$

g. $1.2 A$

Correct answer: $0.4 A$

By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors R_3 & R_4 are in series, and they are in parallel with R_2 , and they are in series with R_1 :

$$R_{series} = R_3 + R_4 = 10 \Omega + 20 \Omega = 30 \Omega$$

The parallel resistance for R_{series} & R_2 is:

$$R_{parallel} = \frac{R_2 * R_{series}}{R_2 + R_{series}} = \frac{10 \Omega * 30 \Omega}{10 \Omega + 30 \Omega} = 7.5 \Omega$$

The equivalent resistance is:

$$R_{eq} = R_{parallel} + R_1 = 7.5 \Omega + 5 \Omega = 12.5 \Omega$$

The current due to the voltage source in the branch of voltage source:

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$$I = \frac{V_S}{R_{eq}} = \frac{20\text{ V}}{12.5\ \Omega} = 1.6\text{ A}$$

Now, the current in the branch of R_2 resistor is found by using current divider rule:

$$I_{R_2} + I_{R_3+R_4} = I$$

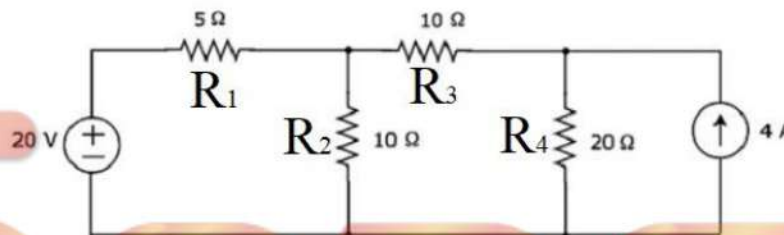
$$I_{R_2} = I - I_{R_3+R_4}$$

$$R_2 I_{R_2} = (R_3 + R_4) I_{R_3+R_4}$$

$$R_2 (I - I_{R_3+R_4}) = (R_3 + R_4) I_{R_3+R_4}$$

$$I_{R_3} = \frac{R_2 I}{R_2 + R_3 + R_4} = \frac{10\ \Omega * 1.6\text{ A}}{10\ \Omega + 10\ \Omega + 20\ \Omega} = 0.4\text{ A}$$

77. The current (in the branch of $20\ \Omega$ resistor/ in the branch of R_4) due to voltage source only is: (superposition)



a. 0.4 A

b. $\frac{40}{39}\text{ A}$

c. 1.6 A

d. $\frac{40}{13}\text{ A}$

e. 1.2 A

f. $\frac{80}{39}\text{ A}$

g. $\frac{20}{13}\text{ A}$

Correct answer: 0.4 A

By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors R_3 & R_4 are in series, and they are in parallel with R_2 , and they are in series with R_1 :

$$R_{series} = R_3 + R_4 = 10\ \Omega + 20\ \Omega = 30\ \Omega$$

The parallel resistance for R_{series} & R_2 is:

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$$R_{parallel} = \frac{R_2 * R_{series}}{R_2 + R_{series}} = \frac{10 \Omega * 30 \Omega}{10 \Omega + 30 \Omega} = 7.5 \Omega$$

The equivalent resistance is:

$$R_{eq} = R_{parallel} + R_1 = 7.5 \Omega + 5 \Omega = 12.5 \Omega$$

The current due to the voltage source in the branch of voltage source:

$$I = \frac{V_s}{R_{eq}} = \frac{20 V}{12.5 \Omega} = 1.6 A$$

Now, the current in the branch of R_2 resistor is found by using current divider rule:

$$I_{R_2} + I_{R_3+R_4} = I$$

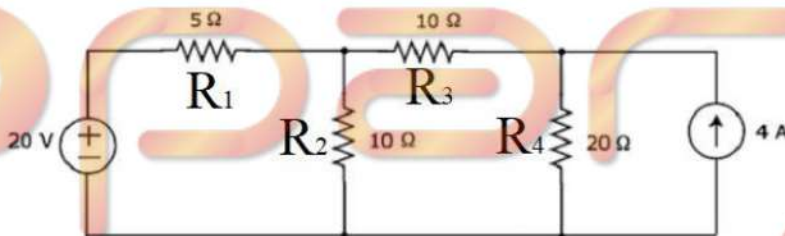
$$I_{R_2} = I - I_{R_3+R_4}$$

$$R_2 I_{R_2} = (R_3 + R_4) I_{R_3+R_4}$$

$$R_2 (I - I_{R_3+R_4}) = (R_3 + R_4) I_{R_3+R_4}$$

$$I_{R_2} = \frac{R_2 I}{R_2 + R_3 + R_4} = \frac{10 \Omega * 1.6 A}{10 \Omega + 10 \Omega + 20 \Omega} = 0.4 A$$

78. The current (in the branch of voltage source/ in the branch of 5Ω resistor/ in the branch of R_1) due to current source only is: (superposition)



a. $1.2 A$

b. $\frac{20}{13} A$

c. $1.6 A$

d. $\frac{40}{39} A$

e. $\frac{80}{39} A$

f. $0.4 A$

g. $\frac{40}{13} A$

Correct answer: $\frac{80}{39} A$

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By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors R_1 & R_2 are in parallel, and they are in series with R_3 , and they are in parallel with R_4 :

$$R_{parallel} = R_1 + R_2 = 5 \Omega + 10 \Omega = 15 \Omega$$

The series resistance for $R_{parallel}$ & R_3 is:

$$R_{series} = \frac{R_3 * R_{parallel}}{R_3 + R_{parallel}} = \frac{10 \Omega * 15 \Omega}{10 \Omega + 15 \Omega} = 6 \Omega$$

The equivalent resistance is:

$$R_{eq} = \frac{R_{series} * R_4}{R_{series} + R_4} = \frac{6 \Omega * 20 \Omega}{6 \Omega + 20 \Omega} = \frac{60}{13} \Omega$$

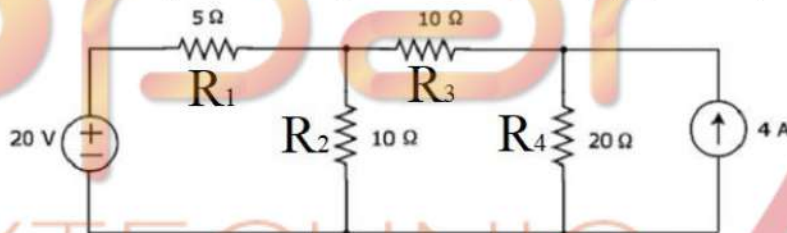
The current due to the current source in the branch of $(R_1 // R_2 + R_3) = R_{series}$ is found by using the current divider rule:

$$I_{R_{series}} = \frac{R_4 I}{R_{series} + R_4} = \frac{20 \Omega * 4 A}{6 \Omega + 20 \Omega} = \frac{40}{13} A$$

The current due to the current source in the branch of voltage source is found by using the current divider rule:

$$I_{R_1} = I_{V_S} = \frac{R_2 I_{R_{series}}}{R_1 + R_2} = \frac{10 \Omega * \frac{40}{13} A}{5 \Omega + 10 \Omega} = \frac{80}{39} A$$

79. The current in the branch of R_2 due to voltage source only is: (superposition)



- a. 0.4 A
- b. $\frac{40}{13} A$
- c. $\frac{20}{13} A$
- d. 1.6 A
- e. $\frac{40}{39} A$
- f. $\frac{80}{39} A$
- g. 1.2 A

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Correct answer: $\frac{40}{39} A$

By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors R_1 & R_2 are in parallel, and they are in series with R_3 , and they are in parallel with R_4 :

$$R_{parallel} = R_1 + R_2 = 5 \Omega + 10 \Omega = 15 \Omega$$

The series resistance for $R_{parallel}$ & R_3 is:

$$R_{series} = \frac{R_3 * R_{parallel}}{R_3 + R_{parallel}} = \frac{10 \Omega * 15 \Omega}{10 \Omega + 15 \Omega} = 6 \Omega$$

The equivalent resistance is:

$$R_{eq} = \frac{R_{series} * R_4}{R_{series} + R_4} = \frac{6 \Omega * 20 \Omega}{6 \Omega + 20 \Omega} = \frac{60}{13} \Omega$$

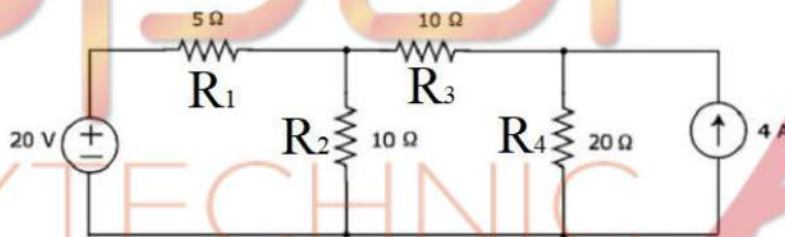
The current due to the current source in the branch of $(R_1 // R_2 + R_3) = R_{series}$ is found by using the current divider rule:

$$I_{R_{series}} = \frac{R_4 I}{R_{series} + R_4} = \frac{20 \Omega * 4 A}{6 \Omega + 20 \Omega} = \frac{40}{13} A$$

The current due to the current source in the branch of R_2 is found by using the current divider rule:

$$I_{R_2} = \frac{R_1 I_{R_{series}}}{R_1 + R_2} = \frac{5 \Omega * \frac{40}{13} A}{5 \Omega + 10 \Omega} = \frac{40}{39} A$$

80. The current in the branch of R_3 due to voltage source only is: (superposition)



- a. $0.4 A$
- b. $1.2 A$
- c. $\frac{40}{39} A$
- d. $1.6 A$
- e. $\frac{20}{13} A$
- f. $\frac{40}{13} A$
- g. $\frac{80}{39} A$

Correct answer: $\frac{40}{13} A$

By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors R_1 & R_2 are in parallel, and they are in series with R_3 , and they are in parallel with R_4 :

$$R_{parallel} = R_1 + R_2 = 5 \Omega + 10 \Omega = 15 \Omega$$

The series resistance for $R_{parallel}$ & R_3 is:

$$R_{series} = \frac{R_3 * R_{parallel}}{R_3 + R_{parallel}} = \frac{10 \Omega * 15 \Omega}{10 \Omega + 15 \Omega} = 6 \Omega$$

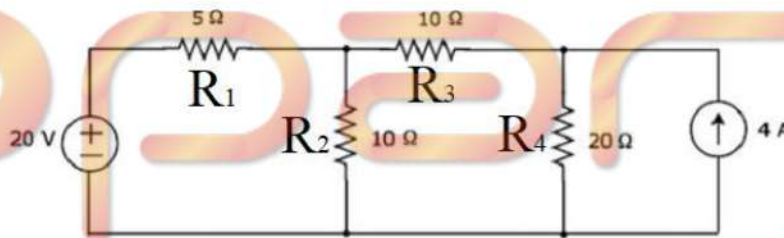
The equivalent resistance is:

$$R_{eq} = \frac{R_{series} * R_4}{R_{series} + R_4} = \frac{6 \Omega * 20 \Omega}{6 \Omega + 20 \Omega} = \frac{60}{13} \Omega$$

The current due to the current source in the branch of $(R_1 // R_2 + R_3) = R_{series}$ (which is the branch of R_3) is found by using the current divider rule:

$$I_{R_{series}} = I_{R_3} = \frac{R_4 I}{R_{series} + R_4} = \frac{20 \Omega * 4 A}{6 \Omega + 20 \Omega} = \frac{40}{13} A$$

81. The current (in the branch of 20Ω resistor/ in the branch of R_4) due to current source only is: (superposition)



- a. $\frac{40}{39} A$
- b. $0.4 A$
- c. $\frac{40}{13} A$
- d. $\frac{80}{39} A$
- e. $1.2 A$
- f. $1.6 A$
- g. $\frac{20}{13} A$

Correct answer: $\frac{20}{13} A$

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By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors R_1 & R_2 are in parallel, and they are in series with R_3 , and they are in parallel with R_4 :

$$R_{parallel} = R_1 + R_2 = 5 \Omega + 10 \Omega = 15 \Omega$$

The series resistance for $R_{parallel}$ & R_3 is:

$$R_{series} = \frac{R_3 * R_{parallel}}{R_3 + R_{parallel}} = \frac{10 \Omega * 15 \Omega}{10 \Omega + 15 \Omega} = 6 \Omega$$

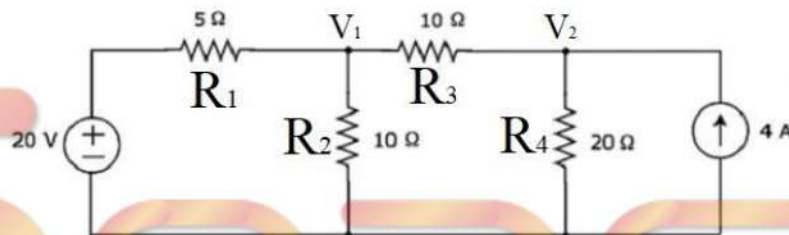
The equivalent resistance is:

$$R_{eq} = \frac{R_{series} * R_4}{R_{series} + R_4} = \frac{6 \Omega * 20 \Omega}{6 \Omega + 20 \Omega} = \frac{60}{13} \Omega$$

The current due to the current source in the branch of R_4 is found by using the current divider rule:

$$I_{R_4} = \frac{R_3 I}{R_{series} + R_4} = \frac{10 \Omega * 4 A}{6 \Omega + 20 \Omega} = \frac{20}{13} A$$

82. The voltage V_1 due to voltage source only is: (superposition)



- a. 8 V
- b. 12 V
- c. 32 V
- d. 40 V
- e. 20 V
- f. 32 V
- g. None of these

Correct answer: 12 V

By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors R_3 & R_4 are in series, and they are in parallel with R_2 , and they are in series with R_1 :

$$R_{series} = R_3 + R_4 = 10 \Omega + 20 \Omega = 30 \Omega$$

The parallel resistance for R_{series} & R_2 is:

$$R_{parallel} = \frac{R_2 * R_{series}}{R_2 + R_{series}} = \frac{10 \Omega * 30 \Omega}{10 \Omega + 30 \Omega} = 7.5 \Omega$$

The equivalent resistance is:

$$R_{eq} = R_{parallel} + R_1 = 7.5 \Omega + 5 \Omega = 12.5 \Omega$$

The current due to the voltage source in the branch of voltage source:

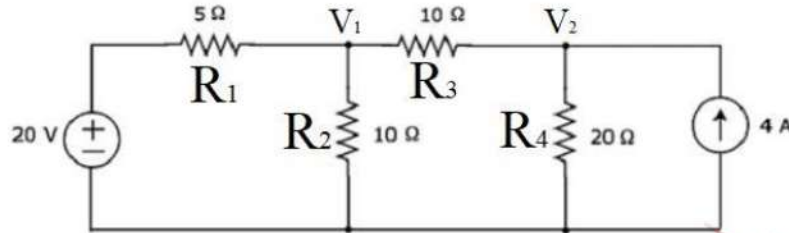
ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

$$I = \frac{V_S}{R_{eq}} = \frac{20\text{ V}}{12.5\ \Omega} = 1.6\text{ A}$$

Now, the voltage V_1 is found by KVL:

$$V_1 = V_S - IR_1 = 20\text{ V} - 1.6\text{ A} * 5\ \Omega = 12\text{ V}$$

83. The voltage V_2 due to voltage source only is: (superposition)



- a. 20 V
- b. 32 V
- c. 32 V
- d. 8 V
- e. 40 V
- f. 12 V
- g. None of these

Correct answer: 8 V

By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors R_3 & R_4 are in series, and they are in parallel with R_2 , and they are in series with R_1 :

$$R_{series} = R_3 + R_4 = 10\ \Omega + 20\ \Omega = 30\ \Omega$$

The parallel resistance for R_{series} & R_2 is:

$$R_{parallel} = \frac{R_2 * R_{series}}{R_2 + R_{series}} = \frac{10\ \Omega * 30\ \Omega}{10\ \Omega + 30\ \Omega} = 7.5\ \Omega$$

The equivalent resistance is:

$$R_{eq} = R_{parallel} + R_1 = 7.5\ \Omega + 5\ \Omega = 12.5\ \Omega$$

The current due to the voltage source in the branch of voltage source:

$$I = \frac{V_S}{R_{eq}} = \frac{20\text{ V}}{12.5\ \Omega} = 1.6\text{ A}$$

Now, the current in the branch of R_2 resistor is found by using current divider rule:

$$I_{R_2} + I_{R_3+R_4} = I$$

$$I_{R_2} = I - I_{R_3+R_4}$$

$$R_2 I_{R_2} = (R_3 + R_4) I_{R_3+R_4}$$

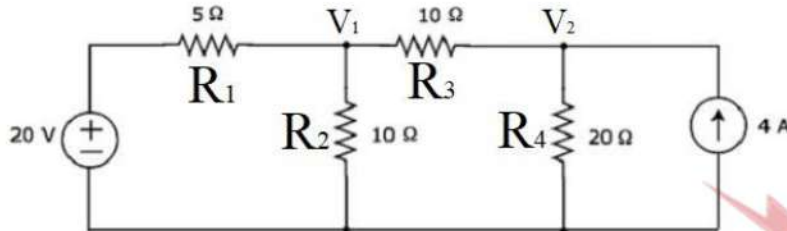
$$R_2 (I - I_{R_3+R_4}) = (R_3 + R_4) I_{R_3+R_4}$$

$$I_{R_3} = \frac{R_2 I}{R_2 + R_3 + R_4} = \frac{10 \Omega * 1.6 A}{10 \Omega + 10 \Omega + 20 \Omega} = 0.4 A$$

Now, the voltage V_2 is found by KVL:

$$V_2 = V_S - (I_{R_1} + I_{R_3} R_3) = 20 V - (1.6 A * 5 \Omega + 0.4 A * 10 \Omega) = 8 V$$

84. The voltage V_1 due to current source only is: (superposition)



- a. 20 V
- b. 32 V
- c. 32 V
- d. 8 V
- e. 12 V
- f. 40 V
- g. None of these

Correct answer: 8 V

By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors R_1 & R_2 are in parallel, and they are in series with R_3 , and they are in parallel with R_4 :

$$R_{parallel} = R_1 + R_2 = \frac{R_1 * R_2}{R_1 + R_2} = \frac{5 \Omega * 10 \Omega}{5 \Omega + 10 \Omega} = \frac{10}{3} \Omega$$

The series resistance for $R_{parallel}$ & R_3 is:

$$R_{series} = R_3 + R_{parallel} = 10 \Omega + \frac{10}{3} \Omega = \frac{40}{3} \Omega$$

The equivalent resistance is:

$$R_{eq} = \frac{R_{series} * R_4}{R_{series} + R_4} = \frac{\frac{40}{3} \Omega * 20 \Omega}{\frac{40}{3} \Omega + 20 \Omega} = 8 \Omega$$

Then, the voltage V_2 is found by using ohm's law:

$$V_2 = I_S R_{eq} = 4 A * 8 \Omega = 32 V$$

The current due to the current source in the branch of R_3 source is found by using current divider rule:

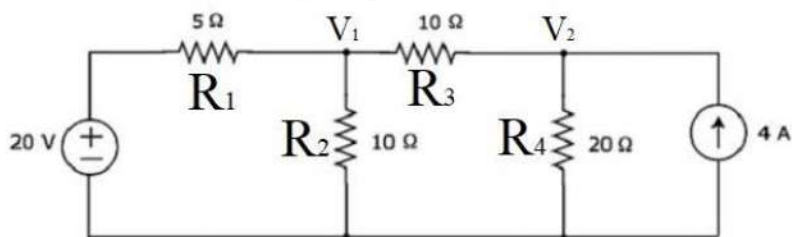
$$I_{R_3} = \frac{R_4 I}{R_{series} + R_4} = \frac{20 \Omega * 4 A}{\frac{40}{3} \Omega + 20 \Omega} = 2.4 A$$

Now, the voltage V_1 is found by KVL:

$$V_1 = V_2 - I_{R_3} R_3 = 32 V - 2.4 A * 10 \Omega = 8 V$$

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

85. The voltage V_2 due to current source only is: (superposition)



- a. 40 V
- b. 32 V
- c. 8 V
- d. 32 V
- e. 12 V
- f. 20 V
- g. None of these

Correct answer: 32 V

By using superposition theorem, current sources become open circuit, and the voltage sources become short circuit except the source that the required response will take from it, so the resistors R_1 & R_2 are in parallel, and they are in series with R_3 , and they are in parallel with R_4 :

$$R_{parallel} = R_1 + R_2 = \frac{R_1 * R_2}{R_1 + R_2} = \frac{5 \Omega * 10 \Omega}{5 \Omega + 10 \Omega} = \frac{10}{3} \Omega$$

The series resistance for $R_{parallel}$ & R_3 is:

$$R_{series} = R_3 + R_{parallel} = 10 \Omega + \frac{10}{3} \Omega = \frac{40}{3} \Omega$$

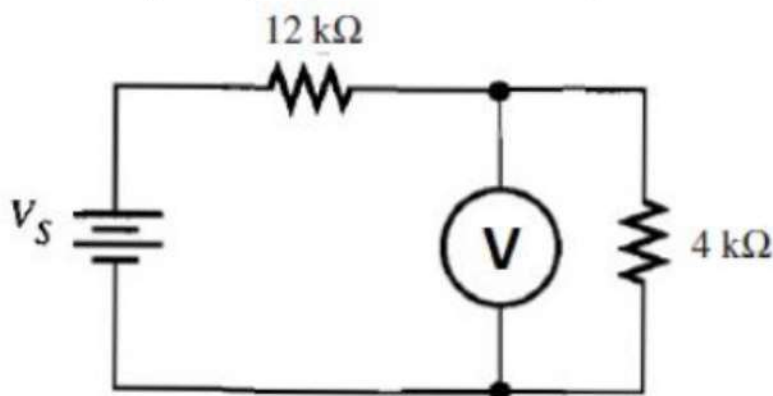
The equivalent resistance is:

$$R_{eq} = \frac{R_{series} * R_4}{R_{series} + R_4} = \frac{\frac{40}{3} \Omega * 20 \Omega}{\frac{40}{3} \Omega + 20 \Omega} = 8 \Omega$$

Then, the voltage V_2 is found by using ohm's law:

$$V_2 = I_S R_{eq} = 4 A * 8 \Omega = 32 V$$

86. For the circuit shown when $V_S = 6 V$, then the voltmeter reading is:

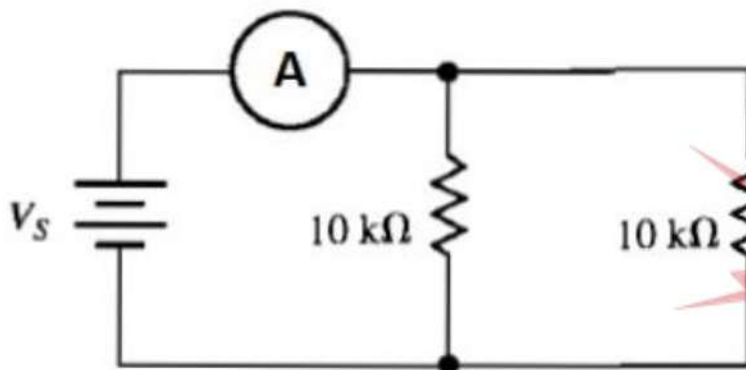


ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

- a. 2.5 V
- b. 3 V
- c. 1.5 V
- d. 2 V

Correct answer: 1.5 V

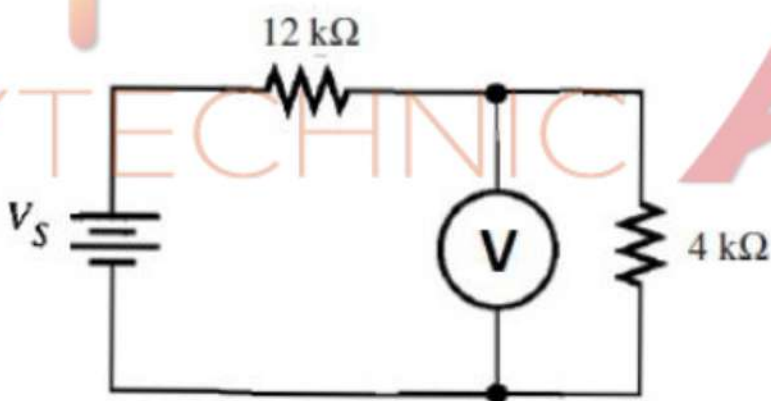
87. For the circuit shown when $V_S = 10\text{ V}$, then the ammeter reading is:



- a. 2 mA
- b. 1.6 mA
- c. 2.4 mA
- d. 1.2 mA

Correct answer: 2 mA

88. For the circuit shown when $V = 6\text{ V}$, then the percentage of error when the voltmeter reading is 1.6 V equal to:

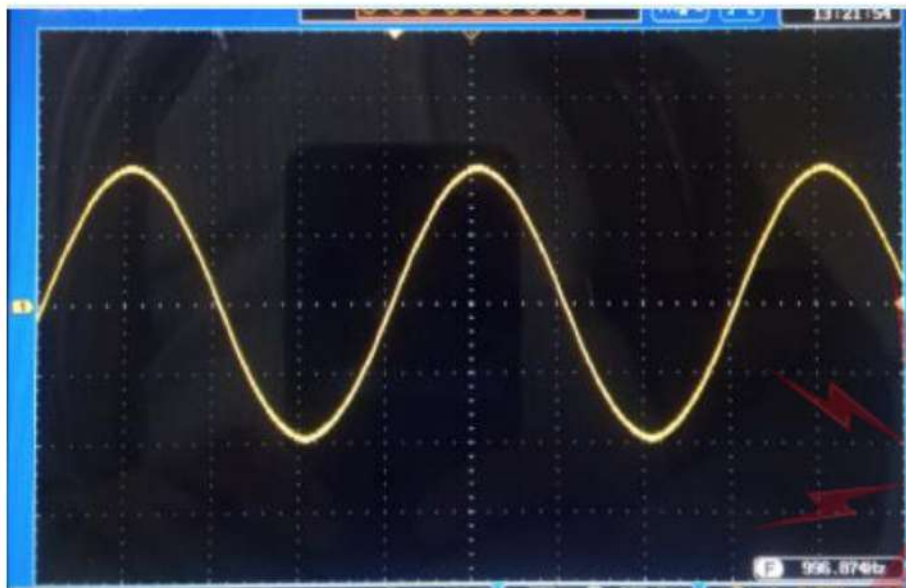


- a. 8%
- b. 6.7%
- c. 12%
- d. 10%

Correct answer: 6.7%

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

89. For the signal shown below, when the scale of VOLT/DIV = 2.0, then the peak (V) of the signal shown is:



- a. 2 V
- b. 4 V
- c. 8 V
- d. 1 V

Correct answer: 4 V

90. For the XY-mode figure shown below, the phase shift between the two signals is approximately:



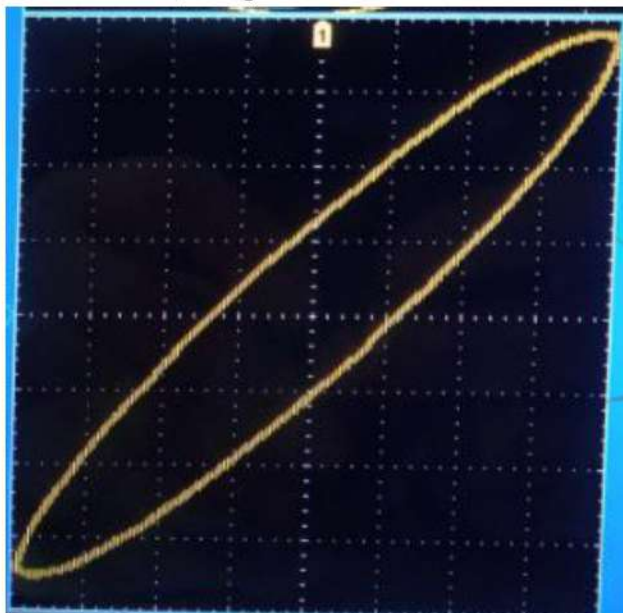
- a. 64°
- b. 38°

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

- c. 52°
- d. 31°

Correct answer: 38°

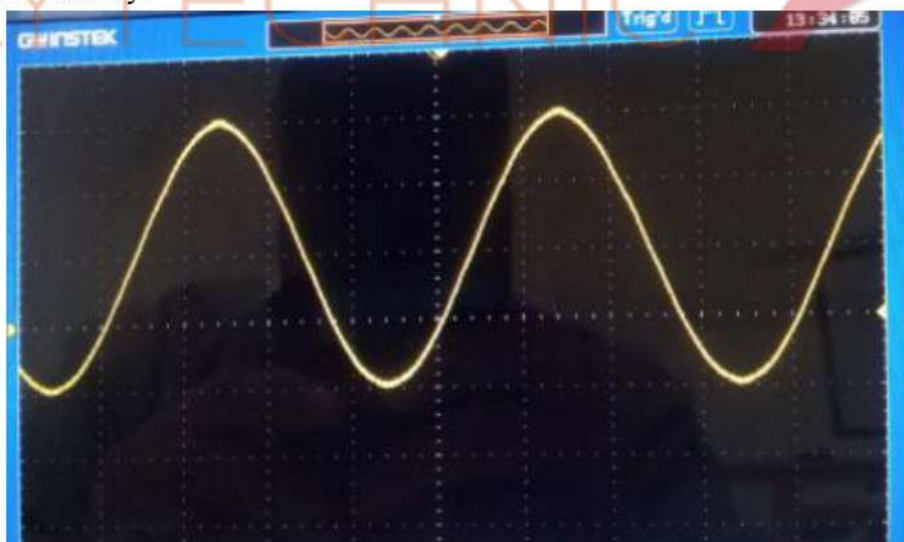
91. For the XY-mode figure shown below, the phase shift between the two signals is approximately:



- a. 64°
- b. 35°
- c. 17°
- d. 72°

Correct answer: 17°

92. For the signal shown below, when the scale of TIME/DIV = 0.5 ms, then the frequency of the signal shown is approximately:



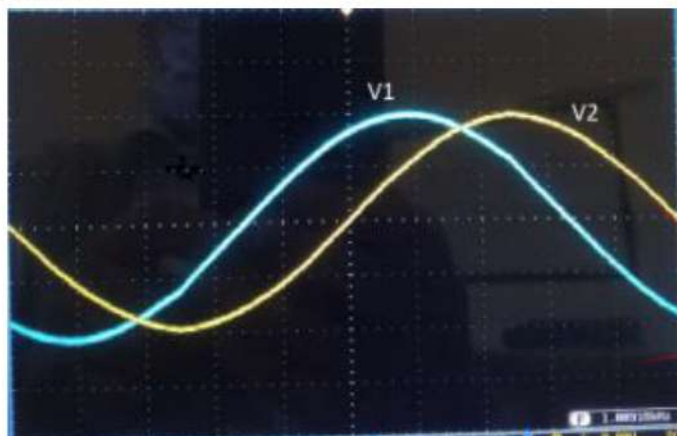
- a. 0.74 kHz

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

- b. 0.5 kHz
- c. 1 kHz
- d. 0.37 kHz

Correct answer: 0.5 kHz

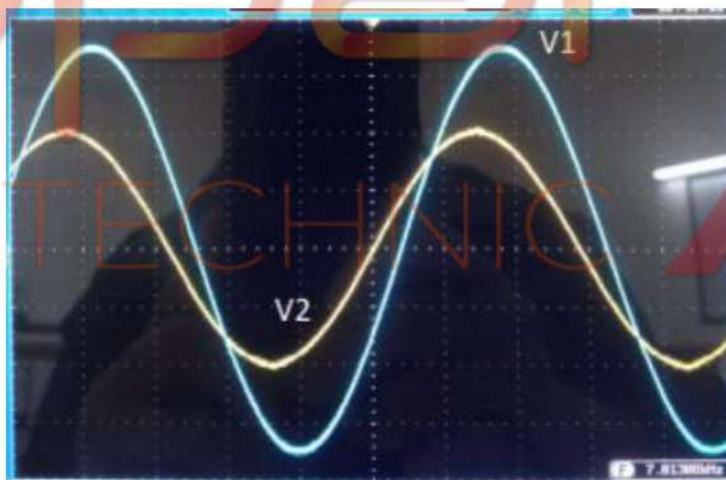
93. For the signals shown below:



- a. V_1 leads V_2
- b. V_1 and V_2 are in phase
- c. V_2 leads V_1

Correct answer: V_1 leads V_2

94. For the signals shown below:

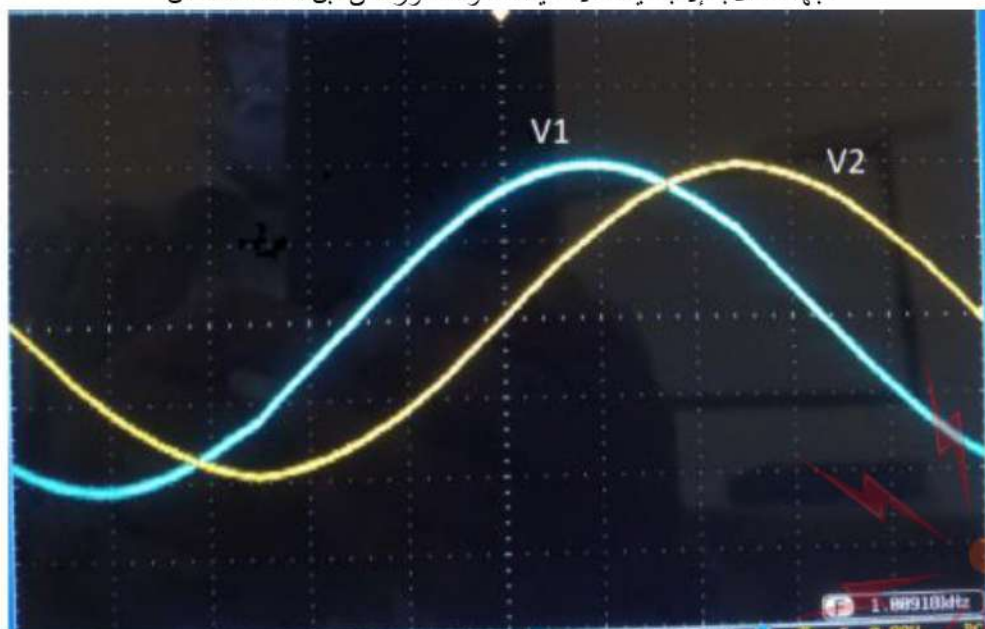


- a. V_2 leads V_1
- b. V_1 and V_2 are in phase
- c. V_1 leads V_2

Correct answer: V_2 leads V_1

95. For the signals shown below, the phase shift between the two signals is approximately:

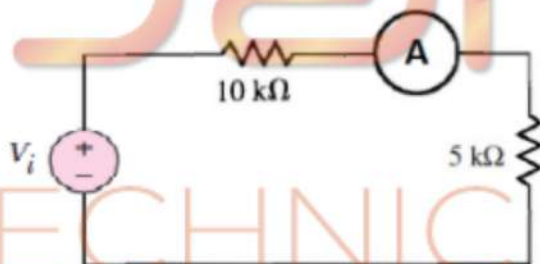
ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



- a. 55°
- b. 64°
- c. 120°
- d. 72°

Correct answer: 55°

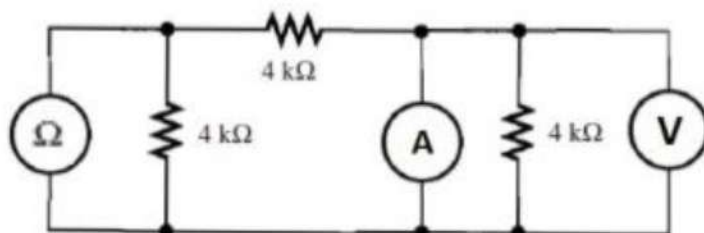
96. For the circuit shown, when the F.G has $V_{i\text{p-p}} = 18\text{ V}$, then the ammeter reading is:



- a. 0.424 mA
- b. 1.2 mA
- c. 0.6 mA
- d. 0.848 mA

Correct answer: 0.424 mA

97. For the circuit shown the ohmmeter reading is:

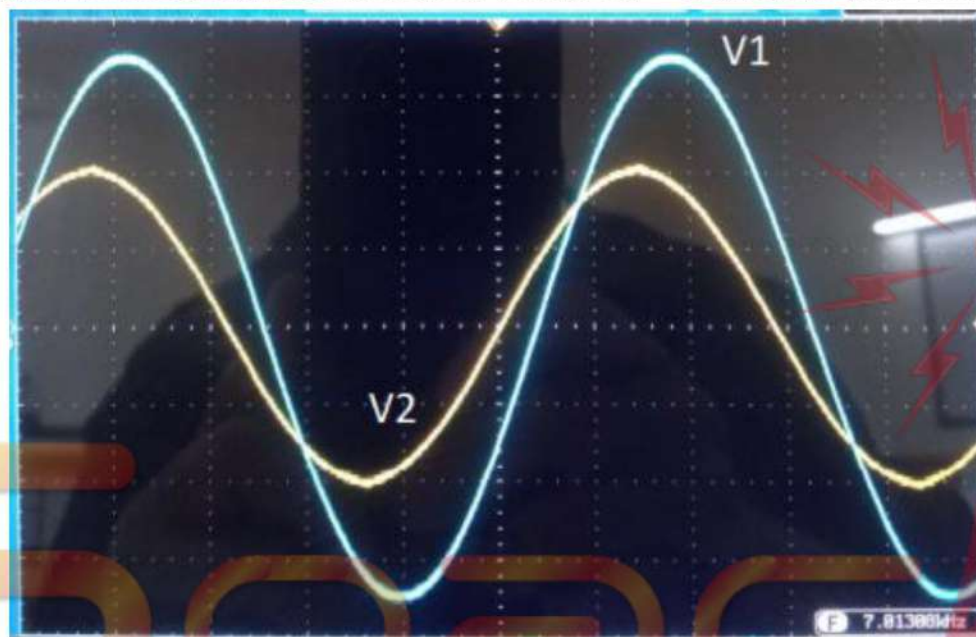


ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

- a. $12\text{ K}\Omega$
- b. $4\text{ K}\Omega$
- c. $2\text{ K}\Omega$
- d. $2.6667\text{ K}\Omega$

Correct answer: $2\text{ K}\Omega$

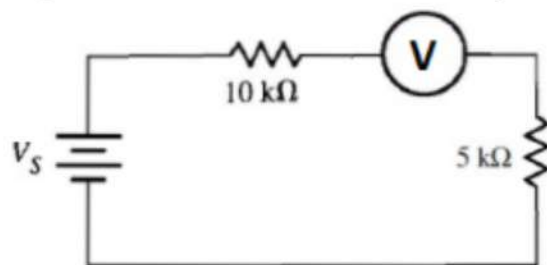
98. For the signals shown below, when the scale of $\text{VOLT/DIV} = 0.5$, then the peak voltage of V_i (V_p) is:



- a. 2 V
- b. 3.5 V
- c. 1 V
- d. 1.75 V

Correct answer: 1.75 V

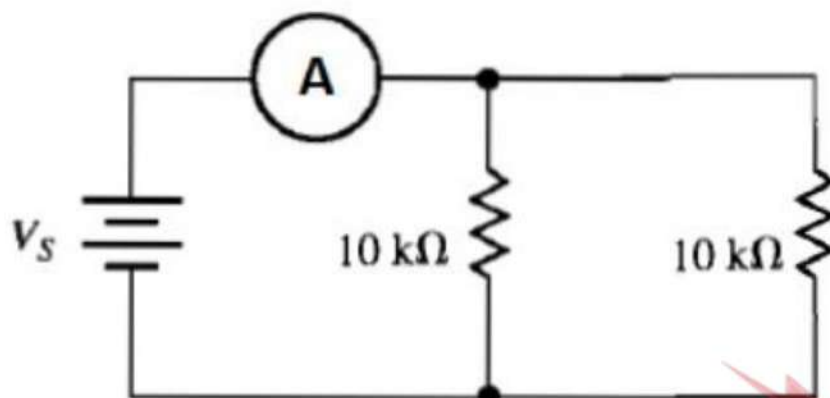
99. For the circuit shown when $V_S = 12\text{ V}$, then the voltmeter reading is:



- a. 8 V
- b. 12 V
- c. 4 V
- d. 0 V

Correct answer: 12 V

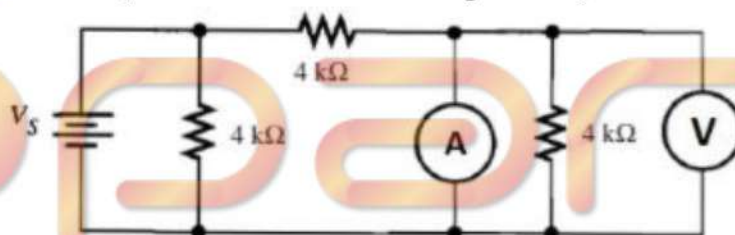
100. For the circuit shown, when the ammeter reading is 2mA, then voltage value of the source is:



- a. 5 V
- b. 20 V
- c. 2.5 V
- d. 10 V

Correct answer: 10 V

101. For the circuit shown, when the ammeter reading is 2mA, then the voltmeter reading is:

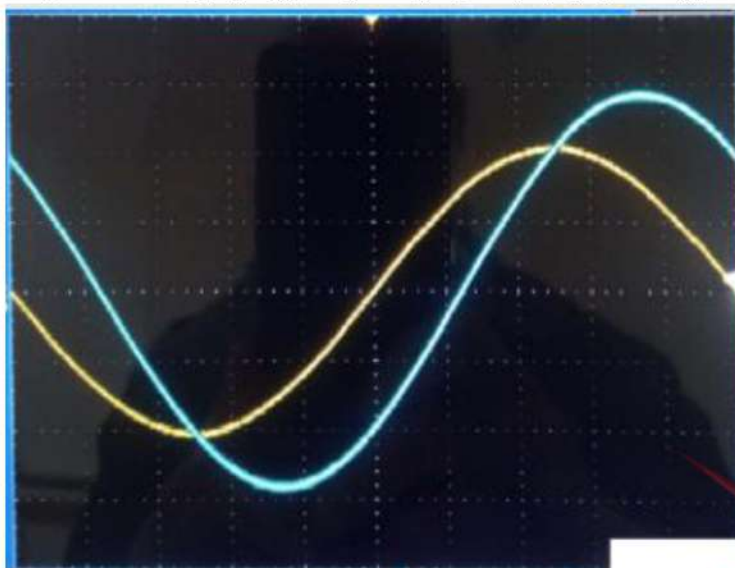


- a. 4 V
- b. 8 V
- c. 0 V
- d. 2 V

Correct answer: 0 V

102. For the signals shown below, when the scale of TIME/DIV = 0.5 m, then the frequency of the signals shown is approximately:

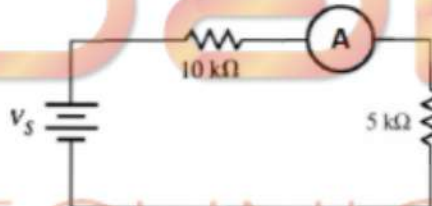
ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



- a. 0.37 kHz
- b. 0.5 kHz
- c. 0.2 kHz
- d. 0.74 kHz

Correct answer: 0.2 kHz

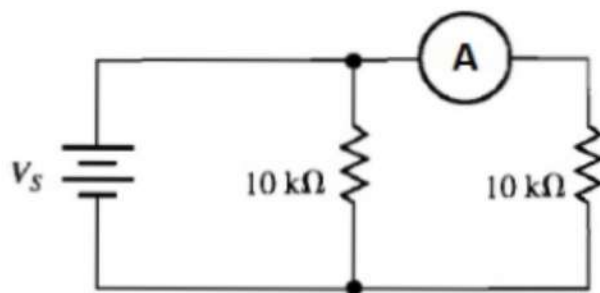
103. Consider the circuit shown where $V_S = 24\text{ V}$. When the ammeter reading is 1.52 mA then the percentage of error is:



- a. 12.5%
- b. 10%
- c. 5%

Correct answer: 5%

104. For the circuit shown when $V_S = 6\text{ V}$, then the ammeter reading is:



- a. 0.3 mA

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

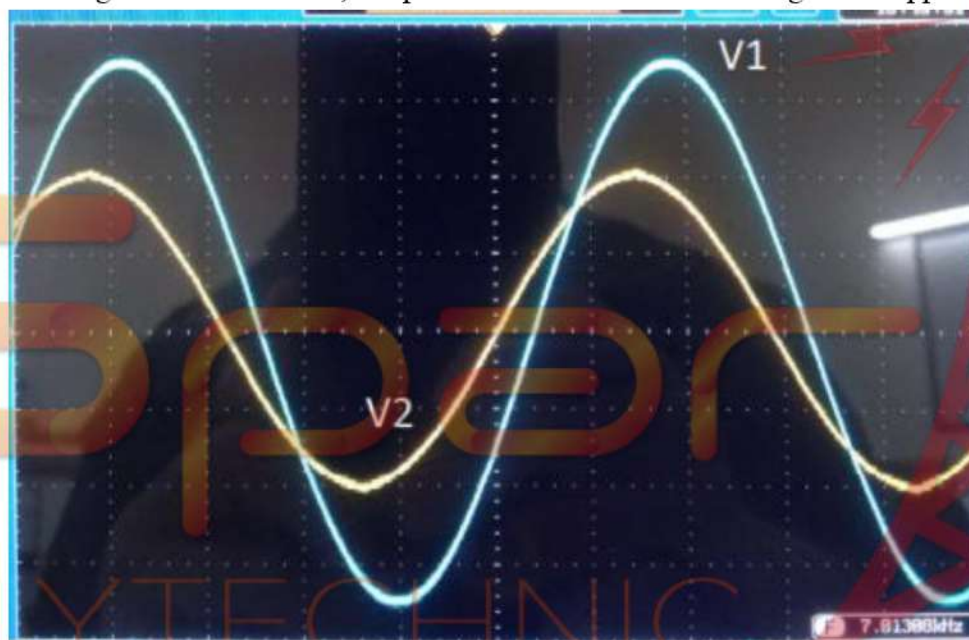
- b. 0.9 mA
- c. 0.6 mA
- d. 1.2 mA

Correct answer: 0.6 mA

105. To display the AC component only for a signal, the coupling setting that we can use:
- a. is GND
 - b. is DC coupling
 - c. is AC coupling
 - d. both AC and DC couplings

Correct answer: is AC coupling

106. For the signals shown below, the phase shift between the two signals is approximately:

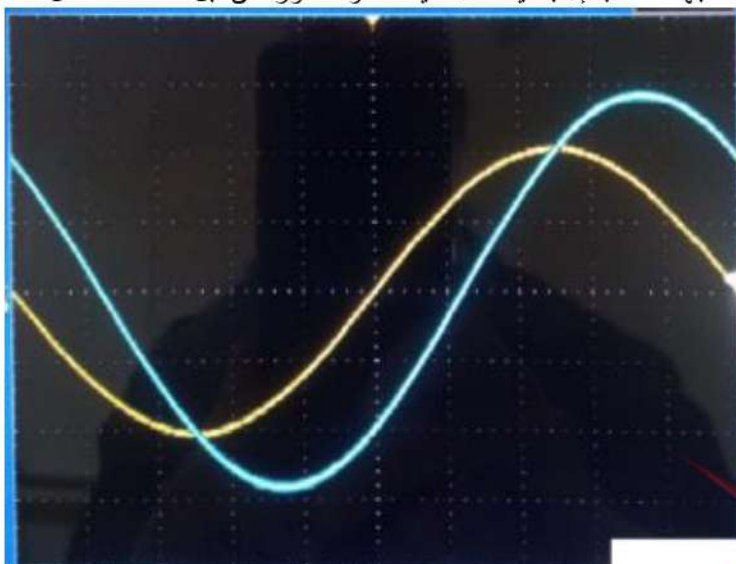


- a. 52°
- b. 122°
- c. 25°
- d. 14°

Correct answer: 25°

107. If the frequency of the signals shown is 2 kHz , then the scale of $\text{TIME/DIV} =$

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



- a. 10μ
- b. 20μ
- c. 50μ
- d. $0.1m$

Correct answer: 50μ

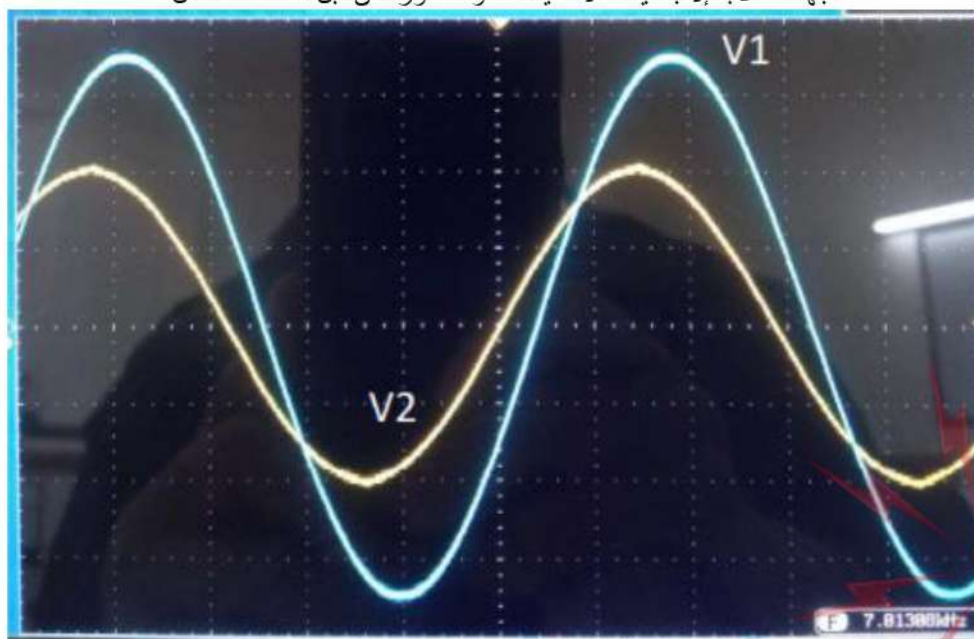
108. If a practical voltmeter (with a large internal resistance) is used to measure the voltage across a resistance, then the value of voltage measured is that when the voltmeter is ideal _____:

- a. larger than
- b. smaller than
- c. the same as

Correct answer: smaller than

109. If the peak voltage of V_1 , (V_{1p}) is 7 V, then the scale of VOLT/DIV =

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



- a. 1
- b. 2
- c. 5
- d. 0.5

Correct answer: 2

110. Consider a load of 8.0 KW worked 0.5 hours. For this load, if the number of rotations is 3000, then the wattmeter constant is:

- a. 1000 rot/kWh
- b. 1500 rot/kWh
- c. 750 rot/kWh
- d. 600 rot/kWh

Correct answer: 750 rot/kWh

111. Consider a load with 4 kW power worked 5 hours. If the wattmeter constant 200 rot/kWh, then the number of the disc rotation for this load is:

- a. 1000 rot
- b. 4000 rot
- c. 8000 rot
- d. 2000 rot

Correct answer: 4000 rot

112. Consider a load worked 4 hours. For this load, if the wattmeter constant is 500 rot/KWh and the number of rotation is 2500, then the load power is:

- a. 0.625 kW
- b. 1.25 kW

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

- c. 5.0 kW
- d. 2.5 kW

Correct answer: 1.25 kW

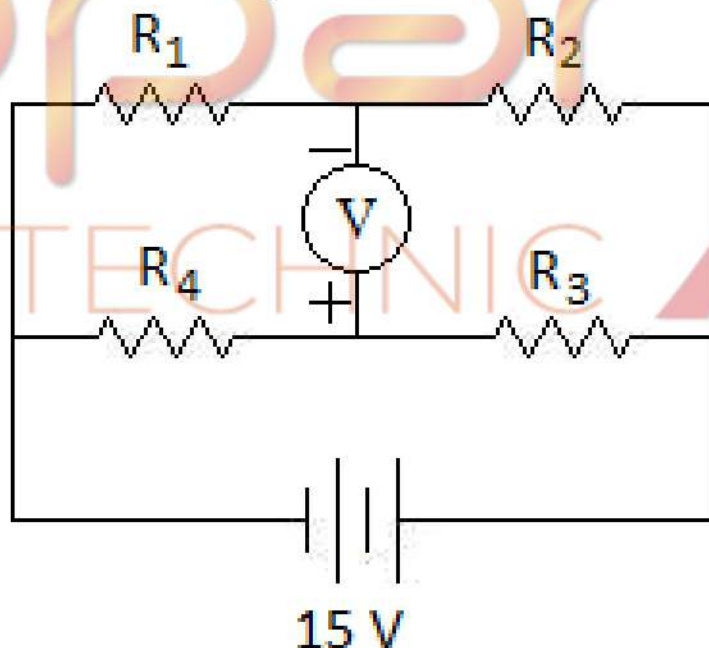
113. Consider a load with 1.6 KW power and 220 nominal voltage. If this load worked 5 hours, then the wattmeter reading is:
- a. 16 kWh
 - b. 4 kWh
 - c. 8 kWh
 - d. 50 kWh

Correct answer: 8 kWh

114. Consider a load worked 20 hours at a power factor 0.8 with 10 A current. For this load, if the wattmeter reading 32 KWh, then the load voltage is:
- a. 380 V
 - b. 220 V
 - c. 200 V
 - d. 150 V

Correct answer: 200 V

115. The reading of the voltmeter is equal to:



- a. $\left(\frac{R_4}{R_3+R_4} - \frac{R_1}{R_1+R_2} \right) V_S$
- b. $\left(\frac{R_1}{R_1+R_2} - \frac{R_4}{R_3+R_4} \right) V_S$

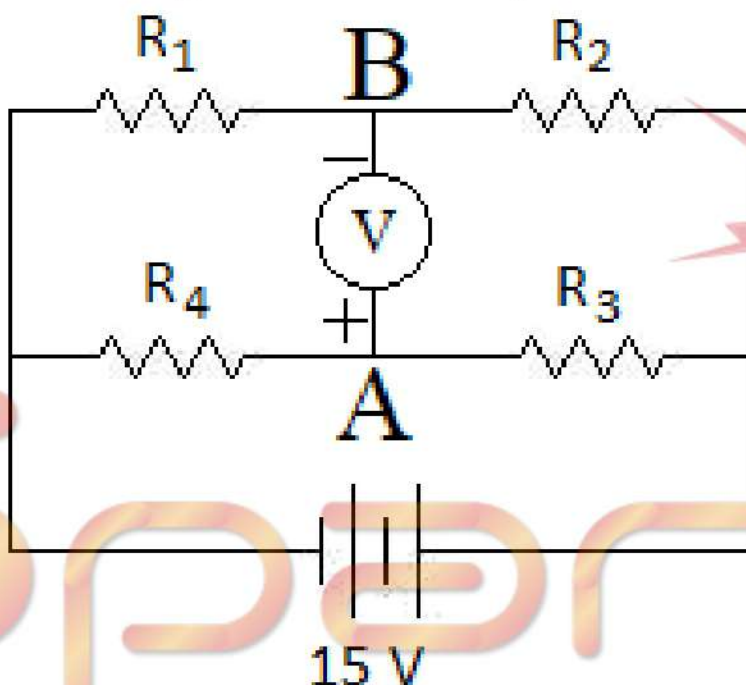
ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك، ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

c. $\left(\frac{R_3}{R_3+R_4} - \frac{R_2}{R_1+R_2} \right) V_S$

d. $\left(\frac{R_2}{R_1+R_2} - \frac{R_3}{R_3+R_4} \right) V_S$

Correct answer: $\left(\frac{R_4}{R_3+R_4} - \frac{R_1}{R_1+R_2} \right) V_S$

The reading of the voltmeter is the voltage difference between the points A & B:



The voltages at points A & B are found using voltage divider rule (VDR):

$$V_A = \frac{R_4}{R_3 + R_4} V_S$$

$$V_B = \frac{R_1}{R_1 + R_2} V_S$$

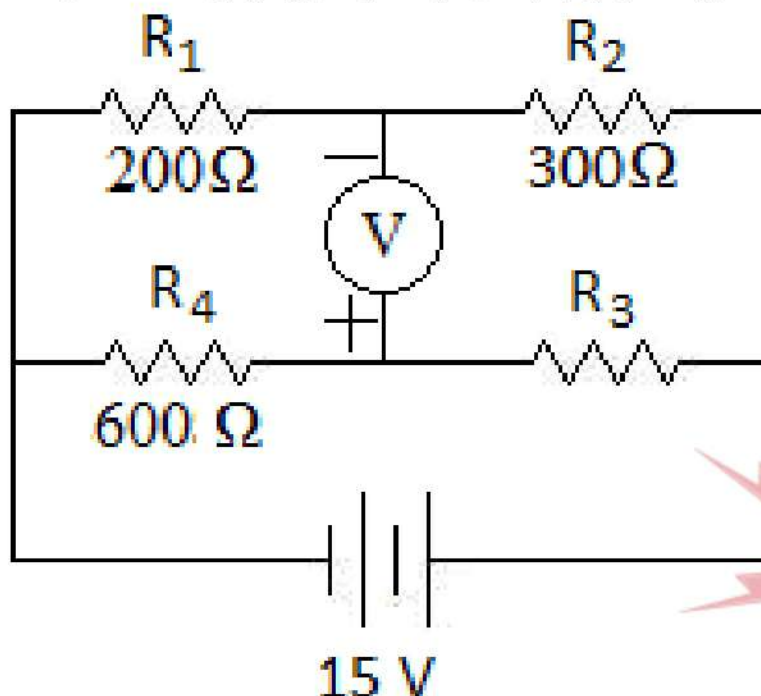
The voltage difference between the points A & B is:

$$\text{The reading of the voltmeter} = V_{AB} = V_A - V_B$$

$$V_{AB} = \frac{R_4}{R_3 + R_4} V_S - \frac{R_1}{R_1 + R_2} V_S = \left(\frac{R_4}{R_3 + R_4} - \frac{R_1}{R_1 + R_2} \right) V_S$$

116. Find R_3 if the reading of the voltmeter is equal to 3 V:

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



- a. 2400Ω
- b. 400Ω
- c. 900Ω
- d. 240Ω

Correct answer: 400Ω

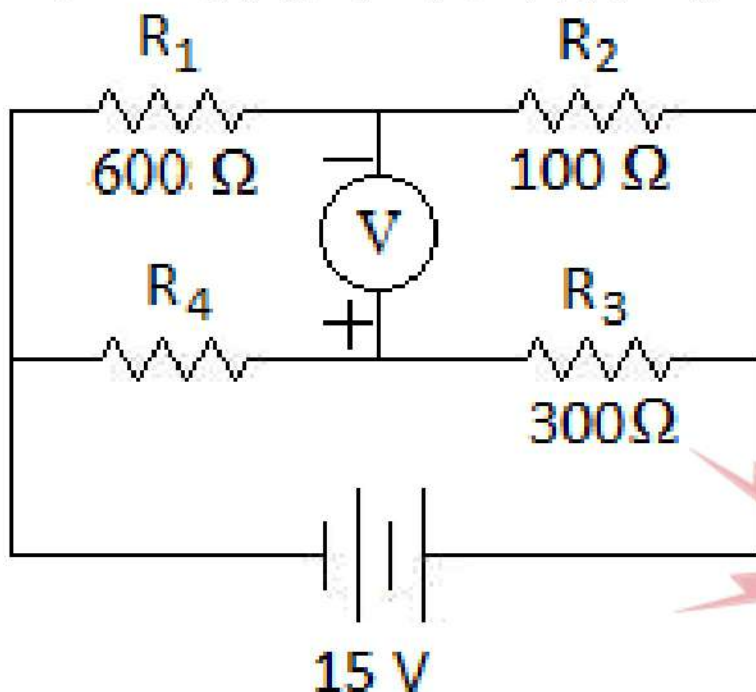
$$V_{AB} = \left(\frac{R_4}{R_3 + R_4} - \frac{R_1}{R_1 + R_2} \right) V_S$$

$$R_3 = R_4 \left(\frac{1}{\frac{V_{AB}}{V_S} + \frac{R_1}{R_1 + R_2}} - 1 \right)$$

$$R_3 = 600 \Omega \left(\frac{1}{\frac{3 V}{15 V} + \frac{200 \Omega}{200 \Omega + 300 \Omega}} - 1 \right) = 400 \Omega$$

117. The reading of the voltmeter is zero, find R_4 :

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



- a. 180Ω
- b. 1800Ω
- c. 50Ω
- d. None of these

Correct answer: 1800Ω

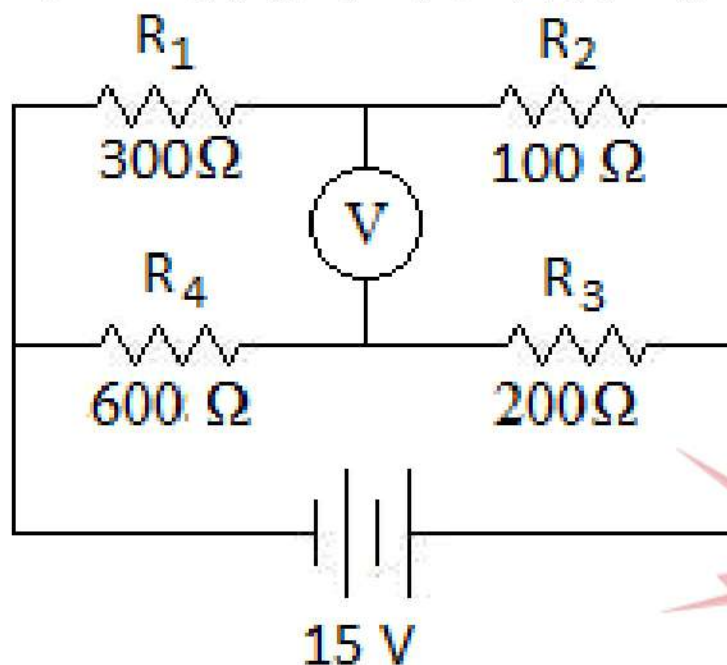
$$V_{AB} = \left(\frac{R_4}{R_3 + R_4} - \frac{R_1}{R_1 + R_2} \right) V_S$$

$$R_4 = \frac{R_3}{\left(\frac{1}{\frac{V_{AB}}{V_S} + \frac{R_1}{R_1 + R_2}} - 1 \right)} = \frac{300 \Omega}{\left(\frac{1}{\frac{0 V}{15 V} + \frac{600 \Omega}{600 \Omega + 100 \Omega}} - 1 \right)}$$

$$R_4 = 1800 \Omega$$

118. The reading of the voltmeter is equal to:

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



- a. 10 V
- b. 7.5 V
- c. 0 V
- d. None of these

Correct answer: 0 V

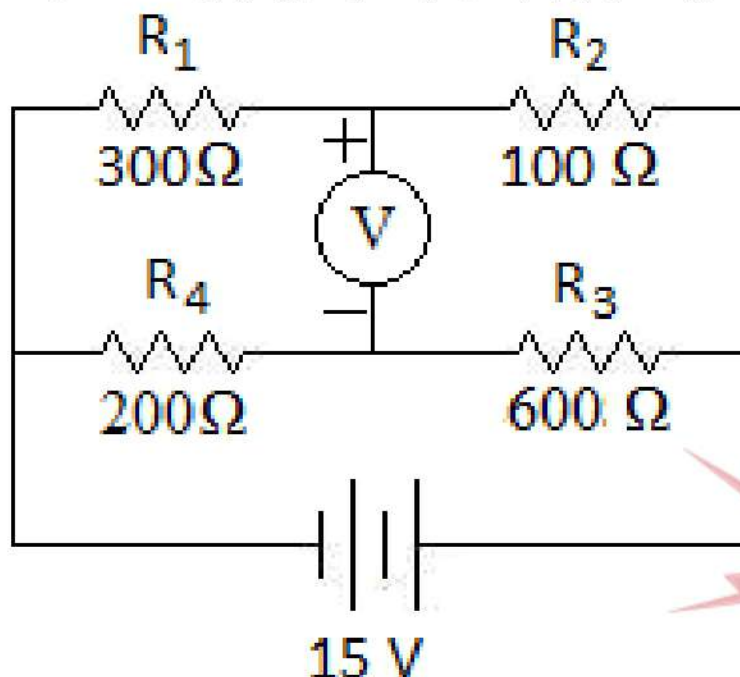
The reading of the voltmeter is the voltage difference between two points and note that the ratio between resistances on each side is the same:

$$\frac{100 \Omega}{200 \Omega} = \frac{300 \Omega}{600 \Omega} = \frac{1}{2}$$

Then, the drop voltage is the same on each branch, so there is no voltage difference between the two points which are connected with voltmeter

119. The reading of the voltmeter is equal to:

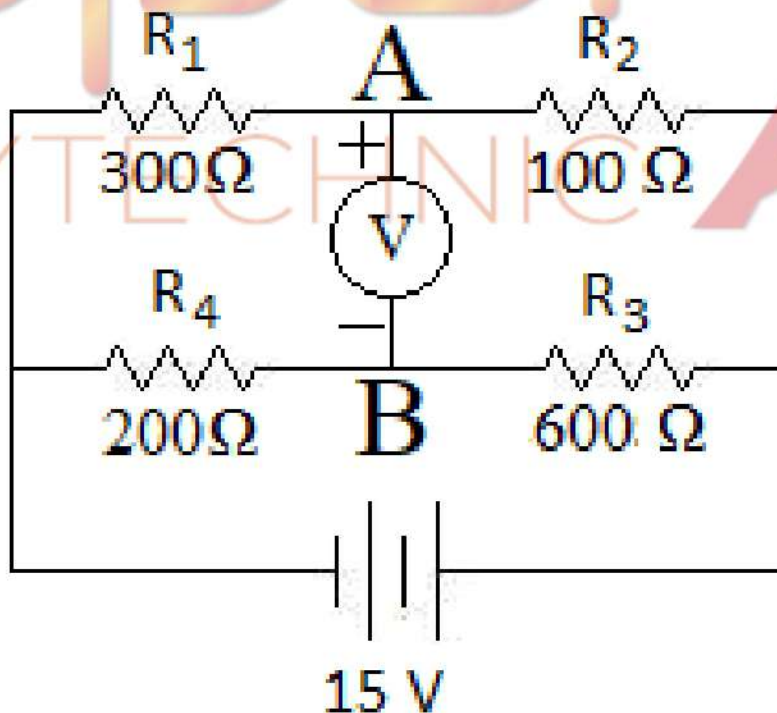
ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



- a. 7.5 V
- b. 10 V
- c. 0 V
- d. None of these

Correct answer: 7.5 V

The reading of the voltmeter is the voltage difference between the points A & B:



The voltages at points A & B are found using voltage divider rule (VDR):

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

$$V_A = \frac{R_1}{R_1 + R_2} V_S$$

$$V_B = \frac{R_4}{R_3 + R_4} V_S$$

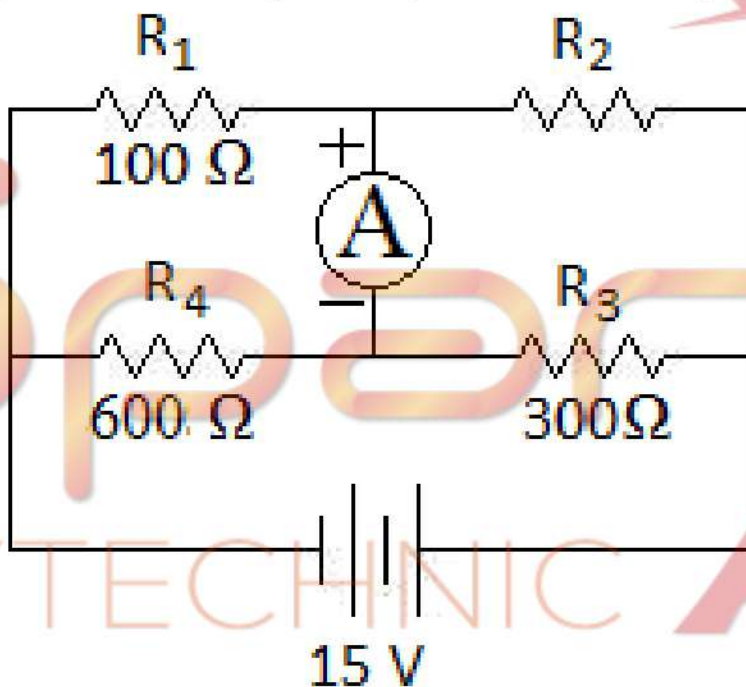
The voltage difference between the points A & B is:

The reading of the voltmeter = $V_{AB} = V_A - V_B$

$$V_{AB} = \frac{R_1}{R_1 + R_2} V_S - \frac{R_4}{R_3 + R_4} V_S = \left(\frac{R_1}{R_1 + R_2} - \frac{R_4}{R_3 + R_4} \right) V_S$$

$$V_{AB} = \left(\frac{300 \Omega}{300 \Omega + 100 \Omega} - \frac{200 \Omega}{600 \Omega + 200 \Omega} \right) * 15 V = 7.5 V$$

120. The reading of the ammeter is equal to zero, then the resistance R_2 is equal to:



Correct answer:

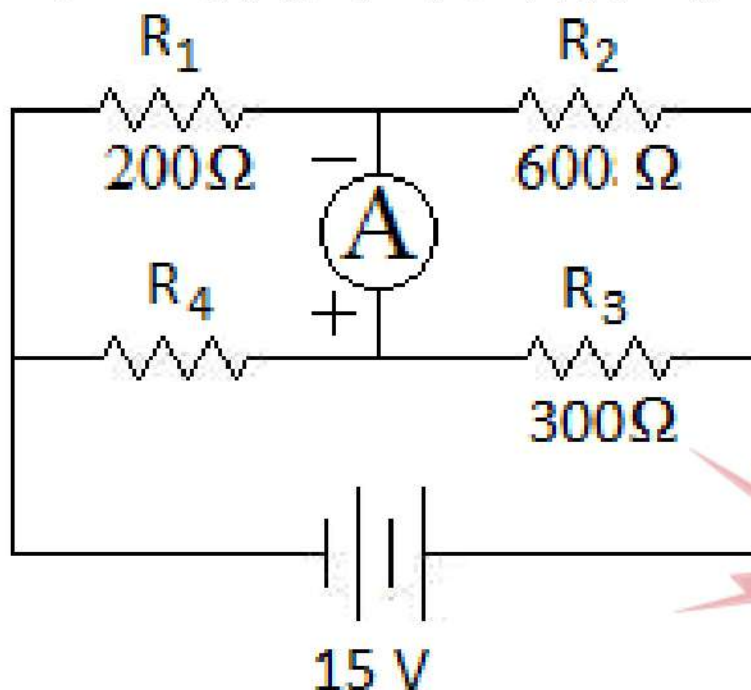
As the reading of the ammeter is equal to zero, then the voltage difference between the two terminals of the ammeter is zero and so the ratio between resistances on each side is the same:

$$\frac{100 \Omega}{600 \Omega} = \frac{R_2}{300 \Omega}$$

$$R_2 = \frac{100 \Omega}{600 \Omega} * 300 \Omega = 50 \Omega$$

121. The resistance $R_4 = 800 \Omega$, then the magnitude of the reading of the ammeter is equal to:

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



- a. 27.778 mA
- b. 8.333 mA
- c. 33.335 mA
- d. 19.445 mA
- e. 13.889 mA
- f. None of these

Correct answer: 19.445 mA

As the equivalent circuit of the ammeter is a short circuit, so the resistances R_1 & R_4 are connected in parallel and also, the resistances R_2 & R_3 are connected in parallel:

$$R_{eq} = R_1 // R_4 + R_2 // R_3$$

$$R_{eq} = \frac{R_1 R_4}{R_1 + R_4} + \frac{R_2 R_3}{R_2 + R_3} = \frac{200 \Omega * 800 \Omega}{200 \Omega + 800 \Omega} + \frac{600 \Omega * 300 \Omega}{600 \Omega + 300 \Omega} = 360 \Omega$$

$$I_T = \frac{V_S}{R_{eq}} = \frac{15 V}{360 \Omega} = \frac{1}{24} A$$

The voltage across the resistors R_2 & R_3 is calculated by using voltage divider rule (VDR):

$$V_{R_2} = V_{R_3} = \frac{R_2 // R_3}{R_2 // R_3 + R_1 // R_4} V_S$$

$$V_{R_2} = V_{R_3} = \frac{\frac{R_2 R_3}{R_2 + R_3}}{\frac{R_2 R_3}{R_2 + R_3} + \frac{R_1 R_4}{R_1 + R_4}} V_S = \frac{\frac{600 \Omega * 300 \Omega}{600 \Omega + 300 \Omega}}{\frac{600 \Omega * 300 \Omega}{600 \Omega + 300 \Omega} + \frac{200 \Omega * 800 \Omega}{200 \Omega + 800 \Omega}} * 15 V$$

$$V_{R_2} = V_{R_3} = 8.333 V$$

So, The voltage across the resistors R_1 & R_4 is calculated by using voltage divider rule (VDR):

$$V_{R_1} = V_{R_4} = V_S - V_{R_2} = V_S - V_{R_3} = 15 V - 8.333 V = 6.667 V$$

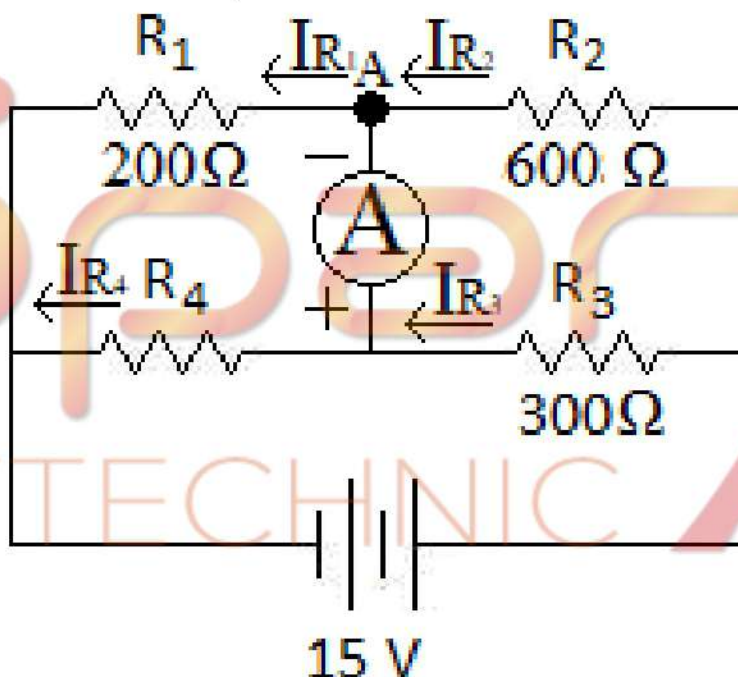
The currents through the resistors are calculated using ohm's law:

$$I_{R_1} = \frac{V_{R_1}}{R_1} = \frac{6.667 V}{200 \Omega} = 33.335 mA$$

$$I_{R_2} = \frac{V_{R_2}}{R_2} = \frac{8.333 V}{600 \Omega} = 13.889 mA$$

$$I_{R_3} = \frac{V_{R_3}}{R_3} = \frac{8.333 V}{300 \Omega} = 27.778 mA$$

$$I_{R_4} = \frac{V_{R_4}}{R_4} = \frac{6.667 V}{800 \Omega} = 8.333 mA$$



By applying KCL at node A, the current passes through the ammeter is calculated:

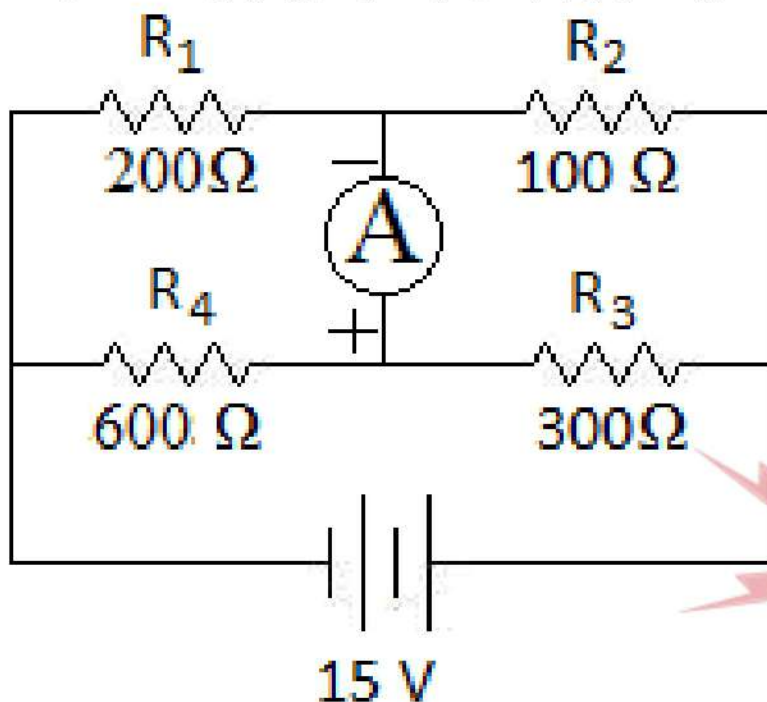
$$\sum I_{in} = \sum I_{out}$$

$$I_{R_3} = I_{R_4} + I_{ammeter}$$

$$I_{ammeter} = I_{R_3} - I_{R_4} = 27.778 mA - 8.333 mA = 19.445 mA$$

122. The reading of the ammeter is equal to:

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



- a. -6.667 mA
- b. 6.667 mA
- c. 3.333 mA
- d. 0 mA

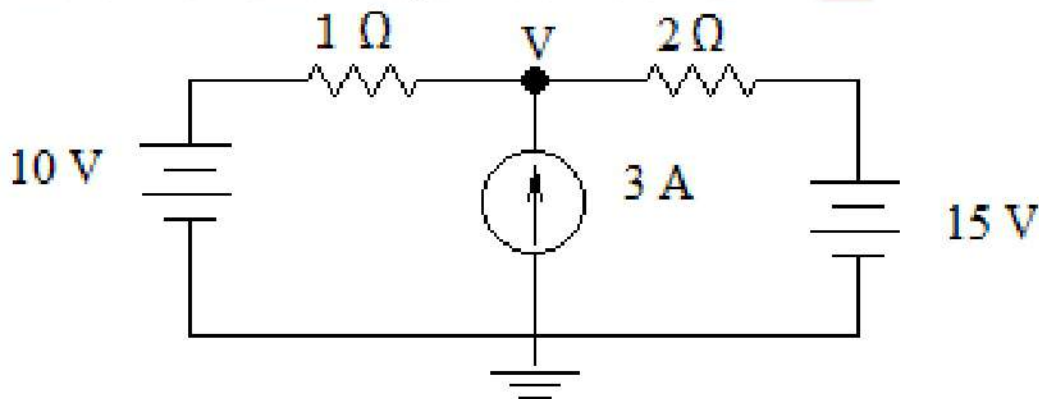
Correct answer: 0 mA

The ratio between resistances on each side is the same:

$$\frac{100\ \Omega}{300\ \Omega} = \frac{200\ \Omega}{600\ \Omega} = \frac{1}{3}$$

Then, the drop voltage is the same on each branch, so there is no voltage difference between the two terminals of the ammeter, then the current that flows through the ammeter is zero

123. Find the value of the node voltage V:



- a. 9.67 V
- b. 8.33 V
- c. 13.67 V

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

d. None of these

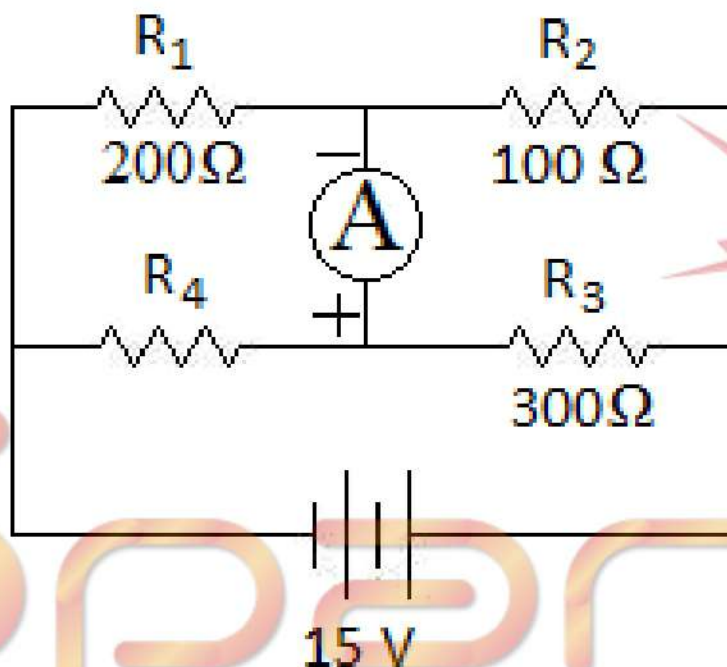
Correct answer: 8.33 V

Using Nodal Analysis Method (NAM), the node voltage is:

$$\frac{V - 10}{1} - 3 + \frac{V - 15}{2} = 0$$

$$V = 13.67 V$$

124. The reading of the ammeter is -13.6363 mA, then the resistance R_4 is equal to:

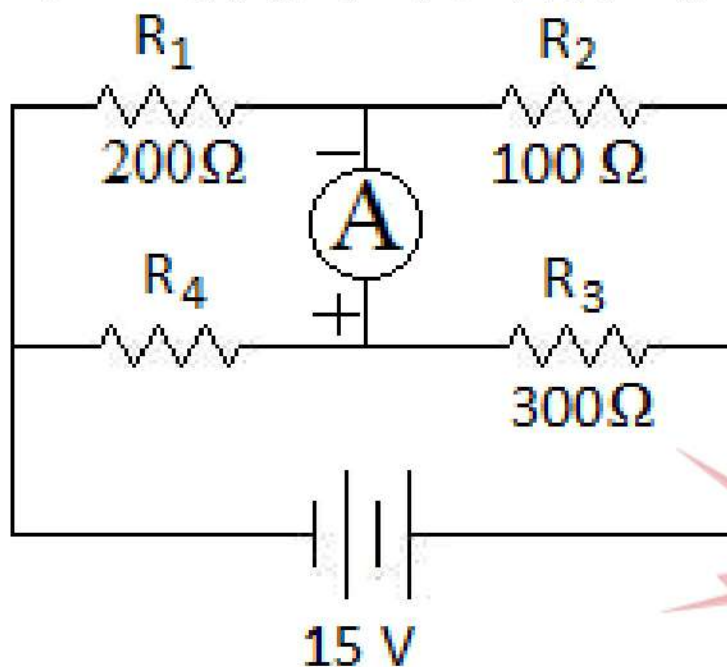


- a. 600 Ω
- b. 150 Ω
- c. Infinity
- d. Zero

Correct answer: Infinity

125. The reading of the ammeter is 15 mA, then the resistance R_4 is equal to:

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



- a. 600Ω
- b. Infinity
- c. 150Ω
- d. Zero

Correct answer: Zero

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Appendix I: Summary of series resonance circuits

Characteristic	Series circuit
ω_o	$\sqrt{\omega_o \omega_o} = \frac{1}{\sqrt{LC}}$
Q	$\frac{\omega_o L}{R} = \frac{1}{\omega_o RC} = \frac{1}{R} \sqrt{\frac{L}{C}}$
B	$\frac{\omega_o}{Q}$
ω_1, ω_2	$\omega_o \sqrt{1 + \left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_o}{2Q}$
$Q \geq 10, \omega_1, \omega_2$	$\omega_o \pm \frac{B}{2}$

